OECD ECONOMIC SURVEYS

SWITZERLAND

1997

SPECIAL FEATURES

- EDUCATION AND TRAINING

OECD ECONOMIC SURVEYS

1996-1997

SWITZERLAND

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

© OECD, 1997.
© Software: 1987-1996, Acrobat is a trademark of ADOBE.
All rights reserved. OECD grants you the right to use one copy of this Program for your personal use only. Unauthorised reproduction, lending, hiring, transmission or distribution of any data or software is prohibited. You must treat the Program and associated materials and any elements thereof like any other copyrighted material.

All requests should be made to: Head of Publications Service, OECD Publications Service, 2, rue Andr´ e-Pascal, 75775 Paris Cedex 16, France.

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

Pursuant to Article 1 of the Convention signed in Paris on 14th December 1960, and which came into force on 30th September 1961, the Organisation for Economic Co-operation and Development (OECD) shall promote policies designed:

- to achieve the highest sustainable economic growth and employment and a rising standard of living in Member countries, while maintaining financial stability, and thus to contribute to the development of the world economy;
- to contribute to sound economic expansion in Member as well as non-member countries in the process of economic development; and
- to contribute to the expansion of world trade on a multilateral, non-discriminatory basis in accordance with international obligations.

The original Member countries of the OECD are Austria, Belgium, Canada, Denmark, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, the United Kingdom and the United States. The following countries became Members subsequently through accession at the dates indicated hereafter: Japan (28th April 1964), Finland (28th January 1969), Australia (7th June 1971), New Zealand (29th May 1973), Mexico (18th May 1994), the Czech Republic (21st December 1995), Hungary (7th May 1996), Poland (22nd November 1996) and the Republic of Korea (12th December 1996). The Commission of the European Communities takes part in the work of the OECD (Article 13 of the OECD Convention).

Publié également en français.

© OECD 1997
Permission to reproduce a portion of this work for non-commercial purposes or classroom use should be obtained through the Centre français d'exploitation du droit de copie (CFC), 20, rue des Grands-Augustins, 75006 Paris, France, for every country except the United States. In the United States permission should be obtained through the Copyright Clearance Center, Inc. (CCC). All other applications for permission to reproduce or translate all or part of this book should be made to OECD Publications, 2, rue André-Pascal, 75775 Paris Cedex 16, France.

Table of contents

Assessment and recommendations 1

I. Recent trends and short-term prospects 15

Overview 15
Renewed fall in output 18
Rising unemployment 22
A stable price level 26
A large and stable current external surplus 30
The outlook to 1998 34

II. Macroeconomic policies 38

Monetary policy 38
Fiscal policy 49

III. Implementing structural reform: a review of progress 61

Reform of unemployment insurance 61
Active labour market policies 62
Working-time flexibility 65
Labour force skills and competences 65
Product market competition 68
Measures to reform business taxation 72

IV. Education and training 79

Attainment and structure of the formal education system 80
Achievement 92
Adult literacy 94
Responsiveness of outputs to client demands 96

The transition from education to work 99
Labour-market outcomes by level of educational attainment 102
Costs of education and training 108
Tertiary education: financing and efficiency 113
Continuing education 119
Summary and conclusions 121

Notes 125
Bibliography 136

Annexes

I. Institutional arrangements in the education and training sector 139
II. Calendar of main economic events 145

Statistical annex and structural indicators 149

Boxes

Text

1. The apprenticeship system 83
2. University financing arrangements 116

Annex

A1. Areas of responsibility and administration 140

Tables

Text

1. Supply and use of current output 21
2. Employment in the 1990s 22
3. Level and structure of recent unemployment 24
4. Price trends 27
5. Current external account 33
6. Short-term projections 37
7. Monetary aggregates 41
8. Government accounts 50
9. Central government budget 53

10. Financial outlook of the Confederation, 1998-2000 59
11. Evolution of unemployment insurance benefits and contributions 63
12. Implementing structural reform: an overview 66
13. Business telecommunications charges 70
14. The main direct taxes related to business activities, 1994 73
15. Corporate tax burden, 1965-94 73
16. Employment in private firms 75
17. Revenue effects of business tax reform 77
18. Percentage of women in the adult working-age population by level of attainment, 1994 82
19. Vocational education by gender, linguistic region and age cohort 85
20. Educational choices after compulsory schooling 86
21. Secondary school drop-out rate 87
22. Tertiary graduation rates, 1994 90
23. Student achievement 93
24. Low literacy and migrant status 96
25. Public views on the relative importance of school subjects 98
26. The employment situation by regions and age groups 101
27. Overall rates of return to education 106
28. Public expenditure on education, 1994 109
29. Basic subsidies to cantonal universities 116

Statistical annex and structural indicators

A. Selected background statistics 150
B. Gross national product (current prices) 151
C. Gross national product (1980 prices) 152
D. Gross value added by main area of activity 153
E. Producer and import prices 154
F. Money supply 155
G. Interest rates and capital markets 156
H. Foreign trade by area 157
I. Foreign trade by commodity group 158
J. Balance of payments 159
K. Labour market: structural and institutional characteristics 160
L. The structure of taxation 161
M. Interest rate margins in banking 162

Figures

Text

1. Potential output and output gap 16
2. Key aspects of economic activity 17
3. Contributions to real GDP growth 20
4. Unemployment and vacancies 23
5. Total and long-term unemployment 26
6. International comparison of consumer prices 28
7. Selected components of inflation 29
8. International competitiveness and trade 31
9. Foreign trade indicators 32
10. Indicators of economic activity 35
11. The adjusted monetary base 39
12. Interest rates 43
13. Exchange rates 44
14. Monetary conditions index and the Taylor rule 48
15. Cyclically-adjusted budget balance 51
16. Budget balances and debt 51
17. Electricity prices in industry, 1995 71
18. Marginal and average tax rates of the federal direct tax on profits 75
19. Educational attainment of the adult working-age population, 1994 81
20. Adult literacy, 1994 95
21. Public confidence in schools 97
22. Relative youth unemployment by type of school system, 1994 100
23. Mean annual earnings by level of educational attainment and gender 104
24. Education-related earnings premia, 1992 105
25. Public expenditure on education, 1993 110
26. Educational expenditure per student in relation to GDP per capita 112
27. Participation in job-related continuing education and training, 1993 120

Annex

A1. The Swiss education system 142

BASIC STATISTICS OF SWITZERLAND

THE LAND

Area (1 000 sq. km)	41.3
Cultivated land, grassland and pastures (1 000 sq. km), 1979/85	15.8
Forest (1 000 sq. km), 1979/85	12.5

Major cities, 1995 estimates (1 000 inhabitants):	
Zurich	353.8
Basle	178.6
Geneva	174.9
Bern	133.3

THE PEOPLE

Population, December 31, 1996, estimates (1 000)	7 085
Number of persons per sq. km	177
Net annual rate of population increase (per 1 000 inhabitants, average 1990-94)	8.1
Civilian employment, 1996 (1 000)	3 789
Percentage distribution:	
Agriculture	4.5
Industry and construction	27.8
Other activities	67.7

Number of foreign workers (1 000), December 1996	869.7
Average increase in the number of foreign workers census, end of August (1 000):	
1962-72	16.8
1973-96	0.9

PRODUCTION

Gross domestic product in 1996 (billion Swiss francs)	360.1
Growth of real GDP, 1990-96 average (annual rate, per cent)	–0.1
Real gross fixed investment in 1996 (in per cent of GDP)	30.0
Growth of real investment, 1990-96 average (annual rate, per cent)	0.1

Value added by origin in 1991 (in per cent of GDP at factor cost):	
Agriculture	3.0
Industry	25.4
Construction	8.1
Services	65.2

THE GOVERNMENT[1]

Public consumption in 1996 (in per cent of GDP)	14.4
Revenue of the Confederation in 1995 (in per cent of GDP)	10.7
Total debt in 1995 (in per cent of GDP)	47.0

Composition of Parliament (in per cent):	National Council	State Council
Radical Democrats	44	17
Democrats-christians	34	16
Socialists	58	5
Central Democratic Union	30	5
Other	34	3
Last elections: 1995		
Next elections: 1999		

FOREIGN TRADE

Exports of goods and services as a percentage of GDP (average 1992-96)	36.0
Commodity exports (fob, million Swiss francs, 1996)	94 174
Percentage distribution:	
By area in 1996	
To OECD countries	81.1
To EU countries	60.7
To OPEC countries	2.8
By categories in 1996	
Raw materials and semi-finished goods	28.4
Capital goods	36.3
Consumer goods	35.3

Imports of goods and services as a percentage of GDP (average 1992-96)	31.8
Commodity imports (cif, million Swiss francs, 1996)	91 967
Percentage distribution:	
By area in 1996	
From OECD countries	91.3
From EU countries	79.0
From OPEC countries	1.5
By categories in 1996	
Raw materials and semi-finished goods	29.5
Energy	3.6
Capital goods	29.1
Consumer goods	37.7

THE CURRENCY

Monetary unit: Swiss franc

Currency unit per US$, average of daily figures:	
Year 1996	1.2361
June 1997	1.4418

1. Confederation, cantons and communes.

Note: An international comparison of certain statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Switzerland by the Economic and Development Review Committee on 24th June 1997.

•

After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 10th July 1997.

•

The previous Survey of Switzerland was issued in July 1996.

Assessment and recommendations

Tight monetary and fiscal conditions as well as massive restructuring...

Switzerland's real GDP fell by 0.7 per cent in 1996, completing a sixth year of broad economic stagnation. The resulting output gap is estimated to be the largest of all OECD countries. The main causes of this long period of stagnation were tight monetary conditions, fiscal consolidation, a slow-down in export market growth and restructuring in several sectors of the Swiss economy, in particular, massive downsizing of the construction sector after the real estate boom of the late 1980s. The collapse in property prices since 1991 required banks to write off a substantial part of their loans and led to more restrictive lending practices as banks tried to improve the quality of their loan portfolios. In addition, high government deficits required measures of budget consolidation as from 1994 which contributed to extending the period of demand weakness. Moreover, business activity is likely to have been negatively affected by the Swiss voters' decision not to ratify the European Economic Area Agreement in the referendum of 6 December 1992. Tight monetary conditions during the 1993 to 1995 period were primarily a result of the real effective appreciation of the Swiss franc by some cumulative 15 per cent, attributable partly to concerns of international investors about the stability of the planned European single currency. A renewed tightening of monetary policy in 1994, in part motivated by the concern that the introduction of VAT at the beginning of 1995 might cause a price-wage spiral if accommodated by too generous money supply, also contributed.

... caused a six year-long economic stagnation...

Although the Swiss franc has weakened considerably since the summer of 1996, this correction came too late to exert a major positive impact on the aggregate economic outcome for 1996. Moreover, Swiss export market growth is estimated to have slowed in 1996 so that the real foreign balance contributed only little to output growth. Domestic demand continued to be weakened by spending restraint on all levels of government in order to reduce budget deficits, and by feeble private consumption; household real disposable incomes declined in 1996 as real wages and employment broadly stagnated, while non-wage incomes fell and income taxes and social security contributions – in particular for health insurance – surged. Construction investment declined further, reflecting the excess capacity built up in the boom of the late 1980s. The continued buoyancy of machinery and equipment investment contributed little to domestic output growth as most of it was imported. The current external surplus remained at about 7 per cent of GDP in 1996.

... with record-high unemployment but stable prices

In line with falling output, labour market performance deteriorated throughout 1996: unemployment averaged 4.7 per cent of the labour force after some temporary improvement in 1995. The unemployment rate rose further to 5.4 per cent (seasonally adjusted) in the first quarter of 1997, a post-war record high. Despite the output gap estimated to have widened to around 4 per cent of potential, and helped by steeply falling import prices at the producer level, consumer price inflation averaged 0.8 per cent in 1996, little changed from 1994 and 1995 (after adjustment for the effect of the introduction of VAT in 1995). Consumer price inflation remained low in early 1997 – at 0.7 per cent over the year to the first quarter – in spite of the recent exchange-rate induced pick-up in import prices. If allow-

ance were made for some upward bias in the consumer price index, the current true rate of inflation might well be zero or even slightly negative.

But the stage is set for economic recovery

With much of the strong exchange rate appreciation during the 1993 to 1995 period now reversed and the European economy gathering strength, exports of goods and services should accelerate and help to turn economic activity around in 1997. But with persistent excess supply of structures, construction investment is likely to remain a drag on domestic demand. Household consumption remains constrained by stagnant real disposable incomes, which will keep the recovery rather modest and insufficient to raise employment. Thus the unemployment rate is projected to rise further in 1997. Supported by easier monetary conditions and the Federal Government's fiscal package, the constraints on economic activity should gradually weaken. Hence, in 1998 growth of GDP could surpass potential output growth for the first time since 1990, and the rate of unemployment could start falling. Although the current exchange rate depreciation will raise prices of imported goods, the remaining large output gap, high cyclical unemployment and falling barriers to competition should keep overall price and wage inflation low.

The past easing of monetary policy...

When a rise in Swiss interest rates triggered renewed upward pressures on the Swiss franc around the middle of 1996, while expectations of economic recovery evaporated, the National Bank increased the supply of reserves to reverse the rise in interest rates. To underline its determination to relax monetary conditions, the National Bank cut the discount rate by ½ percentage point to 1 per cent in September 1996, the lowest since 1978. The three-month Euro-

Swiss franc interest rate fell from 2.5 per cent in the summer of 1996 to 1.7 per cent in the first quarter of 1997, and the effective Swiss franc exchange rate started to decline. Towards the end of 1996, the reversal of the Swiss franc appreciation was accentuated by the progressive strengthening of the US dollar against European currencies, and by the first quarter of 1997 the real (CPI-adjusted) effective Swiss franc exchange rate had fallen to 1¾ per cent above its average level in the 1991-1992 period.

... should be maintained to support the recovery

More generous liquidity provision lifted the adjusted monetary base above its medium-term target path in the course of 1996 and in the fourth quarter of 1996 it was 5.0 per cent above the level recorded a year earlier. It continued to grow strongly in early 1997, exceeding its medium-term target path by some 4 per cent in March 1997. However, since part of the increased liquidity holdings of banks is thought to be due to changes in banks' cash management procedures, current base money statistics may exaggerate the degree of monetary easing. Therefore, monetary policy should continue to take several indicators into account when assessing monetary conditions, to minimise the risk of interpreting structural shifts in the monetary base as endogenous changes in liquidity demand. In particular, in view of the uncertainties surrounding European monetary integration, it seems appropriate to take movements in the exchange rate into account in determining the stance of monetary policy, for example as done since May 1997 when renewed upward pressures on the Swiss franc were liable to threaten the recovery of the economy, and the day-to-day interest rate was allowed to drop to 0.5 per cent. This policy has been consistent with the National Bank's expressed view that at the current juncture inflation risks are minor, and its announced intention to keep the expan-

sion of the adjusted monetary base above its medium-term target path in 1997.

Fiscal policy becomes supportive of economic activity

Following two years of deficit reduction, fiscal policy moved to a broadly neutral stance in 1996, and is set to become more supportive of economic activity in 1997. The deficit/GDP ratio of the consolidated account of the general government (including social security) is projected to rise by 0.6 percentage point to 2.4 per cent of GDP (SF 9.0 billion). Most of this increase is estimated to be structural, reflecting among other things increased expenditure for active labour market programmes; the structural deficit is estimated to be 1½ per cent of GDP in 1997, most of which is attributable to the Confederation. In an effort to support the early stages of economic recovery, the Confederation allocated SF 561 million to a fiscal stimulation programme (*programme de relance conjoncturel*). These funds had been deducted from planned expenditure in the initial budget (2 per cent of credits were blocked in most domains of expenditure). These funds are to be used mainly to encourage cantons and communes to bring forward maintenance expenditures on public infrastructure, thus inducing an increase in public- and some private spending of SF 2.4 billion, which is estimated to raise economic growth by ⅓ percentage point in both 1998 and 1999. Of the funds allocated to the programme, SF 60 million is to be spent on measures to support vocational education and training. This programme appears to be an effective way for the Confederation to stimulate economic activity in the short term without risking a lasting deterioration in its structural budgetary position. In addition, the focus on infrastructure maintenance has the merit of addressing a perceived need: such maintenance expenditure appears to have been delayed in recent years in the context of budget consolidation. But,

combined with the earlier investment bonus (1993-94), this programme risks generating expectations which will lead lower levels of government to delay investment and maintenance expenditures in the early stages of any future economic downturn in the hope of subsequently obtaining additional federal subsidies, thereby aggravating the cycle.

... but budget consolidation in the medium-term remains a priority

The Confederation's budget deficit is projected in the Financial Plan 1998-2000 to decline progressively to SF 4.8 billion (1.2 per cent of GDP) in 2000, largely owing to expenditure restraint. This rate of deficit reduction is recognised by the authorities to be inadequate as it falls well short of what would be required to halt the rise in the Confederation's indebtedness as a percentage of GDP. Accordingly, the Government has proposed a new consolidation plan (Consolidation Plan 2001) which specifies that the deficit must not exceed SF 4 billion in 1999 and SF 1 billion in 2001; thereafter the budget must be balanced over the business cycle. These objectives are to be embodied in the Constitution. The additional reductions in the deficit (SF 1 billion by 1999, SF 3 billion by 2001) from the baseline are to be achieved by reducing expenditures. Lower growth in 1997 than was assumed when the Financial Plan was elaborated means that deficits are likely to be even larger than had been projected. While the Consolidation Plan 2001 does provide for the possibility of delay in achieving these objectives, decisions on the expenditure reductions necessary to achieve them should be made as soon as possible. By adapting fiscal objectives to economic conditions, the Confederation should be able to effect its necessary budget consolidation and at the same time contribute to economic stability. Wisely, the authorities have clearly indicated that they do not intend to repeat the error of the late 1980s, when opportunities for budget consolida-

tion during the expansion phase of the business cycle were missed. Moreover, if the medium-term budget objective is adopted, the Confederation will be obliged to run budget surpluses during the expansion phase of the business cycle to offset the deficits during the downturn phase. This should serve as a useful discipline during economic expansions, when there is often a temptation to use the cyclical improvement in the budget balance to expand expenditure programmes or to cut taxes. Budget balance over the business cycle would also reduce the Confederation's debt/GDP ratio over time, placing its finances on a more solid footing to meet future pressures associated with population ageing.

Labour market reform tightened conditions for access to unemployment benefits

While the projected economic recovery will eventually reduce the cyclical component of unemployment, the task remains to reverse the trend increase in structural unemployment, estimated to have risen from below 1 per cent of the labour force in the 1980s to possibly close to 3 per cent at present. To this end, and to ''revitalise'' the Swiss economy in general, the Federal Government adopted major legislative reforms in 1995 for implementation in 1996 and 1997, which are in accordance with recommendations of the OECD *Jobs Study*. The rather radical reform of unemployment insurance tightened a number of conditions for access to unemployment benefits as from 1 January 1996; in particular, it introduced a more stringent definition of ''suitable work'', raised the penalty for uncooperative behaviour of the unemployed and introduced intermittent pay supplements from unemployment insurance to those who have to accept wages substantially below previous earnings. To meet its objective, it will be crucial to make sure that intermittent pay does not induce a redesign and reorganisation of production and working pat-

terns so as to substitute part-time or low-paid jobs for full-time or high pay jobs.

... and moved away from passive income support to active labour market programmes

The main pillar of unemployment insurance reform is the introduction of the *activation principle*, which, at an early stage of unemployment, makes further receipt of benefits conditional on participation in active labour market programmes (ALMPs). In order to prevent the creation of a ''benefit-dependency culture'', participation in these programmes does not generate new benefit entitlements, as was formerly the case with temporary employment programmes. In this context, the new law has increased the duration of benefit eligibility from 400 days or less under the previous regime to uniformly 520 workdays, irrespective of the length of the contribution period in excess of the minimum period.

Risk of unintended prolongation of unemployment spells

It is too early for a comprehensive evaluation of the effects of the unemployment insurance reform. But in view of the recent rapid rise in unemployment, the planned number of 25 000 ALMP slots will be insufficient to meet the increased demand by job seekers. This implies that some unemployed will be entitled to *passive* benefit payments during a longer period than hitherto, which will cause expenditures of the unemployment insurance fund to increase and result in a higher deficit than experienced in the past. In spite of tight benefit eligibility conditions, it may also raise the rate of structural unemployment due to reduced search efforts and lowered willingness of these unemployed to accept job offers, thereby running counter to the objectives of the unemployment insurance reform. Hence, to make the activation principle fully operational, it appears necessary to raise the number of ALMP slots in

line with the increased level of unemployment. In accordance with the recommendations of the *OECD Jobs Study*, the increase in the maximum duration of unemployment benefits should be reconsidered: it more than quadrupled during the past twenty years and is likely to have contributed to the prolongation of unemployment spells and to higher structural unemployment.

Unclear efficacy of the new competition law

Another major reform implemented in 1996 is the reform of the Cartels Act and the new internal market law; the latter aims at the elimination of restrictions on competition included in cantonal and communal laws, notably in respect of public procurement. Since the new Cartels Act has been in force for less than a year, it is too early to judge whether the factual prohibition of agreements setting prices, production volume or territorial distribution is as effective as the ban on cartels in most other OECD countries. To make the requirement of effective competition as demanding a concept for securing contestable markets as those used commonly in other OECD countries, the government should insist on strict enforcement of the law in practice. Further efforts are also needed to eliminate the inconsistency between the inter-cantonal agreement on public procurement which does not prevent discrimination between local and other Swiss bidders under certain circumstances and the domestic market law which aims at creating a unified national market for all public procurement.

Business tax reform

An important step towards reform of business taxation is currently being considered in Parliament**.** The draft reform law aims to replace the existing federal progressive taxation of business profits by a flat-rate tax of 8.5 per cent. The old law discriminated against small and labour-intensive busi-

nesses. In addition, it has been decided to abolish the corporate capital tax, and to reduce further the stamp duty on new share capital. Another reform element is the alignment of the taxation of capital gains from restructuring of holding companies with EU rules, in order to restore Switzerland's attractiveness as a location for holdings. Also, a more generous tax treatment of the acquisition of companies' own stocks will increase entrepreneurial flexibility. Although the draft reform law remedies important anomalies of business taxation, the task of a thorough reform of the excessively complex federal and cantonal corporate tax system remains, including the abolition of the double taxation of dividends.

Other reform issues

Enhancing work time flexibility remains on the political agenda, even though the recent proposal of the Federal Government to reform the federal labour law (*loi fédérale sur le travail*) was rejected by a referendum on 1 December 1996. And in spite of first steps towards liberalisation, non-competitive practices remain in the telecommunications sector, having permitted substantial price increases in recent years. This should be remedied by the planned reform of the Telecommunications Act, with the objective to abolish the voice and network monopolies, so that from the beginning of 1998 onwards new entrants will be able to compete with Swiss Telecom across the board. And given the high prices of electricity for industry in Switzerland, the time has come to open up the Swiss electricity market and make it compatible with the forthcoming internal electricity market of the EU.

The education and training system performs well

This year's special topic is education and training, one of the structural themes identified for surveillance in the OECD *Jobs Study*. Switzerland's education and training system has an enviable record of success: Swiss children do well in international comparisons of scholastic achievement, there are few drop outs from non-compulsory secondary education and the transition from education to work is smoother than in most other countries. A key factor in the latter two achievements is the apprenticeship system (also known as the dual system because of the way it combines formal instruction with on-the-job training); it offers (post-compulsory) educational pathways to students with a wide variety of abilities and interests. Most students continue to choose the apprenticeship route to completing their secondary education, although since the mid-1980s there has been a trend for more able students to choose instead an academic route leading to university studies. The proportion of students who continue their education at the tertiary level is near the OECD average, although in Switzerland the proportion in non-university tertiary institutions is much higher. This reflects the dominance of the dual system: graduating in this system rarely provides a university entrance qualification.

... and is being reformed to meet future challenges

As in other countries, earnings and employment prospects improve with educational attainment. But in Switzerland, the improvements appear to be particularly marked for graduates of the dual system. Nevertheless, concerns have emerged that the dual system may not be well adapted to meet the growing demand for workers who are flexible and/or possess advanced skills. To meet these challenges, and to stem the flow of able students from the dual system, a new pathway *(maturité professionnelle-hautes écoles spécialisées*) has been created which places more emphasis

on general education and enables participants in the dual system to continue their education to the level of a university degree. This is a valuable reform, making education in the dual system more relevant to changing labour-market requirements and giving bright students better opportunities to realise their potential.

... but costs are high

The major weakness of Switzerland's education and training system is that it is expensive. In part, this results from factors specific to Switzerland, such as regional decentralisation, the multilingual context and a commitment to high quality service. However, it is also true that more is spent on capital equipment than in other countries, teachers' salaries are high in relation to national income per capita, and the student-to-teacher ratio is low. There is little evidence that all this expenditure is essential to achieve the favourable results. Even within Switzerland, differences in expenditure per student between cantons have no effect on student achievement when differences in social structure are taken into account. Capital expenditure could be reduced by reviewing building codes to make them more reasonable (and more comparable with those in neighbouring countries). Educational authorities could do well to reconsider whether teachers'salaries really need to be so high in relative terms and whether there is scope to raise the ratio of students-to-teachers without materially affecting achievement. More generally, more research is required into what contributes to student achievement so that educational resources can be used more efficiently. In addition, consideration should be given to modifying funding arrangements to give the private sector greater opportunity to compete in the supply of educational services.

And the financing of university education is inefficient and unfair

Another shortcoming in the education and training system concerns the financing of universities. Almost all their costs are paid for by the public sector, and there is little assistance with students' living costs. These arrangements will be inefficient if they do not correspond to the breakdown of the benefits of university education into private and social components and insofar as liquidity constrained and/or risk averse students under-invest in tertiary education owing to limited assistance with living costs. Insofar as the latter factor is a greater barrier to students from poorer backgrounds, these arrangements are also a source of social inequity. Another question is whether social equity may be further undermined by the public financing of universities where this provides large implicit subsidies to individuals with higher lifetime earnings than the average taxpayer. One arrangement which could be more efficient and fairer would be to lend students money to contribute towards tuition- and living costs and to require repayment through the tax system conditional on income exceeding a certain amount. Similar arrangements now exist in a number of OECD countries and could also be considered for financing continuing education.

Summing up

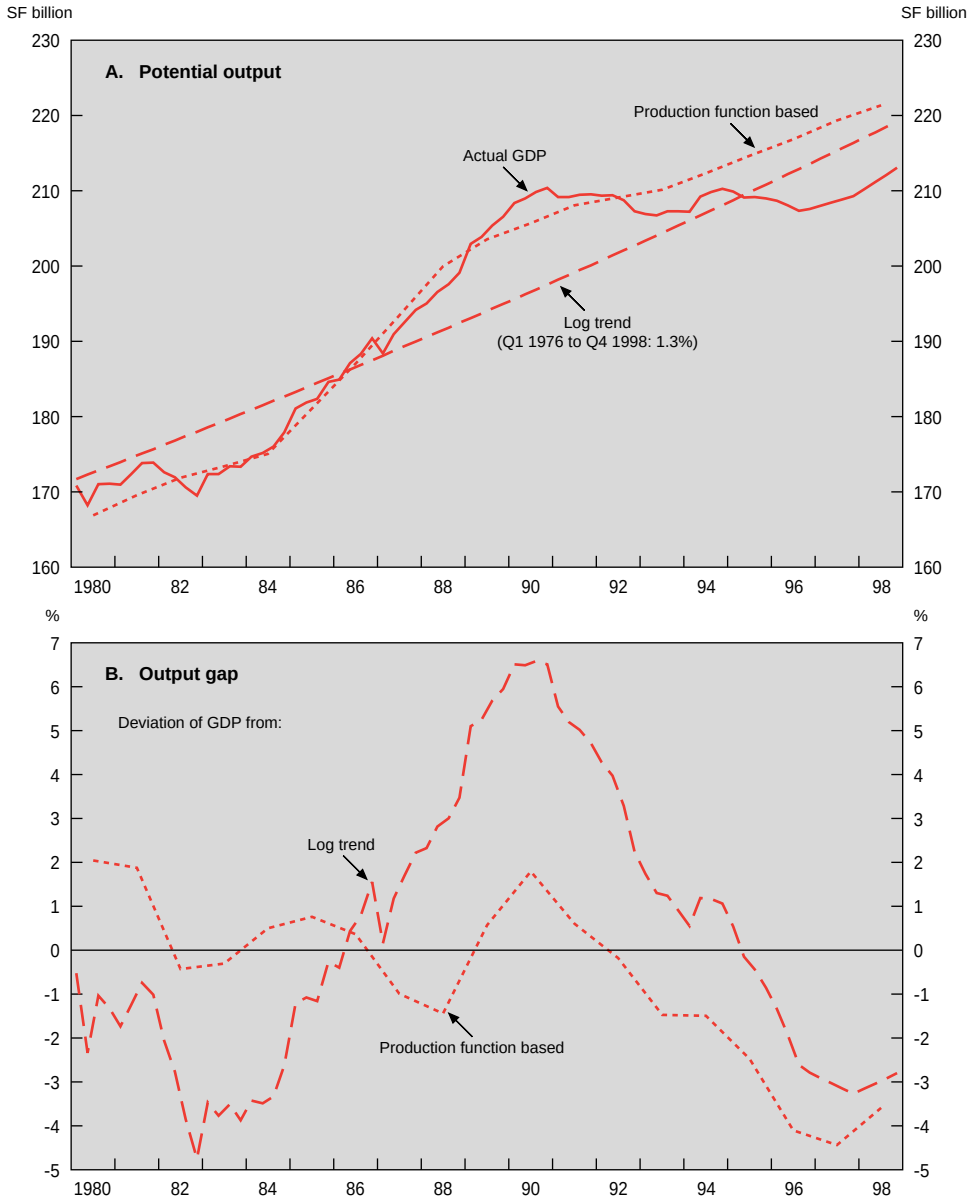
Altogether, with much of the strong appreciation of the Swiss franc since 1993 now being reversed, monetary policy eased, and fiscal policy more supportive of the cycle, the stage is set for recovery from the long period of stagnation. Recent structural reforms, such as the revision of unemployment insurance, the modernisation of the Cartels Act, the new internal market law, the planned move towards more rational business taxation, and the creation of new options of secondary and tertiary education, should help to lift the economy's growth potential and safeguard Switzerland's position among the richest OECD countries.

I. Recent trends and short-term prospects

Overview[1]

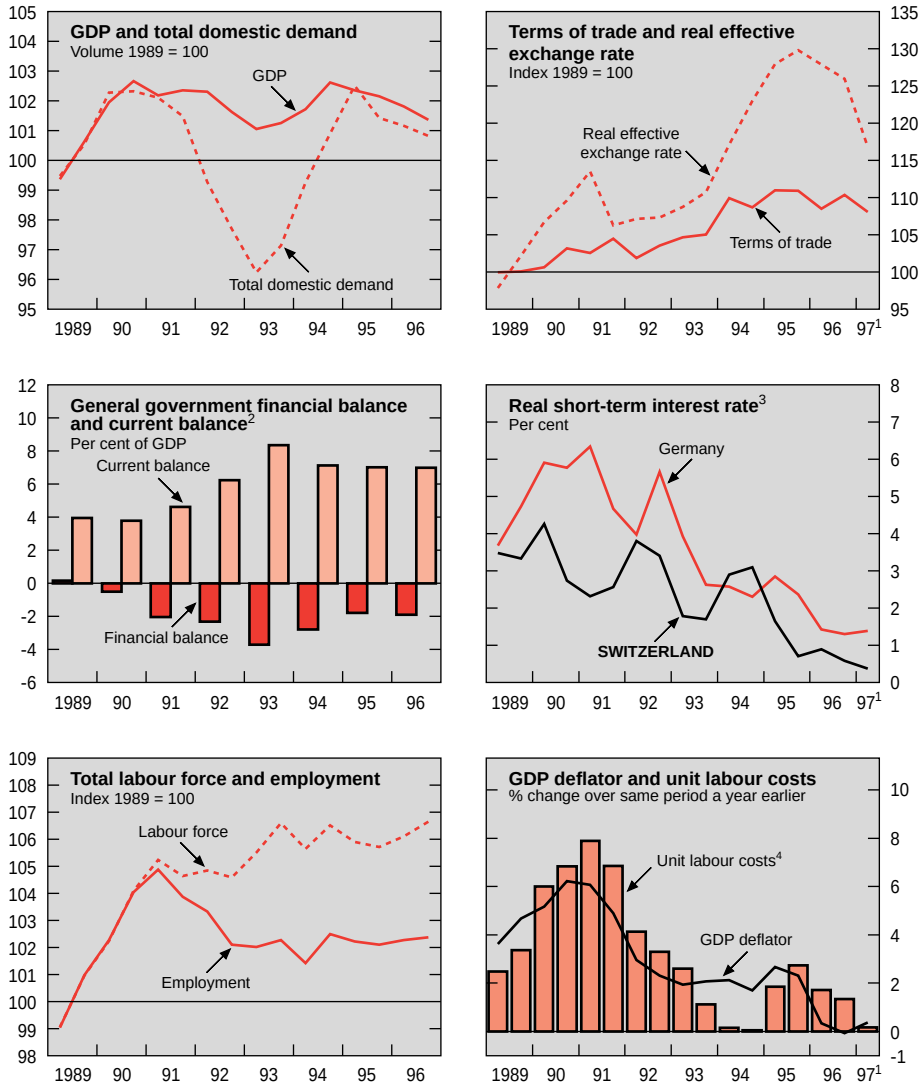
Most of the contractionary forces which had depressed the Swiss economy during the first half of the 1990s continued to exert their influence in 1996: the real effective Swiss franc exchange rate – although no longer rising – remained high on average relative to the early 1990s; with the usual lags at work, its depreciation since the end of 1995 came too late to have a major positive impact on economic activity in 1996. In addition, estimated growth of Swiss export markets fell by nearly 4 percentage points to 6¼ per cent in 1996. On the domestic demand side, large unused capacity in the real estate sector prevailed, which dampened private construction activity and put downward pressures on property prices; the latter effect induced banks to strengthen their balance sheets by restraining the extension of fresh credits, giving rise to signs of a ''credit crunch''. And all levels of government continued to control expenditures in order to reduce budget deficits. All this contributed to the fall in real GDP by 0.7 per cent in 1996, after stagnation in 1995.[2] Output in 1996 was ¾ per cent below the level recorded in 1990, making the past six years the longest period of stagnation since the 1930s.[3] This opened a substantial gap between potential and actual output: based on the OECD's estimate of the Swiss aggregate production function the output gap may have reached 4 per cent of potential output in 1996 and 4½ per cent in the first half of 1997 (Figure 1), in spite of the estimated substantial reduction in potential output growth during the first half of the 1990s. Largely reflecting the protracted weakness of economic activity, the current external surplus remained high, employment stagnated and unemployment rose further, while wage and price inflation stayed low (Figure 2). During the 1991-96 period the growth performance of the Swiss economy was the poorest and the output gap estimate the largest of all OECD countries.[4]

Figure 1. **POTENTIAL OUTPUT AND OUTPUT GAP**[1]



1. Data for 1997 and 1998 are OECD projections.
Source: OECD.

Figure 2. **KEY ASPECTS OF ECONOMIC ACTIVITY**



1. Figures for the first half of 1997 are OECD estimates.
2. The general government financial balance excludes social security.
3. 90-days Euro-Swiss franc rate and German three-month interbank rate deflated by the private consumption deflator.
4. Total economy.

Source: OECD, *National Accounts, Main Economic Indicators*, and Secretariat estimates.

With much of the real exchange rate appreciation of the 1993-95 period now reversed and the European economy gathering strength, the Swiss economy is projected to recover gradually this year and next. This view is supported by National Accounts statistics (new system) for the first quarter of 1997 which indicate – on a seasonally adjusted basis – a small improvement of real GDP compared with the fourth quarter of 1996. However, this recovery is not expected to be sufficiently vigorous to make substantial inroads into unemployment. Wage and price inflation are likely to remain low, given the persisting large output and employment gap.

Renewed fall in output

The decline in output in 1996 was most pronounced in the construction industry (–3½ per cent[5]) which reflects a further steep fall in public construction investment (–5¼ per cent), an accelerating decline in residential investment (–2¾ per cent), and broad stagnation of business construction investment (–½ per cent);[6] the latter had contracted by a cumulative 24 per cent in the 1994-95 period. According to the revised National Accounts, the downturn of construction investment continued in the first quarter of 1997. Excess supply of office space is reported to have remained at a high level, while the number of vacant dwellings even increased in 1996. The fall in public investment reflects ongoing government budget consolidation efforts at all levels of government. Altogether, the cumulative decrease in construction investment volumes amounts to about 10 per cent since 1991, notwithstanding the (largely policy-induced) rise by a good 5 per cent in 1994. Accentuated by declining relative prices, the share of construction in nominal GDP fell to 13½ per cent in 1996, below its previous trough during the deep recession in the mid-1970s and down 4½ percentage points from the previous peak of 18 per cent in 1989[7]. Value added in industry (excluding construction) is estimated to have decreased by about 1 per cent in 1996, and the KOF's (Survey based) measure of industrial capacity utilisation fell to 83.2 per cent in 1996, down from 85.2 per cent in 1995, which was roughly the average during the 1981-90 period; it had peaked at 89.6 per cent in the first half of 1990. Value added in the service sector (including government) – constituting 58½ per cent of GDP – is estimated to have stagnated in 1996.

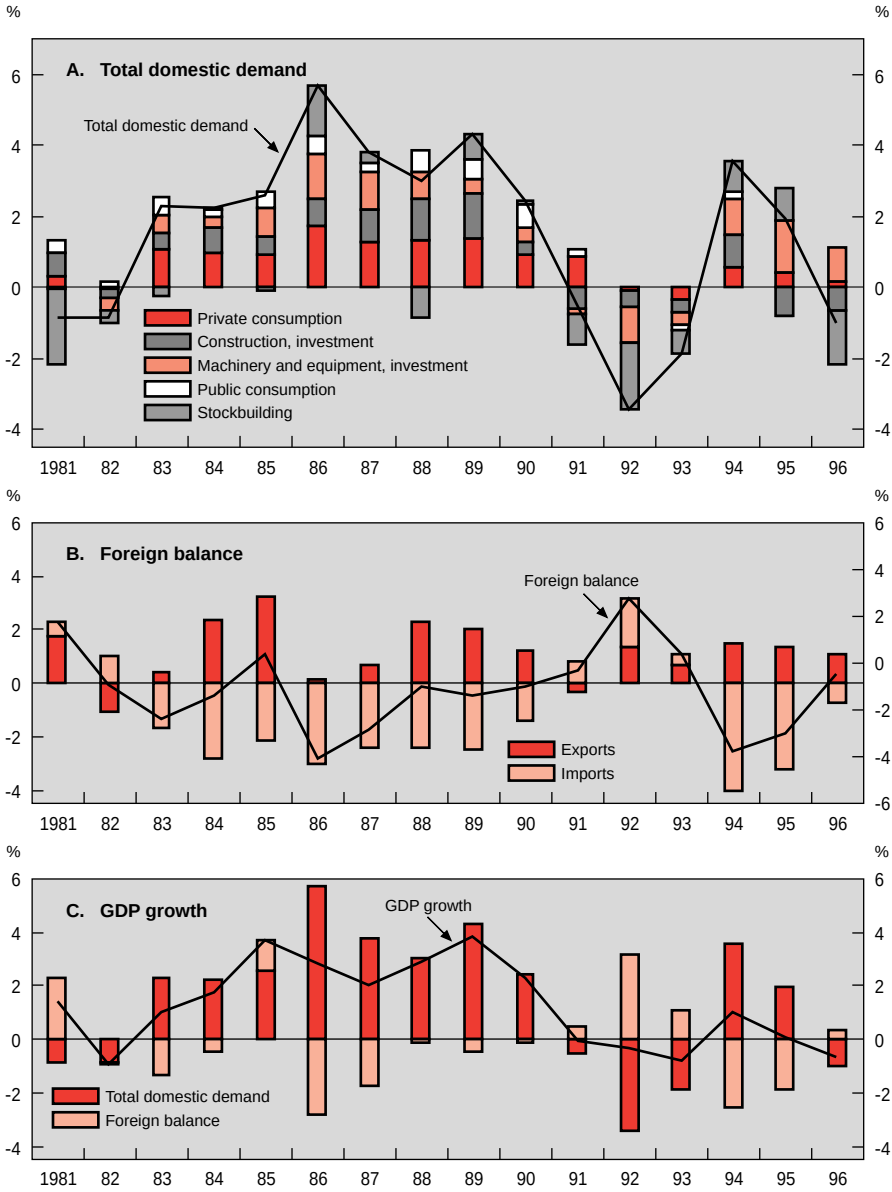
Among individual demand components, the largest negative contribution to domestic demand growth in 1996 is attributable to the residual item which comprises stockbuilding but also statistical errors (Figure 3); it appears that the low stock of orders and the gloomy outlook prompted businesses to curtail inventories. Domestic demand was further weakened by a small decline in public consumption which largely reflects a small reduction of employment in the government sector.

Real household consumption was broadly flat in 1996[8] (Table 1); over the past five years its average annual growth amounted to only 0.2 per cent. In 1996, primary household incomes were curtailed by stagnating employment, modest wage increases in the face of high and rising unemployment, and declining property and other incomes. Disposable household incomes were further eroded by strongly rising income taxes (the usual effect in even-numbered years of the biennial system for collecting federal income taxes), and by a surge in health insurance contributions following the introduction of the new health insurance law[9] at the beginning of 1996. As a result, nominal disposable incomes remained flat, so that real disposable household incomes decreased by about 1 per cent in 1996. Households responded by a reduction in the personal saving ratio from 10½ per cent in 1995 to 9¼ per cent in 1996.

Only the continued buoyancy of machinery and equipment investment supported domestic demand[10] in 1996, but with an import content of between 80 and 90 per cent the resulting contribution to domestic value added is small. Apparently, a major investment motive was the continuing rationalisation of production to cope with the loss in price competitiveness from the recent appreciation of the Swiss franc. This investment was also stimulated by the fall in investment goods prices by about 13 per cent from 1993 to 1996, in part an effect of the exchange rate appreciation. The abolition of the turnover tax on investment goods may also have helped to speed up replacement investment. Firms' efforts to improve competitiveness through increased capital investment appear to have been partly successful as the growth of exports of goods and services – albeit modest by historic standards – only slowed down a little from 1995 to 1996, in spite of the substantial real effective exchange rate appreciation and reduced growth in export markets. With the marked decline in import growth, the real foreign balance contribution to GDP growth became positive in 1996.

Figure 3. **CONTRIBUTIONS TO REAL GDP GROWTH**

As a percentage of GDP in the previous year



Source: OECD, *National Accounts.*

Table 1. **Supply and use of current output**

	1991 at current prices		Percentage volume[1] change from previous period, annual rates, s.a.						
	SF billion	Percentage share of GDP	Average 1991-94	1995	1996	1995		1996	
						I	II	I	II
Private consumption	190.5	57.5	0.4	0.7	0.3	1.5	–0.7	1.2	–0.6
Public consumption	46.6	14.1	0.4	–0.1	–0.2	–0.7	0.0	–1.0	1.1
Gross fixed capital formation	84.8	25.6	–0.7	2.3	1.3	4.1	–2.8	4.8	–1.6
Construction	55.9	16.9	–0.5	–4.3	–3.5	–5.7	–5.0	–2.2	–4.6
Machinery and equipment	28.9	8.7	–0.9	14.1	8.4	22.2	0.8	15.6	2.6
Final domestic demand	321.9	97.2	0.1	1.0	0.5	1.9	–1.2	1.9	–0.6
Change in stocks[2, 3]	4.5	1.4	–0.6	0.9	–1.5	0.7	–0.5	–1.3	0.0
Total domestic demand	326.5	98.6	–0.5	1.9	–1.0	3.2	–2.1	–0.5	–0.7
Exports of goods and services	116.7	35.3	1.9	3.0	2.4	0.7	3.8	2.3	1.2
Exports of goods	89.4	27.0	2.1	4.2	2.5	1.5	3.3	2.8	1.2
Exports of services	27.3	8.3	1.3	–1.9	2.0	–1.4	5.4	0.7	1.8
Imports of goods and services	112.1	33.9	0.7	6.6	1.5	8.5	–0.3	2.4	1.4
Imports of goods	97.8	29.5	0.6	6.3	1.2	8.0	–1.3	2.5	1.1
Imports of services	14.3	4.3	1.6	8.7	3.8	12.3	9.7	2.0	3.5
Change in foreign balance[2]	4.6	1.4	0.5	–1.9	0.3	–1.9	0.9	–0.1	–0.1
Gross domestic product	331.1	100.0	0.0	0.1	–0.7	–0.5	–0.4	–0.7	–0.9
Gross national product	345.4		–0.3	1.8					
Memorandum items:									
GDP deflator			3.0	2.5	0.1	3.6	1.0	–0.3	0.2
Private consumption deflator			3.5	1.8	1.2	1.9	1.8	0.5	1.9
Index of industrial production[4]			0.3	1.9	–2.0	–3.2	4.5	–4.1	–3.9

1. At 1980 prices.
2. As a percentage of GDP of the previous period.
3. This item is partly a residual value; it also contains statistical errors.
4. Excluding mining and quarrying.

Source: Département fédéral de l'économique publique, *La vie économique;* KOF/ETH, *Konjunktur;* OECD.

Rising unemployment

In spite of falling real output in 1996, the number of ''persons in employment'' (''*population active occupée*'') – those who have worked for at least 6 hours per week, including seasonal and frontier workers – remained stable[11] (0.0 per cent). Employment of Swiss nationals improved somewhat (+0.8 per cent) while that of foreigners declined by 1.7 per cent. The four major categories of foreign workers[12] suffered job losses in 1996, with the fall in the employment of seasonal workers by about 14 per cent being the most pronounced. The same statistics recorded a loss of jobs in manufacturing by 2.6 per cent, a fall in employment in the construction sector by 5.4 per cent, but an increase in the number of jobs in the services sector by 1.1 per cent in 1996 (Table 2).

If taken at face value, the total number of ''persons in employment'' would imply a fall in labour productivity by about 3/4 per cent in 1996, which is at odds with numerous reports about the ongoing and widespread restructuring of pro-

Table 2. **Employment in the 1990s**

Percentage change from previous period

	1990	1991	1992	1993	1994	1995	1996
Persons in employment	3.2	2.0	–1.8	–0.9	–0.2	0.4	0.0
Agriculture	..	..	–2.4	1.7	–4.5	3.9	6.8
Manufacturing	..	..	–6.1	–3.5	–1.0	1.6	–2.6
Construction	..	..	–5.5	–5.0	2.2	1.2	–5.4
Services	..	..	0.2	0.4	0.0	–0.2	1.1
Total number of jobs in secondary and tertiary sector[1]	..	..	..	–2.6	–1.1	0.5	–0.2
Manufacturing	..	..	..	–4.8	–4.1	–0.8	–1.6
Construction	..	..	..	–5.9	0.4	–1.4	–3.3
Services	..	..	..	–1.4	–0.3	1.2	0.6
Registered vacancies	–1.7	–38.9	–22.6	–29.7	–9.0	3.3	3.8
Memorandum item:							
Labour productivity[2]	–0.9	–1.2	1.3	–0.2	1.2	–0.1	–0.9

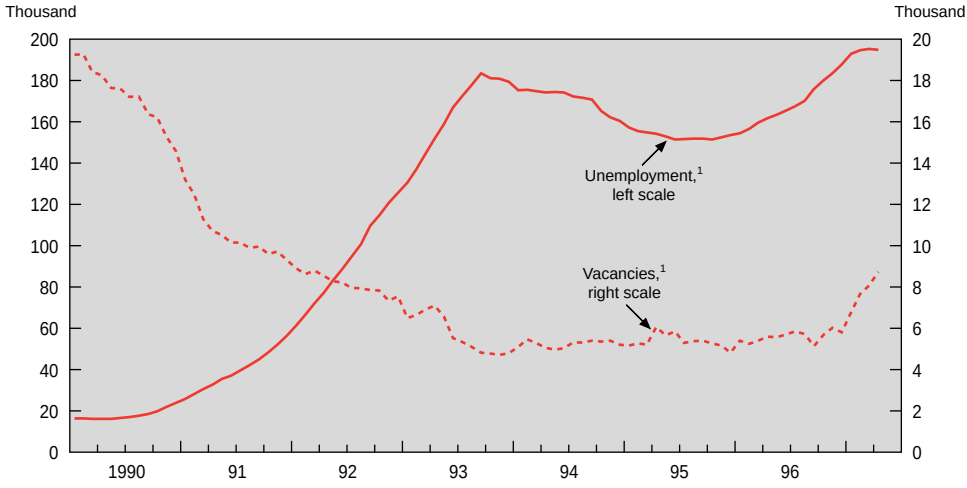
1. Jobs of at least 6 hours per week.
2. Real GDP per person in employment.

Source: Office fédéral de la statistique, *La statistique de la population active occupée;* KOF/ETH, *Konjunktur*.

duction processes, together with the recent surge in capital deepening investment. However, the broad definition of the persons-in-employment statistics together with the continuing trend away from full-time towards part-time employment in 1996 is likely to give too favourable an impression of changes in hours worked.[13] This view is supported by another employment series which is the number of jobs reported by firms in the secondary and tertiary sector. It suggests a fall in total employment (excluding the primary sector) by 0.2 per cent in 1996 and a smaller increase in services employment than reported by the persons-in-employment statistics.

After its decline from a monthly peak of 5 per cent of the labour force in early 1994 to a trough of roughly 4 per cent in the summer of 1995, the seasonally adjusted rate of registered unemployment (the administrative measure) rose again during the second half of 1995. It reached 4.5 per cent at the beginning of 1996, where it remained – with only minor fluctuations – during the first three

Figure 4. **UNEMPLOYMENT AND VACANCIES**



1. Seasonally adjusted.
Source: OECD.

Table 3. **Level and structure of recent unemployment**

	1993		1994		1995		1996		1997 Q1	
	Number	Rate[1]	Number	Rate[1]	Number	Rate[1]	Number	Rate[1]	Number	Rate[1]
Total	163 135	4.5	171 038	4.7	153 316	4.2	168 630	4.7	204 666	5.7
According to region										
German speaking	96 959	3.7	99 463	3.8	86 656	3.3	101 064	3.9	126 333	4.8
French and Italian speaking	66 176	6.5	71 574	7.1	66 661	6.6	67 566	6.7	78 333	7.8
According to gender										
Women	66 571	4.7	73 072	5.2	67 843	4.8	71 801	5.1	82 389	5.8
Men	96 564	4.4	97 966	4.4	85 473	3.9	96 829	4.4	122 277	5.5
According to citizenship										
Swiss	99 631	3.5	102 757	3.7	88 662	3.2	92 735	3.3	110 640	3.9
Foreign	63 504	7.8	68 281	8.4	64 655	8.0	75 895	9.3	94 027	11.6
According to age										
15-24 years	32 098	4.9	30 831	4.7	26 058	3.9	28 168	4.3	35 226	5.3
25-49 years	103 459	4.8	107 410	4.9	96 479	4.4	107 788	5.0	130 833	6.0
50 years and more	27 578	3.5	32 797	4.2	30 779	3.9	32 674	4.2	38 607	4.9
According to sectors										
Agriculture	1 522	1.0	1 590	1.1	1 500	1.0	1 851	1.3	2 622	1.8
Energy, mining	448	1.8	523	2.1	495	2.0	552	2.3	720	2.9
Arts and crafts, industry	40 916	5.5	40 146	5.4	31 010	4.1	32 128	4.3	38 592	5.2
Construction	17 441	5.6	16 455	5.2	13 918	4.4	18 050	5.8	26 708	8.5
Distributive trade, catering, repairs	42 437	5.5	46 889	6.2	44 877	5.9	49 525	6.6	57 332	7.6
Transport and communication	5 183	2.4	5 363	2.4	4 497	2.0	5 057	2.3	6 231	2.8
Banking, insurance, consultancy	20 970	4.9	20 395	4.7	18 010	4.2	20 187	4.7	24 071	5.6
Other services	18 827	3.0	21 650	3.4	21 078	3.4	21 717	3.5	24 781	3.9
Government	3 784	2.5	4 362	2.8	5 909	3.8	7 078	4.6	8 831	5.7
Not specified[2]	11 607	–	13 663	–	12 025	–	12 486	–	14 778	–
Total	163 135	4.5v	171 038	4.7	153 316	4.2	168 630	4.7	204 666	5.7
		Share in %		Share in %		Share in %		Share in %		Share in %
According to duration										
0-6 months	85 667	52.5	73 651	43.0	71 078	46.3	81 417	48.3	100 197	49.0
7-12 months	49 243	30.2	48 148	28.2	38 287	25.0	43 167	25.6	49 858	24.4
More than 1 year	28 225	17.3	49 239	28.8	43 951	28.7	44 046	26.1	54 611	26.7
According to function										
Specialists	80 132	49.1	83 246	48.7	73 093	47.7	80 390	47.7	99 673	48.7
Auxiliaries	64 327	39.4	65 981	38.6	60 188	39.3	67 890	40.3	82 365	40.2
Apprentices/students	9 936	6.1	11 995	7.0	10 839	7.1	10 588	6.3	11 141	5.4
Other	8 740	5.4	9 816	5.7	9 196	6.0	9 762	5.8	11 487	5.6

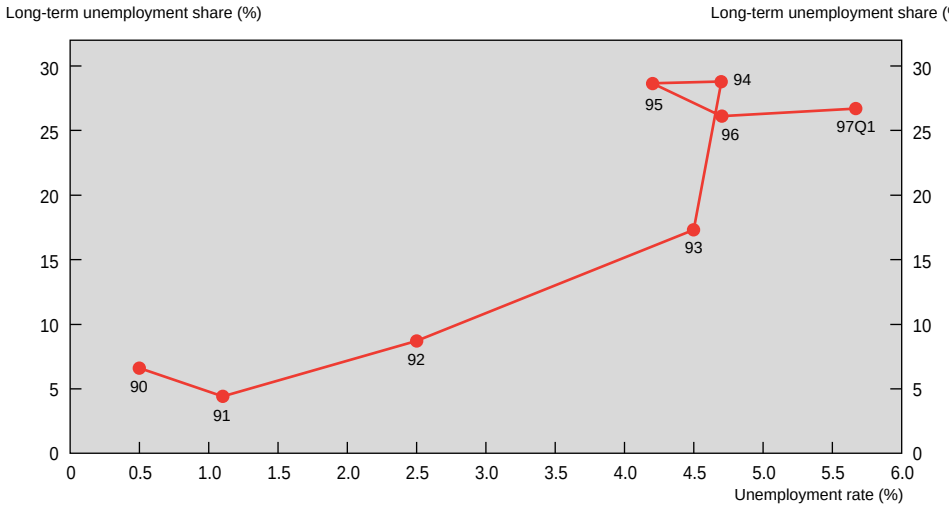
1. Official unemployment statistics include those who are partially unemployed. With the exception of sectoral unemployment rates, official unemployment rates are expressed as a percentage of the work-force in the 1990 population census; sectoral unemployment rates are expressed as a percentage of the work-force in the corresponding period.
2. Persons seeking employment for the first time or re-entering the work-force.

Source: Office fédéral de l'industrie, des arts et métiers et du travail (unemployment); Office fédéral de la statistique.

quarters of the year. But with the labour force increasing again in 1996 (+½ per cent), the unemployment rate rose rapidly from the autumn of 1996 onwards. It averaged 4.7 per cent[14] in 1996, up from 4.2 per cent in 1995.[15] Registered unemployment rose further in 1997, to 5.4 per cent of the labour force in the first five months, a figure last recorded in the 1930s. A small improvement to a seasonally-adjusted rate of 5.3 per cent of the labour force was registered in May 1997. At around 200 000 in the first quarter of 1997, the absolute number of registered unemployed was more than ten times its level in 1990 (Figure 4). An encouraging feature of recent developments is the increase in the number of vacancies (+3.8 per cent) in 1996; it appears to have risen further in the first five months of 1997, albeit from a low level.

As in the past, the foreign workforce was hit particularly hard by weak labour demand: the rate of unemployment for foreigners averaged 9.3 per cent in 1996 and jumped to 11.6 per cent of the foreign labour force in the first quarter of 1997 (Table 3); it stood at 10.9 per cent in May. This is likely to reflect the large share of unskilled workers in the foreign labour force, and their concentration in industries which have been suffering particularly from the current economic crisis, such as construction, the hotel trade and catering. The rate of male unemployment remained lower than that of women, but the differential has narrowed since 1995. Also, French and Italian speaking parts of the country were relatively more affected by unemployment than German speaking cantons, but the latter have experienced a sharper deterioration since 1995. Among age groups, prime age workers suffered most from joblessness: for example, the unemployment rate of the age group 30-34 years reached 7¼ per cent in the first quarter of 1997. Unemployment among the very young (15-19 years) remained below the 2 per cent mark in 1996, largely reflecting the effect of the apprenticeship system which ensures a smooth transition from school to employment (see Chapter IV below). However, the unemployment rate of this very young age group jumped to just under 3 per cent in early 1997. The number of long-term unemployed (more than one year) remained broadly stable in 1996, but surged in early 1997. Hence, the fall in the share of the long-term unemployed in total unemployment from 1995 to 1996 is almost entirely due to the sharp increase of total unemployment (Figure 5).

Figure 5. **TOTAL AND LONG-TERM UNEMPLOYMENT**[1]



1. Long-term unemployment is defined as individuals looking for work for one year or more.
Source: Office fédéral de la statistique; OECD.

A stable price level

The strength of the Swiss franc exchange rate and the large and widening output and employment gaps have kept Switzerland's price and wage inflation very low in since 1994. Average price inflation as measured by the consumer price index (CPI) was 0.8 per cent in 1996, about the same as in 1994 and 1995 (after adjustment for the introduction of VAT in 1995, which entailed an estimated step increase in the CPI of 1 to 1¼ per cent in 1995). Consumer price inflation remained under control in early 1997 – it was 0.6 per cent during the twelve months to May 1997 – in spite of the recent weakening of the exchange rate, which induced a modest pick-up in import prices at the consumer level in late 1996 and in early 1997 (Table 4). Thus, Swiss consumer price inflation remained among the lowest in the OECD and distinctly below the OECD average (Figure 6). If allowance is made for the – unknown – inherent upward bias[16] in the CPI stemming from product substitution, unmeasured quality improvements, new goods and outlet substitution, the current true rate of inflation may well be zero or even negative.

Table 4. **Price trends**

Percentage changes from previous year

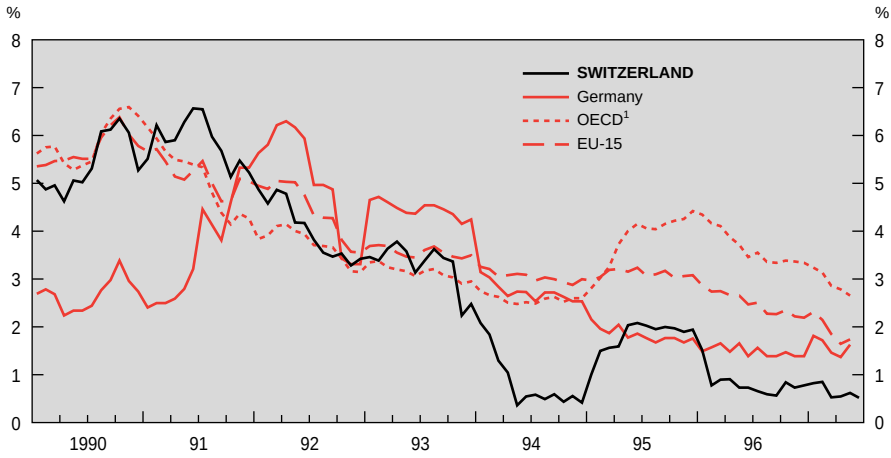
	May 1993 Weight in %	1993	1994	1995	1996	1996		1997
						Q3	Q4	Q1
Total supply index[1]	100.0	0.2	–0.3	0.0	–2.3	–2.8	–2.4	–1.4
Producer price index	70.2	0.4	–0.5	–0.1	–1.8	–2.1	–2.0	–1.4
Import price index	29.8	–0.3	–0.1	0.3	–3.7	–4.7	–3.4	–0.9
Consumer price index	100.0	3.3	0.9	1.8	0.8	0.6	0.8	0.7
Goods	44.2	1.7	0.2	0.3	0.2	–0.0	0.8	0.7
Non-durables	25.6	1.4	0.3	0.8	0.9	0.6	1.9	1.8
of which:								
Food, beverages, tobacco	16.3	–0.1	0.5	0.8	–0.1	–0.4	0.1	0.4
Energy	3.2	1.2	–3.2	5.5	5.3	4.6	9.0	6.3
Services	55.8	4.6	1.4	3.0	1.3	1.1	0.8	0.7
of which: Rents	22.1	5.1	0.6	1.1	1.3	1.1	0.9	1.0
Home produced goods and services	74.6	3.6	1.3	2.5	1.0	0.9	0.7	0.7
Imported goods and services	25.4	2.2	–0.5	–0.3	0.1	–0.2	1.0	0.9
Export prices[2]		0.9	–1.7	–2.8	–0.0	0.4	1.5	4.2[3]
Import prices[2]		–1.1	–5.7	–4.3	1.4	0.3	2.6	6.2[3]

1. The total supply price index is the weighted average of producer and import price indices.
2. Change in the price deflator of goods and services, national account basis, 1980 = 100 until 1996.
3. Revised series (1990 = 100).

Source: Département fédéral de l'économique publique, *La vie économique;* Banque nationale suisse, *Bulletin mensuel*.

Figure 6. **INTERNATIONAL COMPARISON OF CONSUMER PRICES**

Percentage change from same period of previous year



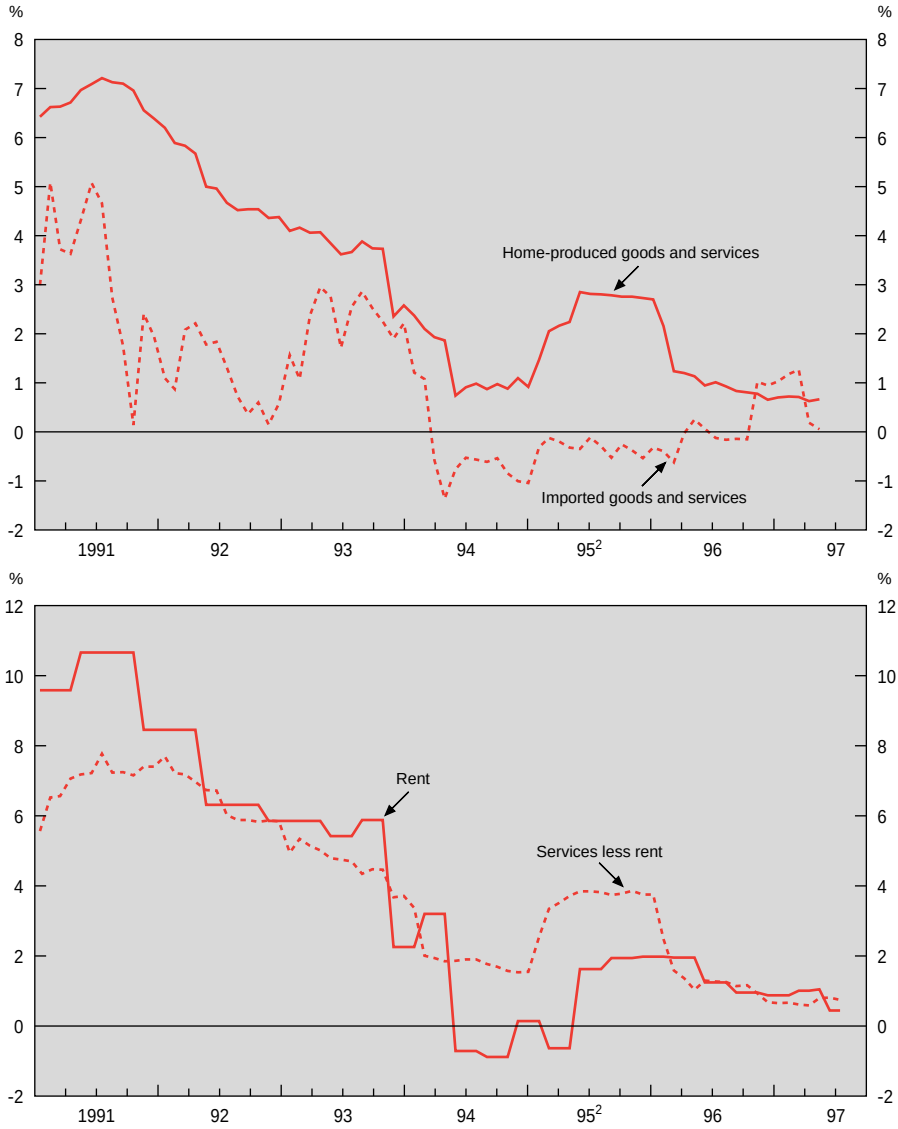
1. Excluding Turkey.
Source: OECD.

The prices of all major groups of consumer goods were almost stable in 1996 and early 1997, with energy prices being an exception; the latter were heavily affected by the rise in world market prices of crude oil. Service price inflation has also come down from the relatively high rates registered until 1993. Among these, rent increases – with a share of 22 per cent in the Swiss CPI an important source of inflation in the late 1980s and the early 1990s – declined to around 1 per cent, reflecting the depressed market for dwellings and stable mortgage rates[17] (Figure 7). Low consumer price inflation was underpinned by falling prices on the producer and wholesale level since 1994 and throughout 1996, in spite of the recent reversal of the Swiss franc appreciation. The GDP deflator, the most comprehensive indicator of domestic price pressures, was virtually unchanged (+0.1 per cent) in 1996, as the rise in the private consumption deflator was offset by falling prices for fixed investment.

Stable prices were underpinned by continuing wage moderation in 1996. As in 1995, the average nominal wage rate rose by 1.3 per cent in 1996, which resulted in a small average real wage gain of ½ per cent, after a loss of the same

Figure 7. **SELECTED COMPONENTS OF INFLATION**[1]

Year-on-year change



1. Weights in CPI as from May 1993 are: 0.75 for home-produced goods and services, 0.25 for imports; 0.29 for services less rent and 0.22 for rent.
2. Introduction of VAT.

Source: OECD, *Main Economic Indicators.*

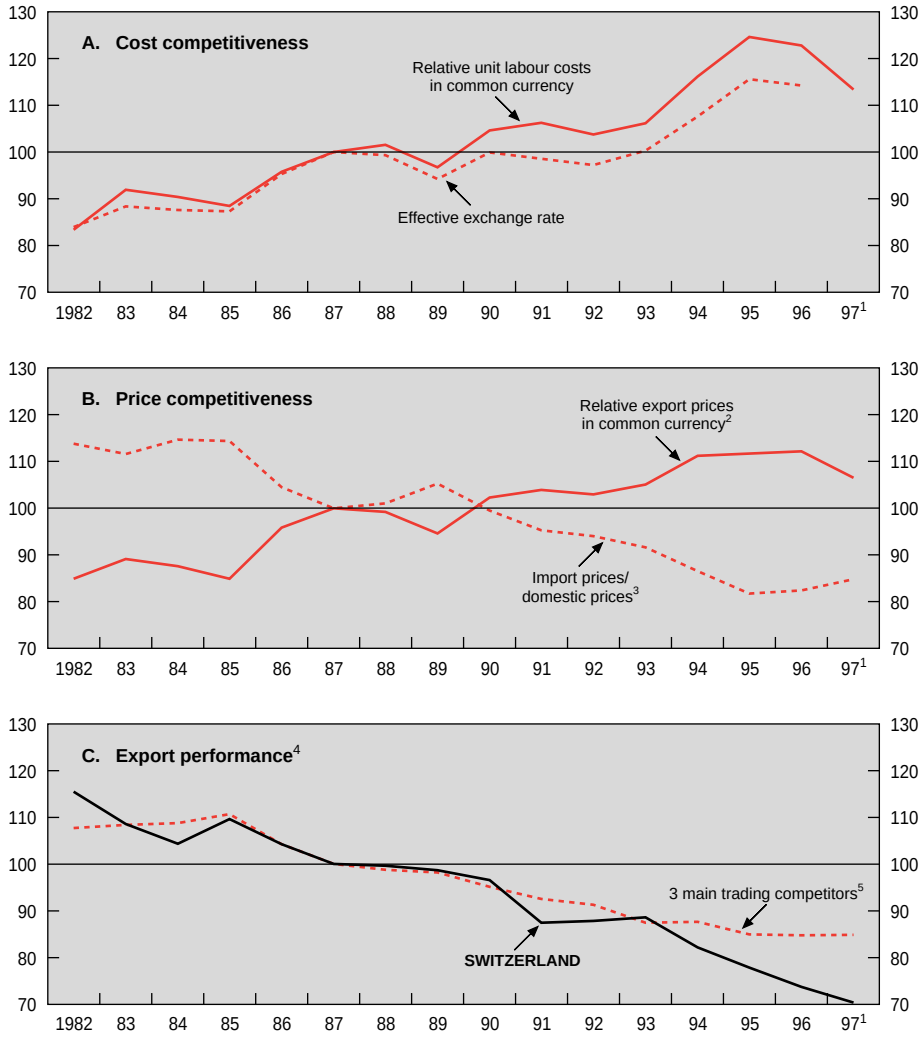
magnitude in 1995. In 1996, real wage gains were highest – 1¾ per cent – in transport and communications, while small real wage losses were recorded in the primary sector and in several services industries, among them the government sector. However, with the estimated decline in labour productivity, aggregate unit labour costs may have increased by some ¾ per cent in 1996. On the basis of wage settlements concluded in the autumn of 1996, nominal wage growth is set to decline further, possibly to a rate of ½ per cent in 1997.

A large and stable current external surplus

Swiss exporters continued to face a difficult environment in 1996: growth of their export markets (for goods) decelerated from about 10 per cent in 1995 to a little above 6 per cent in 1996, and the (nominal) effective appreciation of the Swiss franc by some 15 per cent since 1992 implied a substantial loss in price and cost competitiveness. Although estimates of relative export unit values in common currency suggest that the loss in price competitiveness came to a halt in 1996, its cumulative increase by 8½ per cent since 1992 remained substantial. Moreover, the real franc appreciation as expressed by the change in relative unit labour costs in common currency – a proxy for cost competitiveness – is estimated to have exceeded 20 per cent from 1992 to 1996 (Figure 8). In view of these unfavourable conditions, the growth of merchandise export volumes by 2.1 per cent in 1996, after 4.1 per cent in 1995, is quite remarkable. For the fourth consecutive year, price concessions of exporters – at the expense of a squeeze in profit margins – were required to achieve this result: in spite of rising unit labour costs merchandise export unit values remained broadly unchanged in 1996, after decreases in 1994/95. Nevertheless, Switzerland's export performance – measured as the difference between the growth rates of merchandise exports and export markets – deteriorated further in 1996, and by more than that of its major competitors. However, the sharp turnaround in the franc exchange rate since late 1996 and the gradual recovery in important export markets seem to have had their first positive effects on Swiss exports: in the first four months of 1997, merchandise export volumes rose by a seasonally adjusted annual rate of 12½ per cent.[18] Exporters seized the opportunity to improve their profit margins and raised prices: in the fourth quarter of 1996 – and for the first time since the beginning of 1994 – export unit values were higher than twelve months earlier; in the first four

Figure 8. **INTERNATIONAL COMPETITIVENESS AND TRADE**

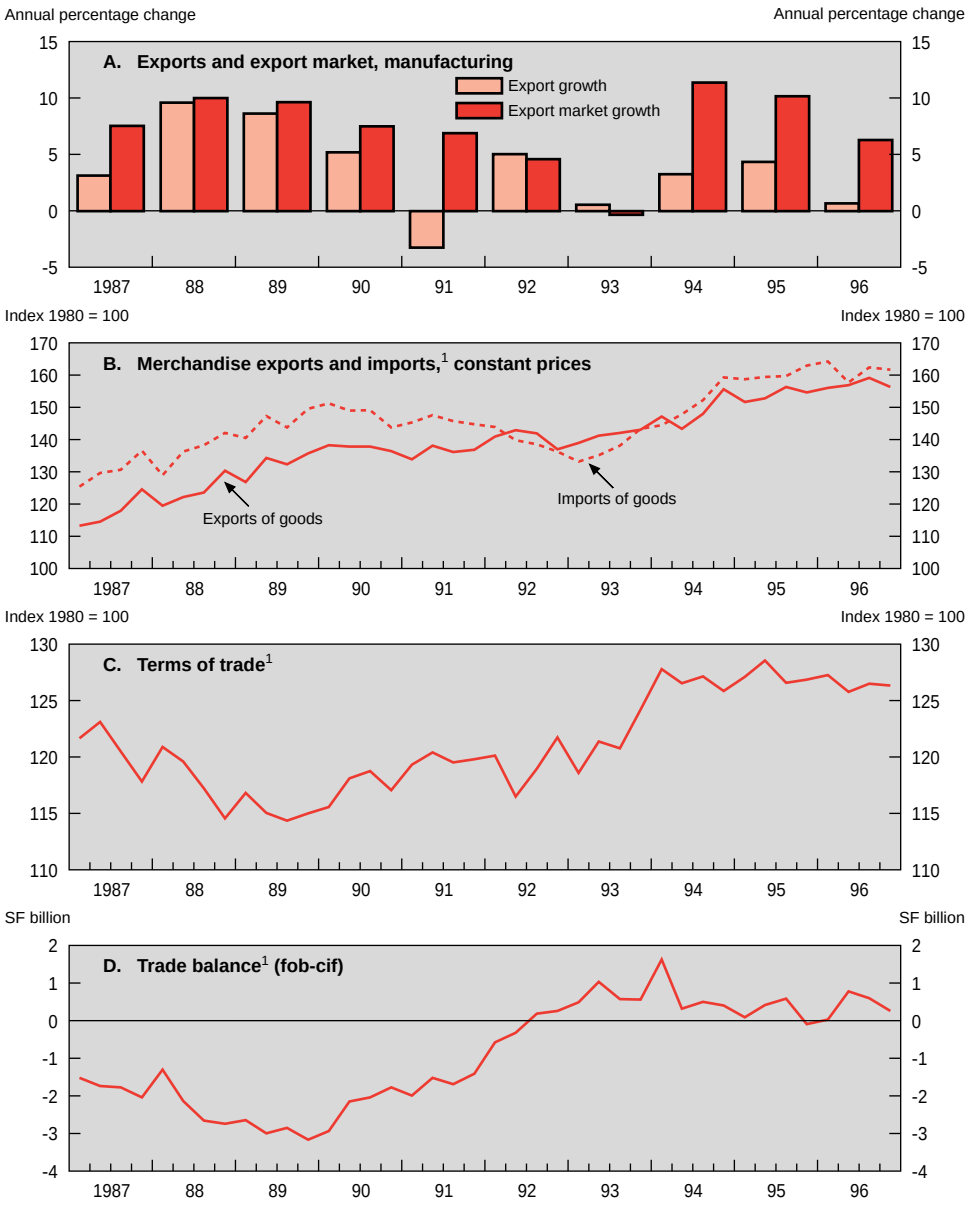
Index 1987 = 100



1. Figures for 1997 are OECD projections.
2. Manufactures.
3. Import prices of total goods and services divided by deflator of total domestic demand.
4. Ratio between export volumes of manufactures and export markets for manufactures.
5. Germany, France and Italy.

Source: OECD.

Figure 9. **FOREIGN TRADE INDICATORS**



1. Seasonally adjusted.
Source: OECD.

months of 1997 they rose by a seasonally adjusted annual rate of 9½ per cent. Import unit values also surged in late 1996 and in early 1997, in line with the exchange rate depreciation.

As in the preceding two years, exports of investment goods were most resilient, growing by about 3 per cent in volume terms in 1996, in spite of weak

Table 5. **Current external account**

SF billion

	1991	1992	1993	1994	1995	1996	Q1 1997[1]
A. Receipts							
Exports of goods (fob)	87.9	92.1	93.3	95.8	96.2	98.6	101.9
Supplementary export items	3.0	3.3	3.5	3.6	3.6	3.6	3.9
Services, receipts	28.4	29.6	31.7	31.4	30.9	31.8	34.4
Tourism	11.0	11.5	11.3	11.4	11.2	10.8	12.3
Other services	17.4	18.1	20.5	20.0	19.7	21.0	22.1
Factor income, receipts	39.5	36.9	37.0	36.7	37.4	37.7	41.2
Labour	1.2	1.3	1.4	1.5	1.5	1.5	1.5
Capital	38.2	35.6	35.6	35.3	35.9	36.2	39.7
Current transfers, receipts	3.4	3.6	3.7	3.5	3.5	3.6	3.6
Total receipts	162.2	165.5	169.2	171.0	171.6	175.2	185.0
B. Expenditures							
Imports of goods (cif)	95.0	92.3	89.8	92.6	94.5	96.7	104.6
Supplementary import items	4.4	4.5	4.5	4.5	4.3	4.6	4.9
Services, expenditures	13.7	14.5	14.9	15.3	16.0	16.7	11.4
Tourism	8.2	8.7	8.8	8.8	9.1	9.6	4.1
Other services	5.4	5.9	6.1	6.5	6.9	7.1	7.3
Factor income, expenditures	26.8	25.2	23.5	25.9	23.4	23.7	25.4
Labour	8.8	8.8	8.5	8.3	8.3	8.4	8.1
Capital	17.9	16.4	14.9	17.6	15.1	15.3	17.3
Current transfers, expenditures	7.1	7.7	7.7	8.2	8.4	8.5	8.4
Total expenditures	147.0	144.3	140.5	146.5	146.6	150.2	154.6
C. Balance							
Trade balance	–7.1	–0.2	3.5	3.2	1.8	1.9	–2.7
Supplementary trade items, net	–1.4	–1.2	–1.0	–1.0	–0.7	–1.0	–1.0
Services, balance	14.7	15.1	16.8	16.1	14.9	15.0	23.1
Tourism	2.8	2.8	2.4	2.7	2.1	1.2	8.3
Other services	11.9	12.2	14.4	13.4	12.8	13.9	14.8
Net factor income	12.7	11.7	13.5	10.8	13.9	14.0	15.8
Labour	–7.6	–7.5	–7.2	–6.9	–6.9	–6.9	–6.7
Capital	20.3	19.2	20.7	17.7	20.8	20.9	22.5
Current transfers, net	–3.7	–4.2	–4.0	–4.7	–4.9	–4.9	–4.8
Balance on current external account	15.2	21.3	28.8	24.4	25.0	25.1	30.4
As a percentage of GDP	4.6	6.3	8.4	6.9	6.9	7.0	8.5[2]

1. At annual rate, estimates, not seasonally adjusted.
2. Based on revised National Accounts.

Source: Banque nationale suisse, *Bulletin mensuel*.

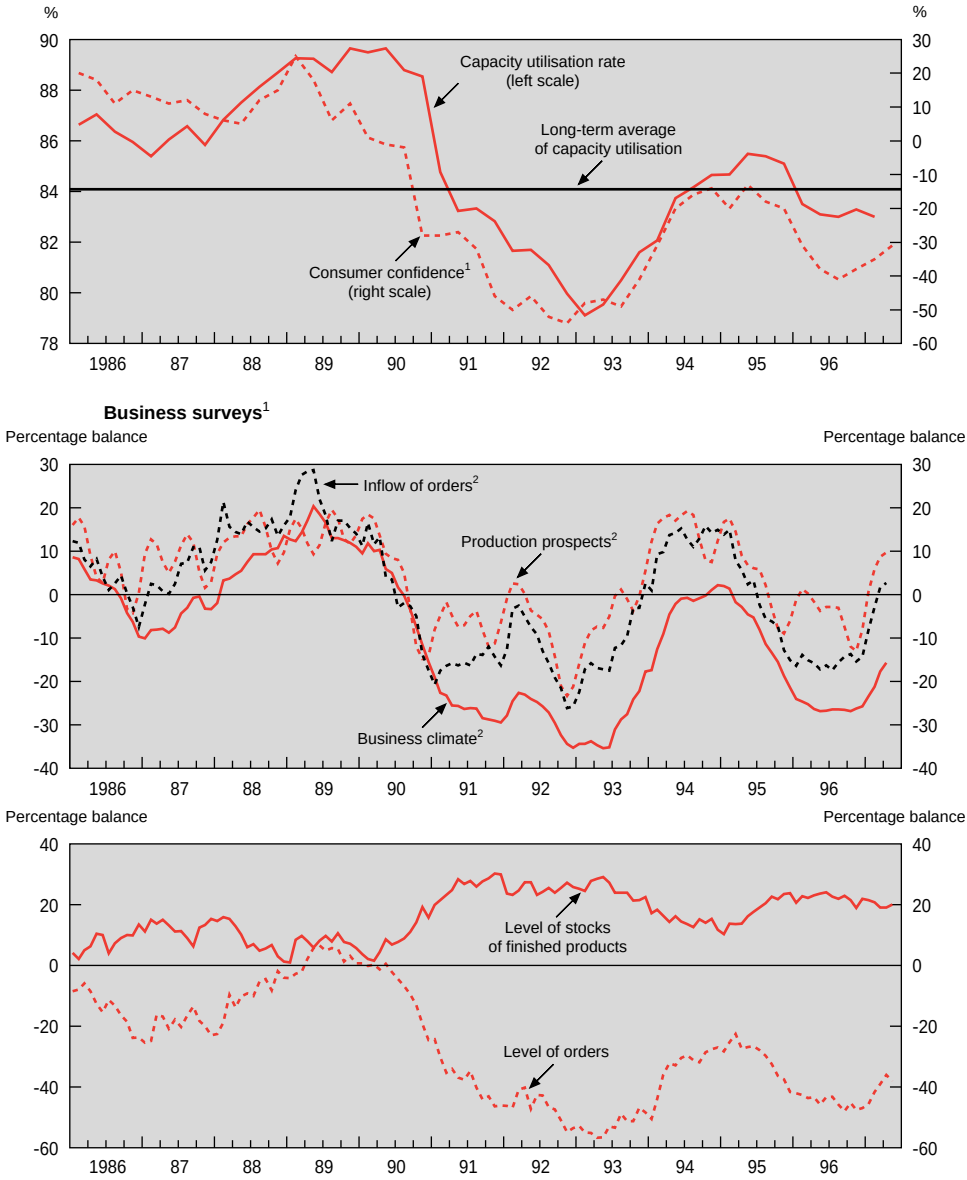
investment demand in major trading partner countries. In contrast, consumer goods export volumes stagnated in 1996. Exports of both categories showed clear signs of improvement in early 1997, growing at annual rates of 15 and 11 per cent, respectively in the first four months of 1997. The regional disaggregation of exports largely reflects the different cyclical positions of destination countries: exports (in current prices) to European OECD countries broadly stagnated. Deliveries to Germany – which account for nearly one quarter of Swiss exports – fell by 2.5 per cent in 1996. In contrast, exports to North American countries and to the United Kingdom were buoyant in 1996. The growth of import volumes slowed down from 6.1 per cent in 1995 to 1.5 per cent in 1996, but accelerated in the first four months of 1997. Their positive growth at a time of shrinking domestic demand in part[19] reflects the increased price competitiveness of imports relative to domestically produced goods (Figure 8, Panel B). The trade account posted another surplus in 1996, for the fourth year in a row, amounting to SF 1.6 billion in 1996, up from SF 1.1 billion in 1995 (Figure 9).

The surplus in non-factor services increased somewhat in 1996, in spite of the steep fall in the surplus from tourism; net tourism revenues have proved relatively price sensitive in the past and are likely to have again suffered from the strong Swiss franc. The non-tourism part of the non-factor services surplus rose primarily due to strong bank commission revenues. Both the deficit on labour incomes – mainly reflecting the employment of ''frontier workers'' – and the large surplus on investment income (some 5¾ per cent of GDP) remained broadly unchanged, leaving the surplus on factor incomes at about SF 14 billion in 1996 (Table 5). The current external surplus also remained about constant at SF 25 billion (7 per cent of GDP) in 1996.

The outlook to 1998

At mid-1997, a number of current economic indicators display distinct signs of cyclical improvement: consumer confidence and the business climate edged up from their depressed levels, the judgement of inflows and levels of orders became more optimistic, and the balance of opinions about the future tendency of production turned positive (Figure 10). But in spite of the ongoing rundown of inventories the level of stocks continues to be regarded as too high and capacity utilisation in industry as unsatisfactory.

Figure 10. **INDICATORS OF ECONOMIC ACTIVITY**



1. Balance of positive and negative responses.
2. 3-month moving average.

Source : KOF/ETH, *Konjunktur*; OECD, *Main Economic Indicators.*

The projections presented below have been made on the technical assumption of nominal exchange rates remaining unchanged at their levels of 1 May 1997. This implies that the nominal effective Swiss franc exchange rate is 6¾ per cent lower in 1997 than in 1996; this translates into a real effective depreciation in terms of relative export unit values of about 5 per cent from 1996 to 1997. It is further assumed that the National Bank keeps short-term interest rates broadly at the current low level in 1997 and tightens only a little in 1998, about in line with the assumed policy of the Deutsche Bundesbank. This implies an unchanged yield curve and may keep the long-term bond rate differential *vis-à-vis* Germany steady at close to 2 percentage points.

The general government deficit is projected to increase from 2.1 per cent of GDP in 1996 to 2½ per cent in 1997, which changes the fiscal policy stance from broadly neutral in 1996 to slightly expansionary in 1997, and in spite of persisting restraint on government consumption (primarily wages), such a trend may continue into 1998. This is the result of the Federal Government's latest fiscal package which amounts to SF 730 million (0.2 per cent of GDP) and which consists of extra spending of SF 560 million (over three years) mainly for maintenance work of the infrastructure and of net annual tax cuts of SF 170 million, ensuing from the reform of business taxation. The programme is designed to induce additional spending at the cantonal and communal level and is projected to induce extra growth of real GDP by one-third of a percentage point in 1998.

With much of the earlier real effective appreciation of the Swiss franc now reversed and export market growth improving, exports of goods and services should accelerate and help to turn economic activity around this year. But with persistent excess supply of structures, construction investment is likely to remain a drag on domestic demand in 1997 (Table 6). Household consumption is restrained by stagnant real disposable incomes this year, which will keep the recovery rather modest and insufficient to raise employment. Thus, the unemployment rate is projected to rise further. Supported by easier monetary conditions and the Federal Government's fiscal package, the constraints on economic activity should gradually weaken. Hence, in 1998, growth of GDP could surpass potential output growth for the first time in eight years and the rate of unemployment could start falling. Although the recent exchange rate depreciation will raise prices of imported goods, the remaining large output gap and high cyclical unemployment should keep overall price and wage inflation low.

Table 6. **Short-term projections**

	Current prices SF billion 1993	Share in GDP %	Percentage changes			
			1995	1996	1997[1]	1998[1]
Demand and output (volume)[2]						
Private consumption	203.0	59.2	0.7	0.3	0.4	0.9
Public consumption	49.7	14.5	–0.1	–0.2	–0.7	0.0
Gross fixed capital formation	77.0	22.5	2.3	1.3	0.2	2.4
Construction	51.2	14.9	–4.3	–3.5	–2.3	1.5
Machinery and equipment	25.8	7.5	14.1	8.4	3.5	3.5
Final domestic demand	329.7	96.2	1.0	0.5	0.2	1.2
Change in stocks[3, 4]	–4.2	–1.2	0.9	–1.5	–0.2	0.1
Total domestic demand	325.6	95.0	1.9	–1.0	0.0	1.3
Exports of goods and services	125.3	36.5	3.0	2.4	3.7	4.8
Imports of goods and services	108.0	31.5	6.6	1.5	1.8	3.6
Change in foreign balance[3]	17.3	5.0	–1.9	0.3	0.8	0.4
Gross domestic product	342.9	100.0	0.1	–0.7	0.8	1.8
Industrial production[5]			1.9	–2.0	1.5	2.5
Prices						
GDP deflator			2.5	0.1	0.6	1.2
Private consumption deflator			1.8	1.2	1.2	1.4
			Levels			
Unemployment rate			4.2	4.7	5.4	5.0
Current balance ($ billion)			21.5	20.3	17.3	17.3
Per cent of GDP			6.9	7.0	6.9	6.8

1. OECD forecast.
2. At 1980 prices.
3. As a percentage of previous year's GDP.
4. Including statistical adjustments.
5. Excluding mining and quarrying.

Source: Swiss national accounts; OECD.

The main uncertainty attaching to the projections concerns the exposed sector of the economy: speed and size of the adjustment to the lower franc exchange rate could be higher, entailing stronger growth. On the other hand, there is the risk of a resumption of appreciation of the Swiss franc in case of renewed uncertainty about the planned European Monetary Union, which could impede the recovery.

II. Macroeconomic policies

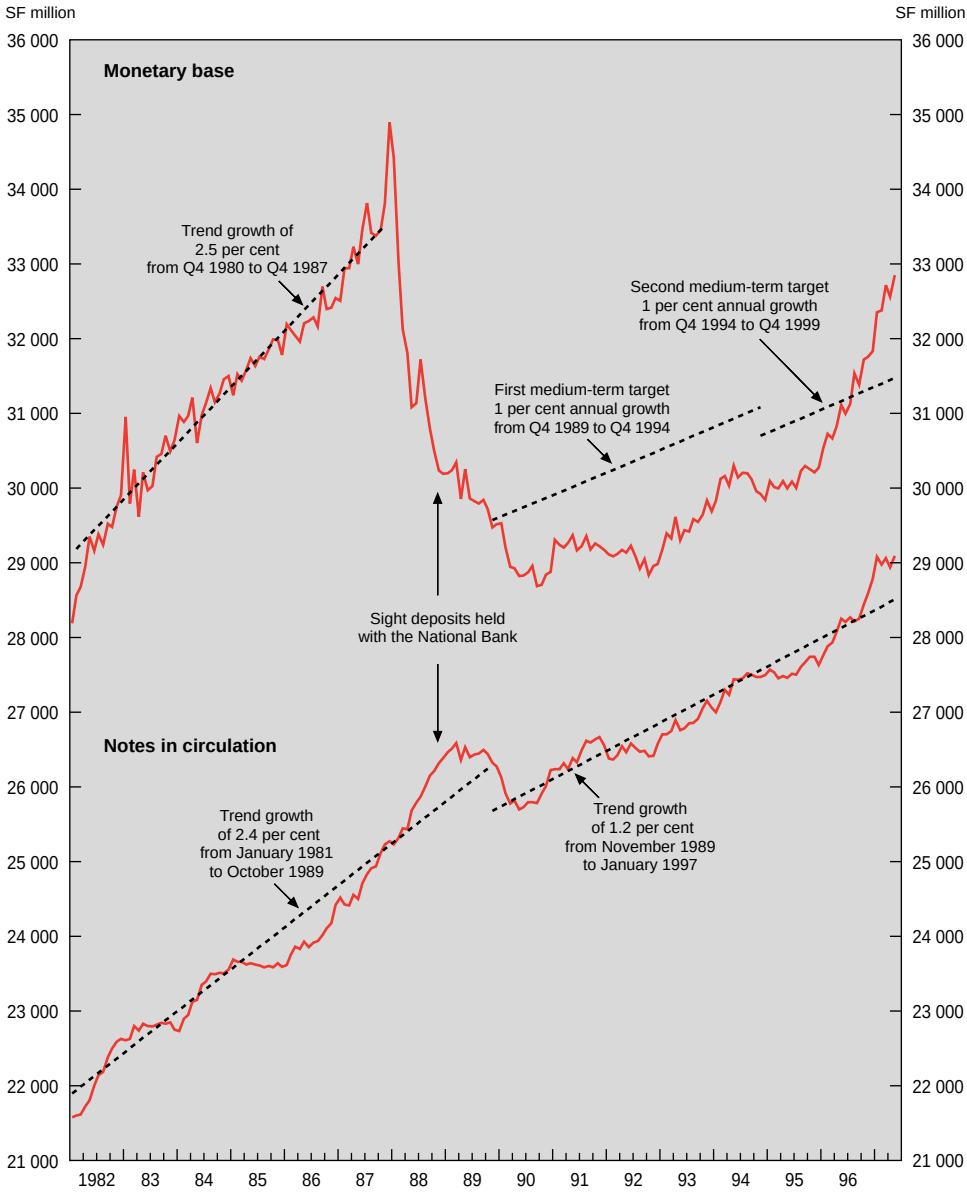
The 1991 to 1996 period of broad economic stagnation – the longest in post-war history – is the result of a series of domestic policy measures combined with adverse external factors such as upward pressures on the Swiss franc exchange rate and slowing export market growth. The easing of monetary policy in response to the 1987 stock market collapse and the structural change in banks' liquidity demand accommodated an acceleration of inflation and an overheating in the real estate market. Subsequent restrictive monetary policy to fight high inflation and a slow-down in exports caused activity to stagnate in 1991. Falling demand for real estate led to substantial excess capacity of structures and threw the construction sector into a severe crisis. In addition, high government deficits called for a turn in fiscal policy towards retrenchment, which acted as a drag on domestic demand via higher taxes and social security contributions combined with government spending restraint. Business activity may also have suffered from the rejection of the European Economic Area Agreement by the people and the cantons on 6 December 1992.

Monetary policy

A decisive factor prolonging the weakness of activity was the appreciation of the Swiss franc from 1993 to 1995 by about 15 per cent in real effective terms[20] (CPI based). To a large extent, the implied tightening of monetary conditions is attributable to the appreciation of the Swiss franc which is related to international investors' perception of Switzerland as a ''safe haven'' to escape the uncertainties of the EMU process, combined with a restrictive monetary policy stance, particularly in 1994. In part, the over-restrictive monetary policy stance in 1994 was motivated by the SNB's concern that the introduction of VAT

Figure 11. **THE ADJUSTED MONETARY BASE**

Seasonally adjusted



Source: Banque nationale suisse, *Bulletin mensuel.*

at the beginning of 1995 might cause a price-wage spiral if accommodated by too generous money supply (Figure 11).

The changing stance of monetary policy

Monetary policy was gradually relaxed in the course of 1995, but this was insufficient to prevent the effective Swiss franc exchange rate from rising further up to a peak in November. Hence, in real effective terms the franc was on average valued another 6 per cent higher in 1995 than in the preceding year. By the end of 1995 the National Bank announced some further moderate monetary easing to allow the adjusted monetary base[21] to grow somewhat above 1 per cent during 1996, for narrowing its gap with the medium-term growth path.[22]

However, the actual growth of base money turned out significantly stronger right from the beginning of 1996 and in May this aggregate met its medium-term target level for the first time in the 1990s. Much of its acceleration is due to the stronger expansion of currency in circulation which responded to the decline in interest rates on saving deposits, its key opportunity cost variable. Saving deposit interest rates had followed the decrease in money market rates during 1995 with a lag of about two quarters. The three-month Euro-Swiss franc interest rate fell further from about 2 per cent in late 1995 to around 1.6 to 1.7 per cent in the period from February to April 1996. But from May onwards banks' sight deposits held with the SNB, which had declined by some 70 per cent[23] from 1988 to 1992 and remained broadly flat during the following three years, began to soar: they grew by seasonally-adjusted annual rates of about 20 per cent in the second quarter and some 50 per cent in the third quarter of 1996, in spite of sluggish economic activity and weak domestic credit expansion (Table 7). Until the summer, this unexpected surge in liquidity demand was not fully accommodated by extra base money supply because the SNB initially interpreted the banks' increasing demand for base money as first signs of economic recovery (Banque nationale suisse, 1997, p. 32). As a consequence, the three-month Euro-Swiss franc interest rate rose by nearly 1 percentage point from its trough in February to 2.5 per cent in June and July 1996, narrowing the short-term interest rate differential *vis-à-vis* major currencies. The three-month Swiss franc-Deutschemark interest rate differential for example narrowed by 1 percentage point to ¾ per cent from early 1996 to June/July, and the effective Swiss franc

Table 7. **Monetary aggregates**

Percentage change from previous period, s.a.a.r.

	1991	1992	1993	1994	1995	1996	1996				1997
							Q1	Q2	Q3	Q4	Q1
Notes in circulation	2.2	0.1	1.5	1.9	0.7	2.4	–1.0	7.7	1.5	4.8	8.0
Sight deposits held with SNB[1]	–6.2	–9.9	3.9	1.0	–0.2	14.1	8.8	16.8	48.3	25.6	33.2
Monetary base[1]	1.1	–0.7	1.5	1.9	0.5	3.5	5.4	4.4	4.9	5.5	9.2
Sight deposits and transactions accounts	1.8	2.5	12.9	6.6	8.2	13.8	16.6	10.5	10.4	7.2	16.5
M1[2]	1.9	2.0	10.5	5.6	6.8	11.8	15.6	6.5	9.8	8.1	14.4
Savings deposits	1.4	3.6	21.5	14.2	3.4	12.4	22.9	7.0	1.3	1.5	6.2
M2[2]	1.7	2.8	16.1	10.2	4.9	12.1	19.6	6.8	5.0	4.5	9.9
Time deposits	4.4	0.9	–17.5	–7.5	–5.8	–9.5	–16.8	10.1	13.7	6.3	8.7
M3[2]	2.7	2.1	3.9	5.1	2.2	7.1	11.1	7.4	6.7	4.8	9.7
Domestic credit expansion	6.7	4.4	1.3	1.7	2.3	3.2	5.9	3.5	1.0	–0.1	0.5
Nominal GDP	5.4	2.3	1.2	2.9	2.6	–0.5	–1.8	–2.3	–2.1	4.0	..

1. Change in definition in 1995.
2. Definition of 1995.

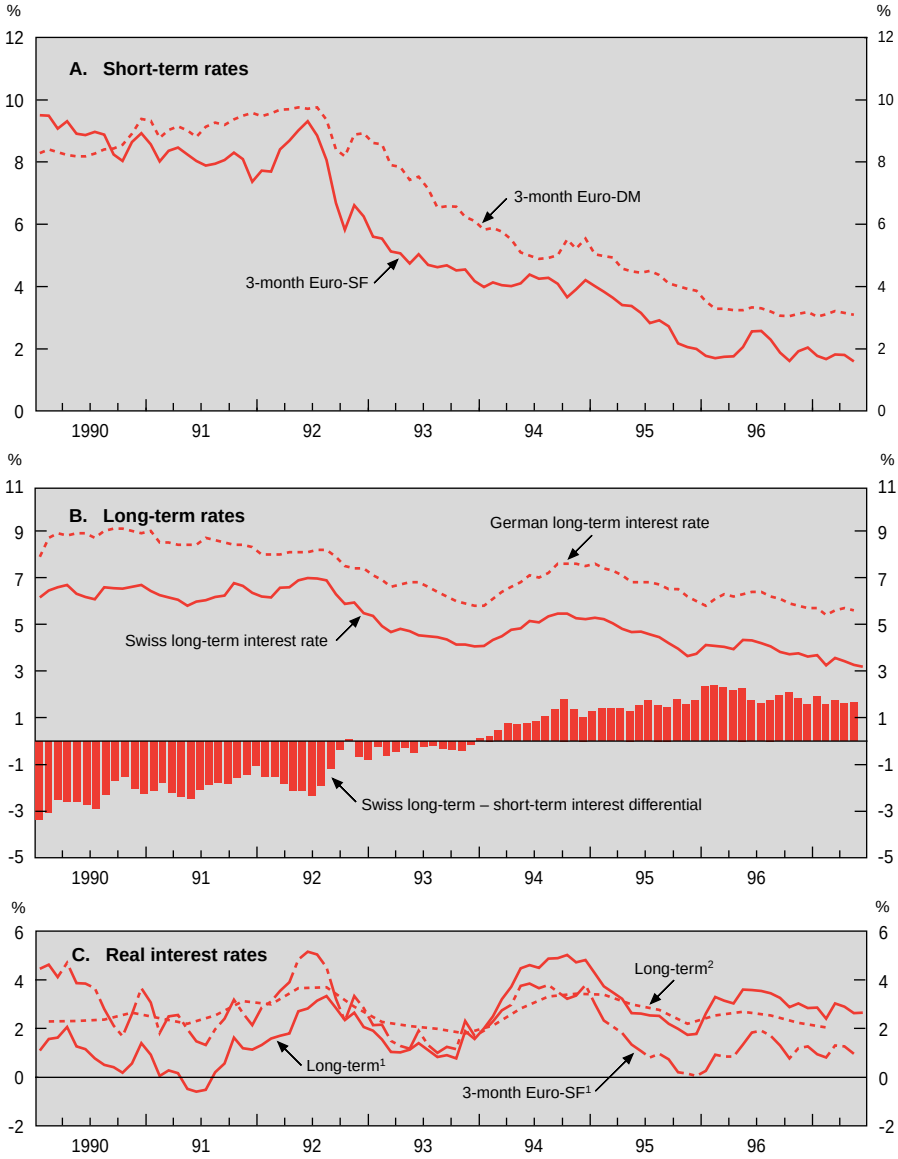
Source: Banque nationale suisse, *Bulletin mensuel*; OECD.

exchange rate, which had depreciated by about 5 per cent from November 1995 to June 1996, began to edge up again in the summer.

During the summer of 1996, it became increasingly clear that the commercial banks' strong demand for central bank money was primarily the result of their move away from a very decentralised to a more centralised postal accounts system.[24] This reduced their postal giro deposit holdings, which can be used to satisfy the banks' liquidity requirements but are not included in the definition of the monetary base. Hence, in order to meet their liquidity requirements, banks had to raise their sight deposits held with the SNB, which put upward pressure on money market interest rates. With the Swiss franc still considered grossly overvalued, interest rates high relative to the actual level of economic activity and expectations of an imminent economic recovery dissipating, the SNB adopted a more aggressive policy of monetary easing from August 1996 on and amply provided the banks with base money through open market operations and foreign exchange swaps. This further lifted the adjusted monetary base, so that in the fourth quarter of 1996 it was 5.0 per cent above the level recorded a year earlier and some 1½ per cent above its medium-term target trajectory. However, since the National Bank attributes some SF 400 million of the incremental demand for base money to the reshuffling of bank liquidity, base money statistics exaggerate the degree of monetary easing (Banque nationale suisse, 1997, p. 32). Indeed, if the monetary base is adjusted for this reshuffling of bank liquidity, the resulting year-on-year growth rate is 3.7 per cent in the fourth quarter of 1996, just enough to bring base money into line with the SNB's medium-term target path. Base money growth remained strong in the first quarter of 1997, amounting to an annualised rate of 9¼ per cent over the final quarter of 1996. It was fuelled by unexpectedly strong demand for currency.[25] Uncertainty about the causes of this surge in the monetary base weakens its significance as a monetary indicator.

To underline its determination to ease monetary conditions, the National Bank cut the discount rate by ½ percentage point to 1 per cent in September 1996 (although the SNB no longer discounts bills), the lowest since 1978. This helped to bring the three-month Euro-Swiss franc interest rate down by ½ percentage point from its temporary peak in the summer of 1996 to 2 per cent at the end of the year, and further to 1.6 per cent in February 1997 (Figure 12). An increase to 1.8 per cent was, however, recorded in March and April, possibly induced by the strong base money demand of banks and the increase in US money market rates.

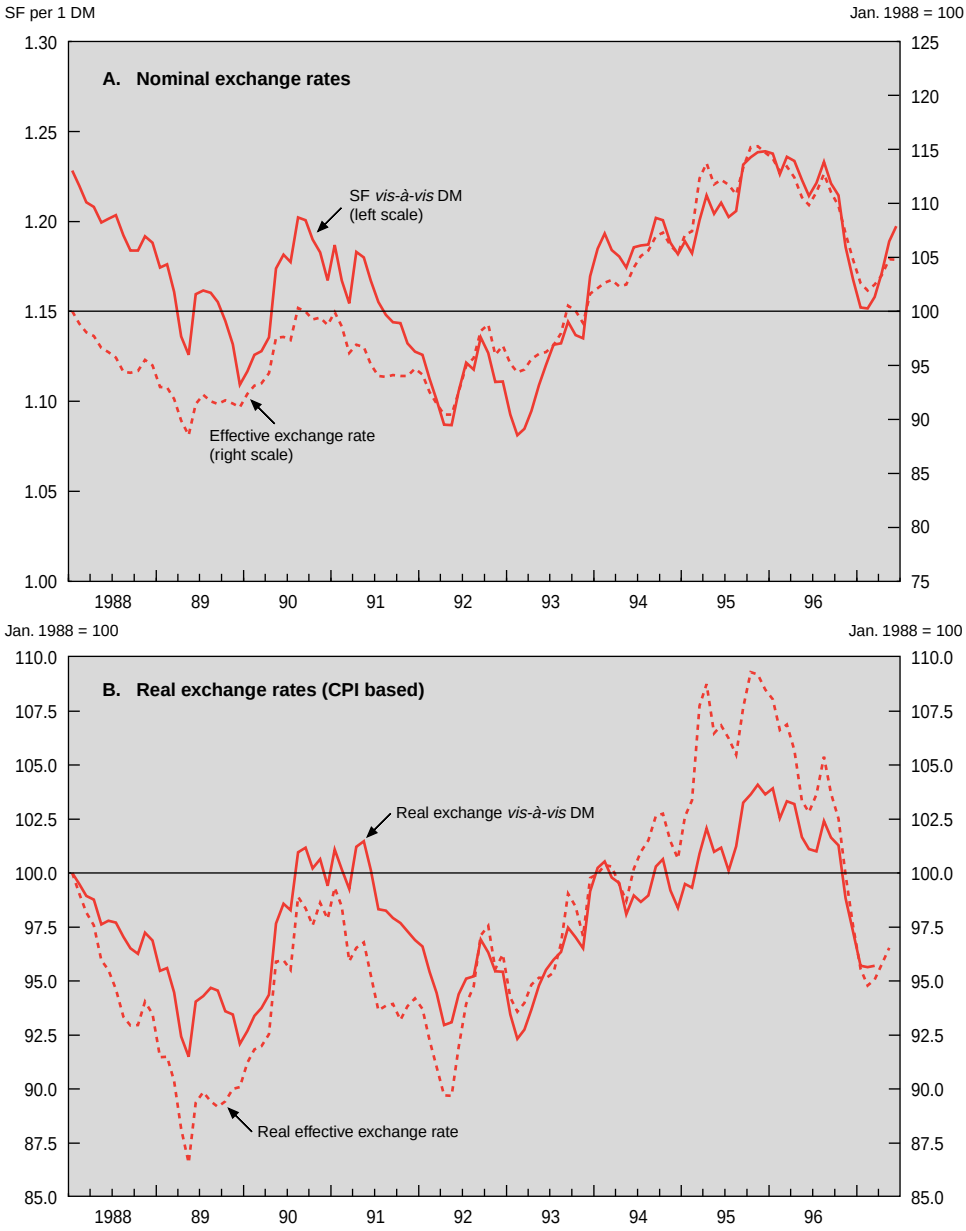
Figure 12. **INTEREST RATES**



1. Nominal rate less the change in the consumer price index from the previous year.
2. Nominal rate less low frequency component of the annualised change in the seasonally adjusted GDP deflator.

Source: OECD, *Main Economic Indicators* and estimates.

Figure 13. **EXCHANGE RATES**



Source: OECD, *Main Economic Indicators.*

The money market rate fell to 1.2 per cent in June 1997. With the negative short-term interest rate differential *vis-à-vis* the Deutschemark and other major currencies widening again during the second half of 1996, the effective Swiss franc exchange rate fell from August 1996 onward. Towards the end of 1996, the reversal of the Swiss franc appreciation was accentuated by the progressive strengthening of the US dollar against European currencies. As a result, the depreciation of the real (CPI-weighted) effective Swiss franc exchange rate amounted to some 10 per cent over the twelve months to the end of 1996 and fell further in early 1997 (Figure 13). In the first quarter of 1997 it was just about 1¾ per cent above its average level in the 1991-1992 period – the first two years of the recent economic slump. Against the Deutschemark the Swiss franc also weakened, and in the first quarter of 1997 it was in real terms about 1½ per cent below its average level of 1991 and 1992.

Is the current stance of monetary policy about ''right''?

Other indicators of monetary conditions confirm that monetary conditions have eased: although the long-term Confederation bond rate followed the temporary increase of short-term interest rates from May to July 1996 after their decline earlier in the year, it fell further in the second half of 1996 and into 1997, to some 3¼ per cent in June 1997. With its decrease being less pronounced than that of money market rates, the long-short interest rate differential rose to above 2 percentage points in early 1996, the highest since the late 1970s indicating a considerable easing of monetary policy relative to 1993. It fell somewhat below this level, hovering in the range of 1½ to 2 per cent in the second half of 1996 and in early 1997. Ex-post real long-term interest rates (CPI deflated), which remained at or above 3 per cent throughout most of 1996, fell to around 2½ per cent in early 1997, remaining higher than in the early phase of the current period of stagnation and significantly above their long-term average of about 1¼ per cent. But it can be argued that current low rates of inflation may underestimate inflation expectations and give an exaggerated impression of the true level of real long-term interest rates. To take an alternative, longer-term approach to modelling expected inflation a Hodrick-Prescott filter[26] was applied to the GDP deflator and used to gauge real bond rates. The exercise confirms the observation that current real long-term bond rates are higher than in the two preceding recessions

of the mid-1970s and of the early 1980s, and about double their long-term average of 1 per cent. This may indicate some measure of financial restraint.

The notion of monetary relaxation in 1995 is also supported by the rise in the growth rates of the broader monetary aggregates M1 and M2 since May 1995, which reflects the reshuffling of funds from time deposits into sight deposits and transactions accounts, which are part of M1, and into savings deposits, which are included in M2. It is also supported by the SNB's four cuts in the discount rate from 3.5 to 1.5 per cent in 1995. The growth of sight deposits and transactions accounts as well as saving deposits was most pronounced in early 1996 and fell thereafter. The acceleration of broader monetary aggregates in 1995-96 reflects the normal, temporary reallocation of portfolios in response to a transition to lower interest rate levels due to monetary policy easing; a comparable episode was the temporary surge of M1 and M2 in 1993, when short-term interest rates dropped by about three percentage points while overall monetary conditions remained rather tight. Care needs to be taken when interpreting movements of M1, which for many years was the National Bank's preferred indicator among the broader monetary aggregates, as recent research casts doubts on its stability (Peytrignet, 1996). The greater stability of M2 (in the definition of 1995) appears to have induced the SNB to accord it now greater weight as a supplementary indicator to the monetary base (Banque nationale suisse, 1996).

Growth of the broad aggregate M3 also accelerated, but this was not accompanied by a surge in bank lending (Table 7). Credit expansion was weakened not only by sluggish domestic demand for credits but also by more restrictive lending practices of banks in an effort to improve the quality of their loan portfolios. This has become necessary as banks were hit hard by the collapse in property prices since 1991; regional banks were particularly affected by this collapse since more than half of their assets are mortgage loans. Altogether, Swiss banks had to write off more than SF 42 billion[27] in the period from 1991 to 1996 (some 12 per cent of GDP in 1996).

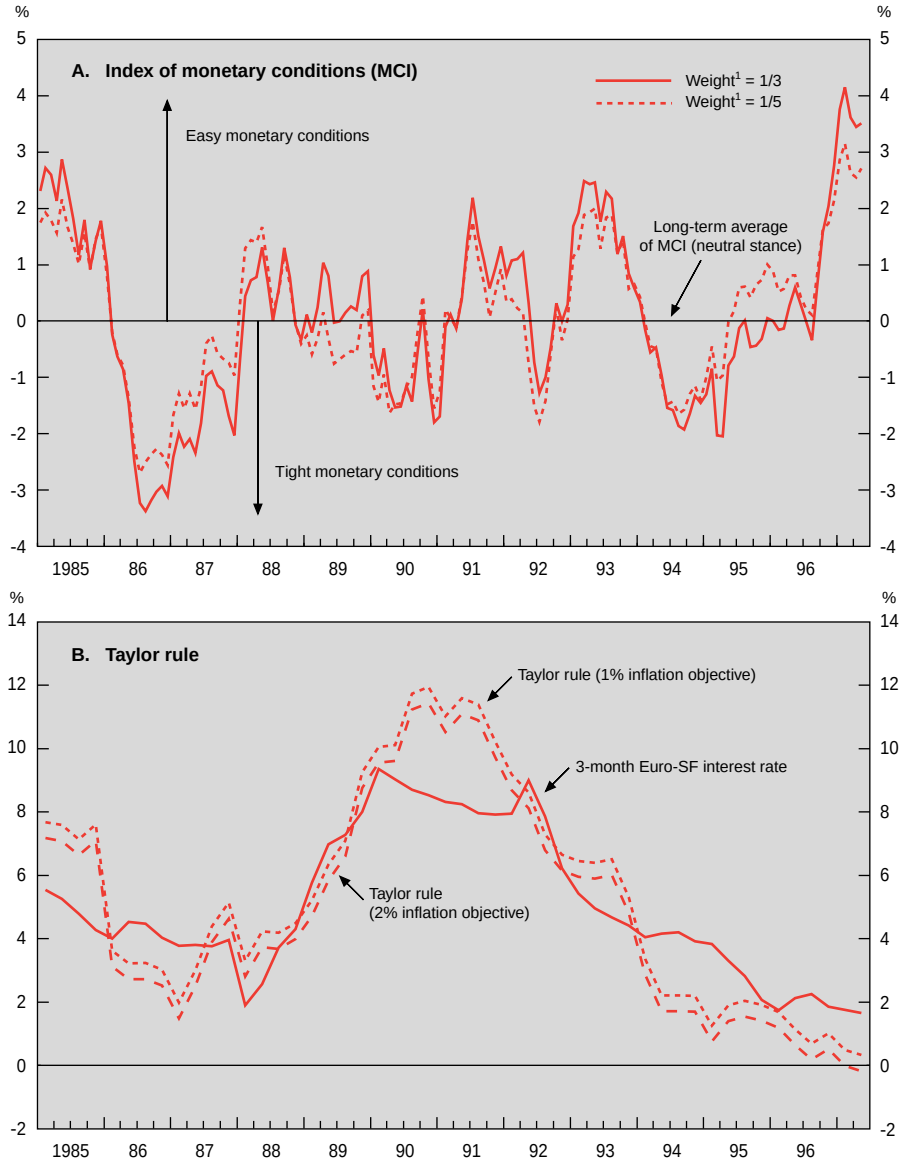
Given that none of the monetary aggregates provides complete information on monetary conditions and fully satisfactory guidance for monetary policy, it is pertinent to review alternative monetary indicators and their properties. One which recently has attracted considerable attention in Switzerland is the ''monetary conditions index (MCI)''. It is used by the central banks of Canada, Finland, Norway and Sweden to gauge the implications of monetary conditions on aggre-

gate demand, and in its simplest form equals the (weighted[28]) average of deviations in the relevant real short-term interest rate and the real effective exchange rate from their long-term averages.[29] The MCI calculated by the Secretariat (shown in the upper panel of Figure 14) implies tight monetary conditions in 1994, with a substantial easing from early 1995 onwards. However, since this indicator concentrates on the short-term implications of monetary policy for real aggregate demand rather than for future inflation, it is of less immediate interest for policy makers focusing on price stability rather than output stabilisation.

The ''Taylor rule'' (Taylor, 1993) offers another auxiliary indicator of monetary conditions: it determines a normative target interest rate as a function of a ''normal'' (*i.e.* long-term average) real short-term interest rate, the output gap and the difference between the actual inflation rate and an inflation objective.[30] Thus, the Taylor rule puts equal weight on the deviations of actual inflation from an inflation target and actual output from its trend value in determining the desirable short-term interest level, implying a responsibility of monetary authorities not only for price stability, but also for the full utilisation of resources. Figure 14 (lower panel) shows the normative short-term interest rate based on the Taylor rule for Switzerland, using OECD Secretariat estimates of the output gap and the differential between actual inflation and two alternative inflation objectives.[31] The target short-term interest rate based on the Taylor rule has declined precipitously from its 1990 peak, reflecting both the reduction in inflation and the widening output gap. Its difference to the actual short-term interest rate also clearly mirrors the tightening of policy in 1994 and the easing in 1995. However, given the large present output gap, the Taylor rule suggests short-term interest rates of close to zero or below by mid-1997, which appears less plausible.[32]

To sum up, the stance of monetary policy has been eased since the second quarter of 1995 and particularly since the middle of 1996. But the strong expansion of base money in 1996 and into 1997 gives an exaggerated impression of the current degree of monetary relaxation. The more aggressive easing of monetary policy since the middle of 1996 is consistent with the National Bank's view that at the current juncture inflation risks are minor and the SNB's announced intention to keep the expansion of the monetary base above its medium-term target path in 1997. Also, monetary policy should continue to take other indicators than the monetary base into account when assessing monetary conditions, to minimise

Figure 14. **MONETARY CONDITIONS INDEX AND THE TAYLOR RULE**



1. Weight on real exchange rate.
2. Inflation rates used for 1995 are adjusted for the estimated effect of VAT on the consumer price index.

Source: OECD.

the risk of interpreting structural shifts in the monetary base as endogenous changes in liquidity demand. However, various alternative indicators discussed in this report often provide ambiguous signals. Since in the case of Switzerland the link between inflation and the growth of broader monetary aggregates (M2 in particular) remains relatively stable, the latter are particularly relevant short-term indicators, complementing the role of the monetary base as a medium-term inflation anchor. But, given the uncertainties surrounding European monetary integration it seems appropriate to stand ready to also take exogenous shocks to the exchange rate into account in determining the stance of monetary policy.[33]

Fiscal policy

Overview[34]

Fiscal policy was counter-cyclical during the first years of the recession/ stagnation of the 1990s. The cyclically adjusted balance on the consolidated account of the general government (including social security[35]) swung from a surplus of 1.3 per cent of GDP in 1989 to a deficit of 2.6 per cent of GDP in 1993; the corresponding swing in the actual budget balance was from a 0.8 per cent surplus to a 3.7 per cent deficit (Table 8). In view of the rapid build up in public debt, from 33 per cent of GDP in 1989 to 43 per cent in 1993, and the expectation that economic recovery was imminent, the top priority of fiscal policy at this time became fiscal consolidation. The deficit was slashed in the next two years, falling to 1.8 per cent of GDP in 1995, with most of this reduction estimated to have been structural.[36] With the economy failing to recover as expected, the stance of fiscal policy swung to broadly neutral in 1996. The deficit remained at 1.8 per cent of GDP. Almost one-half of the 1996 deficit is estimated to have been structural (Figure 15). There were larger increases in the deficits of the Confederation and the cantons, which jointly finance the deficit of the unemployment insurance system, but these were largely offset by the fiscal consolidation of communes, which are estimated to have reached budget balance. Fiscal policy is set to become slightly expansionary this year, with all levels of government projecting a rise in their deficits: the most recent projections are for the combined deficit to rise to 2.4 per cent of GDP in 1997. Most of this increase is estimated to be cyclical: the structural deficit is projected to rise slightly to

Table 8. **Government accounts**

SF million and percentage changes[1]

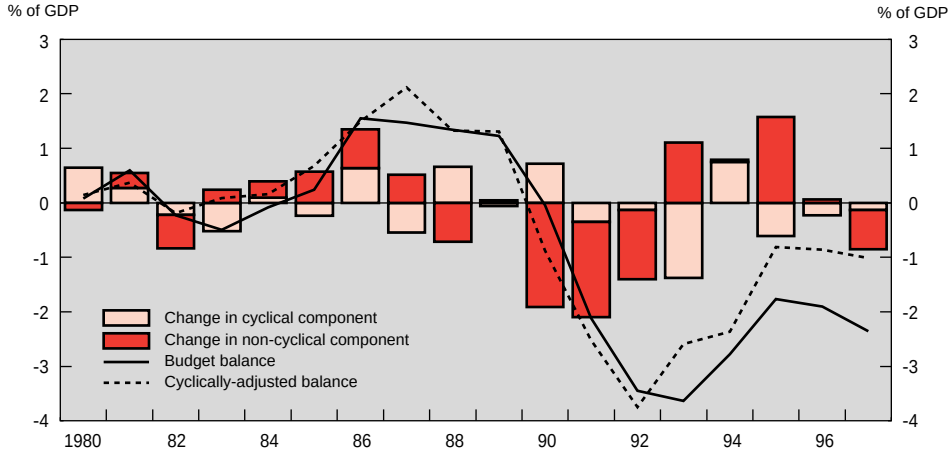
	1993 SF million	Outturns			Budgets[2]	
		1994	1995	1996[3]	1996	1997
Confederation[4]						
Expenditure	40 600	1.8	–2.0	8.2	3.7	0.6
Revenue	31 401	10.3	4.1	6.8	10.3	–1.3
Balance	–9 199	–6 706	–4 467	–5 377	–4 981	–5 768
Cantons						
Expenditure	52 392	0.2	–0.8	4.6	2.4	2.4
Revenue	47 002	3.8	2.8	4.3	5.8	1.5
Balance	–5 390	–3 707	–1 964	–2 200	–2 600	–3 100
Communes						
Expenditure	37 087	2.1	1.5	1.4	1.3	1.1
Revenue	35 892	3.0	1.9	3.4	4.7	1.8
Balance	–1 195	–893	–750	+0	–750	–500
General government[5]						
Expenditure	110 244	1.2	–0.7	3.4	1.0	1.1
Revenue	94 459	6.1	3.3	3.3	6.1	0.2
Balance	–15 785	–11 307	–7 180	–7 577	–8 331	–9 368
As a percentage of GDP	–4.6	–3.2	–2.0	–2.1	–2.3	–2.6
Social security						
Expenditure	35 007	6.8	7.5	–1.3	–	–
Revenue	38 078	1.8	5.9	–0.1	–	–
Balance	+3 071	+1 377	+740	+1 200	+750	+400
Consolidated account of general government and social security						
Expenditure	134 978	2.0	1.1	2.0	–	–
Revenue	122 264	4.5	3.9	2.1	–	–
Balance	–12 714	–9 930	–6 440	–6 377	–7 581	–8 968
As a percentage of GDP	–3.7	–2.8	–1.8	–1.8	–2.1	–2.4

1. For 1994-97, expenditures and revenues are percentage changes and balances in SF million.
2. Initial budget on initial budget.
3. Estimates for cantons, communes and social security.
4. These accounts have been constructed using the definitions of the ''*Statistique financière révisée*''. This makes the Confederation accounts comparable with those of the cantons and communes, making possible the aggregation of the accounts of all three levels of government to obtain estimates for general government finances.
5. Not including social security.

Source: Administration fédérale des finances.

Figure 15. **CYCLICALLY-ADJUSTED BUDGET BALANCE**

General government[1]

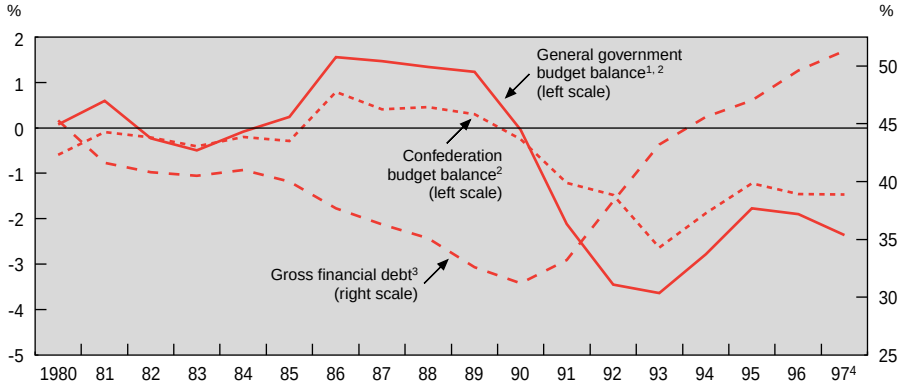


1. Statistique financière révisée basis. Including social security.

Source: Office fédéral des questions conjoncturelles.

Figure 16. **BUDGET BALANCES AND DEBT**

As per cent of GDP, transactions basis



1. Including social security.
2. Definition of the Statistique financière révisée.
3. Of the Confederation, cantons and communes.
4. Projections.

Source: Administration fédérale des finances.

1 per cent of GDP. Gross general government debt continues to increase, having reached almost 50 per cent of GDP in 1996 (Figure 16). As a percentage of GDP, it has now risen by three-fifths since 1990. The Confederation accounts for almost 48 per cent of the general-government debt, with the cantons and communes accounting for 31 per cent and 21 per cent, respectively. The increase in debt during the 1990s has been most marked for the Confederation, which experienced a near doubling in its debt-to-GDP ratio.

The Confederation's finances in 1996[37]

The Confederation's deficit rose by SF 1.1 billion in 1996 to SF 4.4 billion (1.2 per cent of GDP) (Table 9). Expenditure leapt by SF 3.3 billion (8.2 per cent), in large part owing to higher expenditure for social welfare (+SF 2.5 billion). This increase mainly reflected higher payments for unemployment insurance (+SF 1.4 billion), medical insurance (+SF 0.6 billion) and invalidity insurance (+SF 0.3 billion). The large rise in payments for medical insurance was a one-off event, being inflated by a switch to providing federal refunds mainly during the year in which medical expenditures are incurred rather than in the following year: this change resulted in double payments being made in 1996. The other large increase in expenditure was for agricultural support (+SF 0.4 billion). Receipts rose by SF 2.2 billion (5.9 per cent), mainly reflecting strong growth in revenues from federal income tax, withholding tax and stamp duty. As revenues from these taxes are higher in even-numbered years owing to the way in which they are calculated and collected, a better indication of underlying trends can be obtained by examining the period 1995-96. Average growth in total receipts for this period was 5 per cent, which is a little higher than for the recession periods 1993-94 and 1991-92 but much lower than the average for the 1980s.

Compared with the 1996 Budget, the deficit out-turn was slightly higher (SF 0.3 billion) than projected. This reflected a shortfall of receipts (SF 0.4 billion) which was attributable to lower than projected revenues from the federal income tax and the withholding tax. Although aggregate expenditures diverged little from the budgeted level, there were some large deviations by category. Approval for additional expenditures amounting to SF 1.7 billion was required during the year, to a large extent reflecting higher than projected expenditure for unemployment insurance and for agriculture. This was more than compensated

Table 9. **Central government budget**[1]

SF million

	1993		1994		1995		1996		1997	Percentage changes[3]		
	Outturns	Breakdown in %	Initial[2] budget	Outturns	Initial[2] budget	Outturns	Initial[2] budget	Outturns	Initial[2] budget	1995	1996	1997
Total expenditure	40 600	100.0	42 583	41 341	42 399	40 529	43 972	43 840	44 239	–2.0	8.2	0.6
Social welfare	11 295	27.8	11 488	10 695	10 995	10 212	11 988	12 732	11 977	–4.5	24.7	–0.1
Transport	6 239	15.4	6 507	6 176	6 351	6 202	6 280	6 333	7 038	0.4	2.1	12.1
Defence	5 753	14.2	6 019	5 935	5 952	5 856	5 653	5 580	5 456	–1.3	–4.7	–3.5
Agriculture and food	3 416	8.4	3 354	3 496	3 461	3 547	3 730	3 935	3 978	1.5	11.4	6.6
Research and training	2 970	7.3	3 172	3 081	3 233	3 157	3 233	3 124	3 226	2.5	–1.0	–0.2
Foreign affairs	2 070	5.1	2 140	2 114	2 140	2 052	2 141	2 066	2 101	–2.9	0.7	–1.9
Financing and income tax	5 245	12.9	6 119	6 313	6 408	5 917	6 841	6 099	6 674	–6.3	3.1	–2.4
Other	3 612	8.9	3 784	3 531	3 899	3 586	4 106	3 953	3 789	1.6	10.2	–7.7
Total receipts	32 782	100.0	35 609	36 239	36 319	37 266	39 924	39 477	38 471	2.8	5.9	–3.6
Tax receipts	28 589	87.2	31 498	31 428	32 053	32 129	35 047	34 159	34 603	2.2	6.3	–1.3
Federal income tax	7 884	24.0	9 050	8 960	8 650	8 244	9 400	8 972	9 400	–8.0	8.8	0.0
Withholding tax	1 928	5.9	3 400	3 508	2 900	2 048	4 050	3 318	3 000	–41.6	62.0	–25.9
Stamp duty	2 181	6.6	1 700	2 024	1 850	1 703	1 800	1 977	1 925	–15.9	16.1	6.9
Turnover tax/VAT	9 381	28.6	9 600	9 378	10 700	12 428	11 725	12 068	12 575	32.5	–2.9	7.2
Tobacco tax	1 167	3.6	1 215	1 193	1 360	1 325	1 380	1 388	1 400	11.1	4.8	1.4
Customs duties	1 174	3.6	1 194	1 223	1 196	1 197	1 415	1 358	1 055	–2.1	13.5	–25.4
Fuel duties	3 963	12.1	4 450	4 274	4 470	4 273	4 590	4 384	4 214	0.0	2.6	–8.2
Other	911	2.8	889	868	927	911	687	694	1 034	5.0	–23.8	50.5
Other receipts	4 193	12.8	4 111	4 811	4 266	5 137	4 877	5 318	3 868	6.8	3.5	–20.7
Balance	–7 818		–6 974	–5 102	–6 080	–3 263	–4 048	–4 363	–5 768			
As a percentage of GDP	–2.3		–2.0	–1.4	–1.7	–0.9	–1.1	–1.2	–1.6			

1. Financial account.
2. Initial budget agreed by Parliament.
3. For 1995 and 1996: outturn on outturn of previous period; for 1997: initial budget on initial budget for 1996.

Source: Budget of the Swiss Confederation.

by under-spending in some other domains, most notably for public-debt interest payments (–SF 0.5 billion).

The small difference between the budget out-turn and the 1996 Budget projection is surprising given that the economy turned out to be much weaker than had been assumed. Growth of 2 per cent had been assumed for the budget projections, but the economy actually contracted by 0.7 per cent. Abstracting from unemployment insurance, this should have increased the deficit by more than SF 1 billion. And with additional expenditures for unemployment insurance of SF 0.9 billion, the deficit should have been more than SF 2 billion higher than projected. The fact that the deterioration in the deficit was much smaller is attributable to higher than projected receipts from sources not dependent on the economic cycle and to lower than projected expenditures, notably on debt interest. The revenue sources concerned include stamp duty, receipts not included in the initial budget from granting independence to Swiss control[38] and a higher than projected surplus on the federal civil servant's pension fund. In all, receipts from these sources were SF 1 billion higher than projected.

Measured on the same basis as budget balances for lower levels of government (*i.e.*, using the definitions in the *Statistique Financière Révisée (SFR)*), the Confederation's deficit rose by SF 0.9 billion in 1996 to SF 5.4 billion (1.5 per cent of GDP).[39] Much of this increase is estimated to have been structural. Approximately three-quarters of the 1996 deficit is estimated to be structural.

Budgetary developments in the cantons and communes in 1996[40]

As for the Confederation, the cantons' budget consolidation was interrupted in 1996, with the aggregate deficit estimated to have risen to SF 2.2 billion (0.6 per cent of GDP). This increase was more than accounted for by the rise in payments to the unemployment insurance system.[41] A little over half of the aggregate deficit is estimated to be structural. In contrast, communes were able to continue to reduce their budget deficits as cyclical factors have little effect on their budgets. A variety of factors including better control of expenditures and higher taxes contributed to the estimated elimination of the communes' aggregate deficit.

The 1997 government budgets

The Confederation's budget[42]

The 1997 Budget adopted by Parliament provides for an increase in the deficit of ½ percentage point of GDP to SF 5.8 billion (1.6 per cent of GDP) (Table 9). However, this increase largely reflects changes in accounting practices: for the first time the surplus of the federal civil servants' pension fund is excluded from the Financial Account (it was SF 1 billion in 1996) and loans made to the national railways (SF 0.9 billion in 1997) are included. Abstracting from these factors, there is no change in the deficit compared with the previous year's budget, despite a small decline in revenues. This decline is attributable to the biennial system of collecting taxes which gives rise to weaker revenue growth in odd-numbered years. Over the two years 1996-97, receipts nevertheless are projected to grow at an annual average rate of 3.8 per cent. Expenditure (excluding the loans to the railways[43]) is projected to decline by SF 0.5 billion compared with the 1996 Budget. There are, however, significant differences in developments by expenditure category. Loans to the unemployment insurance fund are projected to rise substantially (+SF 1 150 million). In addition, SF 561 million have been set aside over 1997-99 for the programme to stimulate the economy (*programme de relance conjoncturel*) (see below). These funds had been deducted from planned expenditure in the initial budget (2 per cent of planned expenditure was blocked in most domains[44]). This blockage of credits contributes to weak growth in many expenditure categories, including defence and personnel, where expenditures are projected to fall.[45] A large reduction (SF 931 million, 38 per cent) in payments for medical insurance is also projected, following the substantial one-off increase in 1996 associated with the introduction of new arrangements for the payment of federal refunds (see above). Despite rising debt, interest payments are projected to decline as a result of lower interest rates.

These projections were based on the assumptions that real and nominal GDP would grow in 1997 by 1½ per cent and 3 per cent respectively. Although these rates now appear too optimistic – the Federal Finance Administration (FFA) assumes in its latest projections of the 1997 budget deficit that (real) growth is about 1 percentage point lower – the deficit is still projected to be in line with the 1997 Budget estimates owing to upward revisions in revenue projections for stamp duties, VAT and tobacco taxes and downward revisions in some expendi-

tures, especially interest payments on public debt. On an SFR basis, which excludes the surplus on the civil servants' pension fund but (in contrast to the Financial Account), includes the loans to the railways, the deficit is projected to rise slightly to SF 5.5 billion (1.5 per cent of GDP); about three-fifths of this deficit is estimated to be structural.

The budgets of the cantons and communes[46]

Cantons' 1997 budgets provide for a rise in their aggregate deficit to SF 3.1 billion (0.8 per cent of GDP) (Table 8). Compared with the 1996 budgets, expenditures are projected to rise by 2.4 per cent and receipts to rise by 1.5 per cent. The growth in expenditures is mainly due to increases in loans to the unemployment insurance fund (+SF 850 million) and in current transfers (+SF 750 million). Other expenditures are projected to remain unchanged (personnel, goods and services) or to decline (own investment, investment subsidies). The weak growth in receipts is explained by the fact that income tax revenues are based on incomes in 1995 and 1996, years in which economic activity was depressed. Approximately one-third of cantons' combined 1997 deficit is estimated to be structural.

Compared with the 1996 budgets, communes' 1997 budgets provide for a small decline in the aggregate deficit to SF 0.5 billion (0.1 per cent of GDP); in the event that the 1996 out-turn is better than had been budgeted, as is currently expected, the 1997 out-turn also should be more favourable than budgeted and a surplus is possible. This decline reflects both low expenditure growth (1.1 per cent) and, compared with other levels of government, robust revenue growth (1.8 per cent). The more favourable evolution in communes' finances compared with other levels of government is partly attributable to the virtual absence of cyclical influences on their budgets. Small and medium-sized communes continue to make the most progress in reducing their budget deficits. Large cities, on the other hand, project only modest progress in reducing theirs from relatively high levels.

Abstracting from cantons' loans to the unemployment insurance fund, lower growth than assumed in cantons' and communes' 1997 budgets will not have much effect on their eventual out-turns as their fiscal receipts depend on earnings in earlier years.[47] However, the deterioration in the labour market since cantons decided their 1997 budgets will increase their loans to the unemployment insur-

ance system and hence their aggregate deficit.[48] On the basis of the FFA's latest estimates, the cantons (like the Confederation) are likely to have to lend an extra SF 400 million to the unemployment insurance system in 1997.

The fiscal stimulation programme (programme de relance conjoncturel)

The Confederation's *programme de relance conjoncturel* aims at strengthening demand for a limited period in the early stages of the economic recovery. The programme focuses on bringing forward maintenance expenditures on public infrastructure; the focus on maintenance expenditures as opposed to investment in general ensures that future budgets are not burdened by the costs of maintaining a larger capital stock. Cantons and communes are to be offered subsidies (generally of 15-20 per cent) to advance[49] infrastructure maintenance projects which must be completed by mid-1999. And the Confederation is to advance maintenance expenditure on its infrastructure as well. In addition, subsidies are to be offered to encourage renovation of private buildings to make them more energy efficient. Including the non-application of the 2 per cent blockage of credits to some of the Confederation's investment expenditures, the *programme de relance conjoncturel* will cost the Confederation SF 561 million, most of it being spent in 1998 and 1999. Of this sum, SF 60 million is to be spent on measures (''*arrêté fédéral sur les places d'apprentissage: mesures visant à améliorer l'offre de places d'apprentissage*'') to support vocational and technical education in 1997/98 and 1999/2000. The Confederation estimates that the programme will increase investment (including maintenance) by SF 2.4 billion, which should raise economic growth by ⅓ percentage point in both 1998 and 1999.

The success of the programme depends on lower levels of government being prepared to set aside the funds necessary to bring forward maintenance expenditures. The emphasis on cantons and communes playing a major role reflects the Confederation's view that they are better placed to increase expenditure – in aggregate, the structural deficits of the cantons and the communes are smaller than that of the Confederation – as well as the fact that they too are subject to a constitutional obligation to take the economic cycle into account when deciding budgets. Early indications are that the cantons and communes intend to take advantage of the programme. A backlog of infrastructure maintenance work has

accumulated in recent years in the context of fiscal consolidation efforts; communes in particular have reacted positively to the programme in consultations. However, a concern with this programme, like the earlier investment bonus (1993-94), is that it encourages cantons and communes permanently to factor into expenditure plans the possibility of short-term federal subsidies aimed at stimulating economic activity. In particular, this programme and the earlier investment bonus encourage lower levels of government to delay investment and maintenance expenditures during the early stages of an economic downturn in the hope of subsequently obtaining additional federal subsidies, thereby aggravating the cycle.

The Confederation's Financial Plan, 1998-2000[50]

The Confederation's financial plan of September 1996 for the period 1998-2000 shows a progressive decline in the deficit to SF 4.8 billion (1.2 per cent of GDP) (Table 10). This is to be achieved by constraining expenditure growth to an annual average rate of 2.5 per cent, well below the assumed rate of growth in nominal GDP (3.8 per cent per year). Aggregate expenditure is thus projected to decline from 12.0 per cent of GDP in 1997 to 11.3 per cent in 2000. An important factor contributing to the low growth in expenditure is the general blocking of credits. Low growth in expenditure is projected in all areas except transport (4.3 per cent per year) and debt interest (9.5 per cent per year), where expenditures are boosted respectively by public transport projects (including railways reform and major infrastructure projects, notably Rail 2000 and the alpine transit) and rising debt. Receipts are projected to grow broadly in line with the growth in nominal GDP. Much of the growth in receipts is attributable to VAT. The federal income tax, on the other hand, is projected to decline as a percentage of GDP. On the basis of this outlook, the Confederation's debt would rise to 26 per cent of GDP in 2000 from 24 per cent in 1996.

These projections now appear to be too optimistic. In particular, economic growth in 1997 now seems likely to be about 1 percentage point lower than was assumed when the Financial Plan was elaborated. Projections for federal income tax revenues in 1998-1999 will need to be revised down to take account of lower income growth in 1995-96.[51] Moreover, loans to the unemployment insurance system will probably be larger than projected. The projections are also subject to other important uncertainties, including the impact of the consolidation of the

Table 10. **Financial outlook of the Confederation, 1998-2000**

SF million and percentages

	Budget 1997	Outlook			Growth rate annual average	Structure	
		1998	1999	2000	1997-2000	1997	2000
Total expenditure	44 239	46 829	46 402	47 578	2.5	100.0	100.0
Social welfare	11 977	11 662	11 971	12 165	0.5	27.1	25.6
Transport	7 038	8 806	7 606	7 991	4.3	15.9	16.8
Defence	5 456	5 549	5 629	5 688	1.4	12.3	12.0
Agriculture and food	3 978	4 040	3 957	3 856	–1.0	9.0	8.1
Research and training	3 226	3 323	3 417	3 465	2.4	7.3	7.3
Foreign affairs	2 101	2 102	2 131	2 139	0.6	4.7	4.5
Financing and income tax	6 674	7 464	7 805	8 351	7.8	15.1	17.5
Other	3 789	3 883	3 886	3 923	1.2	8.6	8.2
Total receipts	38 471	40 880	41 185	42 744	3.6	100.0	100.0
Tax receipts	34 603	37 073	37 871	39 961	4.9	89.9	93.5
Federal income tax	9 400	9 300	9 450	10 200	2.8	24.4	23.9
Withholding tax	3 000	3 950	3 800	4 350	13.2	7.8	10.2
Stamp duty	1 925	2 000	2 100	2 175	4.2	5.0	5.1
Turnover tax/VAT	12 575	13 000	13 600	14 300	4.4	32.7	33.4
Tobacco tax	1 400	1 400	1 400	1 400	0	3.6	3.3
Customs duties	1 055	830	810	800	–8.8	2.7	1.9
Fuel duties	4 214	5 070	5 149	5 228	7.5	11.0	12.2
Other	1 034	1 523	1 562	1 508	13.4	2.7	3.5
Other receipts	3 868	3 807	3 314	2 783	–10.4	10.1	6.5
Balance	–5 768	–5 949	–5 217	–4 834			
As a percentage of GDP	–1.6	–1.6	–1.3	–1.2			
Economic assumptions							
GDP growth							
– nominal	3.0	4.0	4.0	4.0	3.8		
– real	1.5	2.0	2.0	2.0	1.9		
Interest rate on the Confederation's debt	4.25	4.25	4.25	4.25	4.25		

Source: Financial outlook of the Confederation from September 1996.

railways' finances, the financing of public transport infrastructure projects (notably, the new alpine transit routes), the effects of the new VAT law which will replace the current arrangement, the modalities for increasing VAT by 1 percentage point to finance social security, business tax reform and the possible increase in profits from the Swiss National Bank (associated with better management of its currency reserves).

The deficit reduction projected in the Financial Plan is considered by the authorities to be inadequate as it falls far short of what is required to halt the rise in the Confederation's indebtedness. Accordingly, the Government has proposed a new budget consolidation plan (Consolidation plan 2001) which specifies that the deficit must not exceed SF 4 billion in 1999 and 2 per cent of receipts in 2001 (about SF 1 billion). These targets, along with a subsequent objective of balance over the business cycle, are to be embodied in the Constitution, obliging the Federal Council and Parliament to respect them: if, in the absence of recession, these targets are not met, corrective measures must be taken so that they are met within the following two years. In the event of recession, the Federal Assembly can extend this delay. The additional reductions in the deficit (SF 1 billion by 1999, SF 3 billion by 2001) from the baseline are to be achieved mainly through expenditure cuts. In addition, increases in the taxes which finance the public-transport-infrastructure projects, the old-age pension scheme (*Assurance vieillesse*) and invalidity insurance (*Assurance invalidité*) are expected to provide additional revenue. In view of Switzerland's weak short-term economic outlook, these objectives would appear to be difficult to attain. Indeed, not exceeding the larger deficits in the Financial Plan 1998-2000 would already be an achievement. Preliminary figures for the Financial Plan 1999-2001 confirm this deteriorating trend compared with the previous Plan.

III. Implementing structural reform: a review of progress

Last year's OECD *Economic Survey of Switzerland* provided a set of detailed policy recommendations for reducing unemployment and for improving the Swiss economy's ability to cope with structural change. Along the lines of the OECD *Jobs Study*, these recommendations were derived from a review of structural features of the Swiss economy which determine the incidence of non-cyclical unemployment (OECD, 1996, Chapter III). The analysis concluded that there are few factors which act as a direct and short-term impediment to higher employment in Switzerland: the wage-formation system appears to respond sufficiently flexibly to changing economic conditions at both the macro- and micro-level; the tax and transfer system and employment protection legislation do not seem to constitute high barriers to hiring, and work disincentives or moral hazard problems from the generous provision of unemployment benefits are mitigated by the rigorous application of work tests and tight controls of benefit entitlements. All this kept the Swiss employment-to-population ratio high and unemployment low by international comparison, in spite of broadly stagnating economic activity during the 1991-96 period.

Reform of unemployment insurance

However, the recent experience of much higher unemployment than recorded before 1992 has put the unemployment insurance system under substantial strain and has given rise to concern that the generous unemployment benefit system, established in the past in successive steps to respond to gradually rising unemployment and its longer duration, could induce prolonged job-search and thereby contribute to a build-up of long-term unemployment. The recognition of

these problems led to a rather radical reform of the unemployment insurance system, which was implemented in two steps.

As from 1 January 1996, a number of conditions for access to unemployment benefits were tightened[52] including *inter alia*:

- a more stringent definition of ''suitable work'', which regards work at a minimum of 70 per cent of previous earnings as suitable (80 per cent for job-seekers with dependants) and raises the penalty for uncooperative behaviour; even a job which pays less has to be accepted, but in this case the job-seeker can claim intermittent pay supplements from unemployment insurance; travelling time to work of up to two hours one way is also defined as acceptable;
- the introduction of a waiting period of five days before the start of unemployment benefit payments; unemployed school-leavers who have never contributed to unemployment insurance have to observe a maximum waiting period of up to 120 days before becoming eligible for benefits;
- the reduction of the maximum duration of compensation for income losses arising from short-time work from 24 to 12 months,[53] and those due to bad-weather from 12 to six months[54] (within a period of two years).

Active labour market policies

On 1 January 1997, the second reform step introduced the ''activation principle'', which shifts the emphasis of unemployment insurance from passive income support to active labour market programmes (ALMPs), rather than reducing the generosity of unemployment and related benefits. It makes, at an early stage of unemployment, further receipt of benefits conditional on participation in ALMPs.[55] The payment of passive benefits is limited to 150 days (7 months), or somewhat longer for persons aged over 50.[56] Following this period, benefit receipt for up to a further 370 days (17 months) is conditional on participation in ALMPs, provided that places are available.[57] Hence, the maximum duration of benefit eligibility has increased from previously 170, 250 or 400 workdays – depending on the number of monthly unemployment insurance contributions – to uniformly 520 workdays, irrespective of the length of the contribution period

in excess of the minimum of six months (Table 11). However, participation in these active labour market programmes does not generate new benefit entitlements, as was formerly the case with temporary employment programmes.[58]

The activation principle aims at facilitating the return to work of the unemployed, especially in view of Switzerland's generous unemployment benefits. It is accompanied by the creation of a nation-wide network of regional placement offices (RPOs) which will increase substantially the number of public employment service (PES) employees performing the functions of brokering and referral to ALMPs. Experience in other OECD countries suggests that the focus on a well-functioning PES is appropriate, as actions to assist the unemployed to return to work taken early in a spell of unemployment tend to be more effective than those taken later.

Table 11. **Evolution of unemployment insurance benefits and contributions**

	Maximum benefits		Maximum duration	Contribution
	Per cent of salary (without/with children)	SF per day/month	Number of days	Per cent of salary
1972	60/85	48/1 248	90/150	SF 12/month minimum
1973	65/70	80/2 080	90/150	
1974	65/70	120/3 120	90/135	
1975	65/70	120/3 210	120[1]/150[2]	
1976	65/70	120/3 120	150/180	
1977[3]	65/70	150/3 900	150/180	0.8
1978	65/70	150/3 900	150/180	0.8
1980	65/70	150/3 900	150/180	0.5
1982	65/70	150/3 900	150/180	0.3
1983	65/70	223/5 800	150/180	0.3
1984[4]	70/80	223/5 800	85/250	0.6
1990	70/80	223/6 800	85/250	0.4
1991	70/80	312/8 100	85/250[3]	0.4
1992	70/80	312/8 100	85/250	0.4
1993	70/80[4]	312/8 100	170/300-400[5]	2.0[6]
1995	70/80	312/8 100	170/250/400	3.0
1997	70/80	312/8 100	520	3.0

1. From 16th April onwards.
2. From 27th November onwards (for particular categories of insured).
3. In a certain number of cantons particularly hit by unemployment, the maximum duration was increased to 300 days (decision of the Federal Council).
4. From 1st April onwards.
5. On 1st January, the Federal Council decided a duration of 300 days as the maximum. From 1st April onwards, this was prolonged to 400 days (Urgent Federal Decree).
6. On 1st January.

Source: Office fédéral de l'industrie, des arts et métiers et du travail.

The 1996 *Survey of Switzerland* expected from these reform measures an improved labour market performance if:

- RPOs focus on reducing the duration of unemployment of job losers rather than catering mainly to job changers;
- careful evaluation of ALMP slots assures that they are of high quality and meet specific needs;
- intermittent pay does not induce re-design and reorganisation of production and working patterns so as to substitute part-time or low pay jobs for full-time or high pay jobs.

In view of the fact that major elements of unemployment insurance reform were implemented only recently, while others are still in the process of implementation, it is too early to make a comprehensive evaluation of the effects of the reforms. In particular, it is far too early to assess whether improved counselling or participation in active labour market programmes has shortened the length of unemployment spells. For example, the fact that more than 40 per cent of the unemployed have no qualifications[59] represents a challenge for the effectiveness of re-training courses. And although at the time of writing (Spring of 1997), 130 of the planned 150 RPOs have already opened, many of them were not fully staffed yet, which has led to complaints from job counsellors that they cannot devote sufficient time to each job-seeker.

First official estimates suggest that the target of 25 000 ALMP slots will be attained in 1997. But in view of recent developments, an increased number of ALMP slots would be needed[60] to match the rise in unemployment, implying that a number of unemployed will be entitled to *passive* benefit payments for up to 520 days compared with 400 days under the previous regime. This will cause expenditures of the unemployment insurance fund to increase, and to raise the unemployment insurance deficit.[61] It also may raise the rate of structural unemployment,[62] due to reduced search efforts and lowered willingness of the unemployed to accept job offers, or through reduced preparedness of insiders to restrain their wage claims, thereby running counter to the objectives of the unemployment insurance reform. Hence, to match the number of job losers and to put the activation principle fully into practice would require raising the number of job counsellors beyond the planned number of 1 850 and increasing the number of ALMP slots further in 1998, in line with the increased level of unemployment.

Working-time flexibility

In accordance with the 1996 *Survey* recommendation to increase the flexibility of working time, the Federal Government proposed a revision of the federal labour law (*loi fédérale sur le travail*) which was approved by Parliament in the spring of 1996. Among other things, the revised law would have extended the limit of ''normal'' working hours[63] from 8 p.m. to 11 p.m. and would have lifted the ban on night work for women in industry, while reinforcing the protection of night-workers, especially pregnant women and nursing mothers. Moreover, the revision of the law would have relaxed the prohibition of employment in the retail sector on Sundays,[64] which is also in line with OECD recommendations to enhance product market competition. However, the Swiss people rejected the proposed revision of the labour law in a referendum on 1 December 1996, leaving in force the labour law of 1964.

Another recommendation of last year's *Survey* aimed at reducing distortions in the choice between full-time and part-time work by introducing lower earnings thresholds for compulsory occupational pension contributions of part-time workers. The underlying rationale is the perception that there appears further scope for expanding part-time work, notwithstanding its already high share in total employment. For example, a recent Swiss Labour Force Survey found that nearly 40 per cent of full-time employed would prefer to reduce the number of hours worked per week by an average of eight hours and accept a proportional cut in their incomes (Brunner, 1996). However, although this issue is on the government's policy agenda, no concrete policy initiative has been undertaken so far (Table 12).

Labour force skills and competences

In spite of its generally high qualification, there is still scope for improving skills and competences of the Swiss labour force. For example, the relatively low literacy level of immigrants is a striking shortcoming. Moreover, student enrolment in scientific and technical education could be encouraged further to improve the skill composition of the labour force. These and other issues of training and education are discussed in some detail in Chapter IV.

Table 12. **Implementing structural reform: an overview**

	Proposal	Actions taken	Assessment/recommendation
I.	**Improve the efficiency of active labour market policies**	Unemployment insurance reform in two steps (1996-1997)	Should be fully implemented during 1997
	• Regional Placement Offices focus on reducing unemployment duration of job losers	Ongoing training of Job counsellors	Align number of job counsellors and ALMP slots to higher unemployment
	• Carefully evaluate ALMP slots	Evaluation programme planned for 1997-99	Make careful evaluation as soon as possible
	• Intermittent pay must not induce substitution of part-time or low pay jobs for full-time or high pay jobs	Intermittent pay must correspond to the ''usual local and occupational'' wage rates and there be no preceding relationship between employer and employees	
II.	**Increase working-time flexibility**		
	• Reduce distortions in the choice between full-time and part-time work from pension scheme	No action	Should proceed in line with proposal
	• Remove obstacles to equal treatment of men and women with respect to working time	Proposed new federal labour law, but rejected by referendum	Prepare less controversial law
III.	**Improve labour force skills and competences**	Creation of Fachhochschulen/Hautes écoles spécialisés	Accelerate implementation
	• Tackle low literacy of new immigrants	Taken up by new ALMP programme	
	• Encourage enrolments of women in scientific and technical education	No action	
IV.	**Enhance product market competition**	New Cartels Act of 1996	Carefully evaluate effect of new Act on product market competition
	• Further liberalise telecommunications sector	Reform of Telecommunications Act decided in March 1997	
	• Derestrict shop opening hours	Partial liberalisation proposed by new federal labour law, but rejected by referendum	
	• Align cantonal procurement rules with internal market law	Reform as from 1 July 1998	
	• Deregulate the markets for electricity and other energy through ''third party access'' and ''unbundling'' of production, transport and distribution	Draft law in preparation	
	• Relax restrictive law (*Lex Friedrich*) on real-estate purchase by foreigners	Liberation of law adopted by Parliament in 1997	

Table 12. **Implementing structural reform: an overview** *(cont.)*

Proposal	Actions taken	Assessment/recommendation
V. Reform business taxation		
• Replace progressive taxation of profits by flat rate tax and abolish tax on firms' capital	Federal draft law submitted to Parliament	
• Cut stamp duty on share issues	Federal draft law submitted to Parliament	

Source: OECD.

Product market competition

The new Cartels Act

Having recognised the undesirability of widespread cartellisation,[65] the Federal Government reformed the Cartels Act within the framework of its *revitalisation programme*. The latter is the government's response to the rejection of Switzerland's participation in the EEA in the referendum of 6 December 1992. The new Cartel Act became effective on 1 July 1996. Unlike the practice in other OECD countries, the new law does not prohibit cartels as such because this would have required a time-consuming amendment to the constitution, and thus a referendum. Instead, the new Cartels Act prohibits the elimination of effective competition and restricts its limitation. The law contains a presumption (refutable in certain cases) that horizontal agreements setting prices, production volume or territorial distribution eliminate effective competition and accordingly prohibits such agreements. Other innovations of the new Cartels Act are the adoption of a tougher stance towards firms holding dominant positions – the Act includes a list of potential abuses of dominant positions – and merger control: mergers will be prohibited if they would eliminate effective competition. The notification limits for mergers are sufficiently high so that relatively small mergers are unlikely to have to be notified to the Competition Commission.[66]

Since the new Cartels Act has been in force for less than a year, it is too early to judge whether it is as effective as the ban on cartels in most other OECD countries. And whether the concept of ''eliminating effective competition'' is as demanding a requirement for blocking a merger as the concepts used commonly in OECD countries – a substantial lessening of competition or creating or strengthening a dominant position – also remains to be seen. Much will depend on the strictness of enforcement of the law in practice.[67]

Other reform issues

Other unfinished business in the field of product market competition are the restrictive laws relating to opening hours in the retail sector. A reform enabling shops to open whenever they like should make possible a substantial increase in employment in this sector. But in several cantons proposals to gradually liberalise opening hours have been rejected by the people; one of the major reasons why the federal labour law was rejected in the referendum in late 1996 (see above)

was that it would have allowed several categories of shops to open on six Sundays a year.

The inter-cantonal agreement on public procurement – in force since the beginning of 1996 – applies the high thresholds agreed in the Uruguay Round also to Swiss bidders. It still needs to be brought into line with the internal market law which will provide for legal instruments to enforce non-discriminatory access to public procurement as from 1 July 1998.

Several steps towards liberalisation in the field of telecommunications have been taken, such as the abolition of the monopoly for the transmission of data through the international telecommunications network and the liberalisation of the market for corporate communications and for voice transmission within closed groups over leased lines. Nevertheless, remaining non-competitive practices in the sector have permitted substantial price increases in recent years (Table 13), combined with only average quality of services. The government intends to separate the postal from the telecommunications activities of the PTT (post and telecommunications). Up to 49 per cent of the shares in Swiss Telecom are planned to be sold on the stock market in the second half of 1998 or later. Parliament has taken the decision to reform the Telecommunications Act, with the objective of abolishing the voice and network monopolies, so that from the beginning of 1998 onwards new entrants will be able to compete with Swiss Telecom across the board.

About three-quarters of the Swiss electricity industry (comprising about 1 200 enterprises involved in production, transportation and distribution) are publicly owned,[68] and electricity production and distribution are subject to a regime of concessions. In practice this means that utilities enjoy a geographical monopoly which is, however, heavily regulated. This has contributed to Swiss electricity prices for industry being among the highest in the OECD (Figure 17) and caused large differences in prices within the country. To open up the markets for electricity (and gas) to competition, the so-called Cattin Report recommended far reaching reforms comprising ''third party access'' (once EU countries offer the same access) and the separation of production, transport and distribution of electricity (''unbundling'') (*Office fédéral de l'énergie*, 1995). Following this line of thinking, a recent report laid out detailed proposals as to how to open up the Swiss electricity market and make it compatible with the internal electricity market of the EU; the latter is expected to be in effect in 1999 (*Bundesamt für*

Table 13. **Business telecommunications charges**

Index, basket of services[1]
1990 = 100

	1991	1992	1993	1994	1995	1996
Switzerland	**103.1**	**115.6**	**116.9**	**119.4**	**132.5**	**126.8**
Australia (Telstra)	109.3	112.3	111.9	108.5	106.0	107.2
Austria	100.1	91.4	88.2	86.8	86.9	86.3
Belgium	98.7	99.0	99.0	101.0	107.9	110.4
Canada (Bell Canada)	95.6	94.0	96.2	98.4	97.9	97.9
Denmark	108.2	107.0	108.1	107.3	103.4	112.5
Finland (HTC)	115.2	106.0	105.2	87.8	93.1	84.4
France	101.5	102.5	102.3	96.6	86.8	69.2
Germany	86.0	84.4	82.9	82.5	83.7	67.7
Greece	106.5	97.4	92.3	113.2	121.1	119.6
Iceland	85.2	85.7	91.1	92.3	98.2	92.4
Ireland	104.1	85.0	86.7	83.0	83.3	77.4
Italy	108.6	106.1	103.4	102.5	95.5	94.5
Japan (NTT)	91.3	91.7	93.2	79.2	82.6	86.7
Netherlands	100.2	104.2	101.9	109.6	111.6	102.7
New Zealand (TCNZ)	113.1	98.2	90.9	88.9	90.7	83.6
Norway	68.3	64.5	65.8	53.9	52.4	47.7
Portugal (TP/TLP)	102.9	93.3	91.3	92.6	107.7	97.8
Spain	116.5	113.3	115.4	114.5	135.3	105.8
Sweden (Telia)	114.8	102.7	101.1	86.7	87.8	89.4
Turkey	133.1	122.8	109.7	107.1	89.7	51.1
United Kingdom (BT)	98.6	99.0	95.7	88.3	83.0	85.5
United States (Nynex)	100.0	101.0	95.2	88.7	88.7	89.2
OECD average[2]	102.6	99.0	97.6	95.2	96.8	90.7
Weighted average[2]	99.8	99.3	96.6	91.2	91.0	87.1

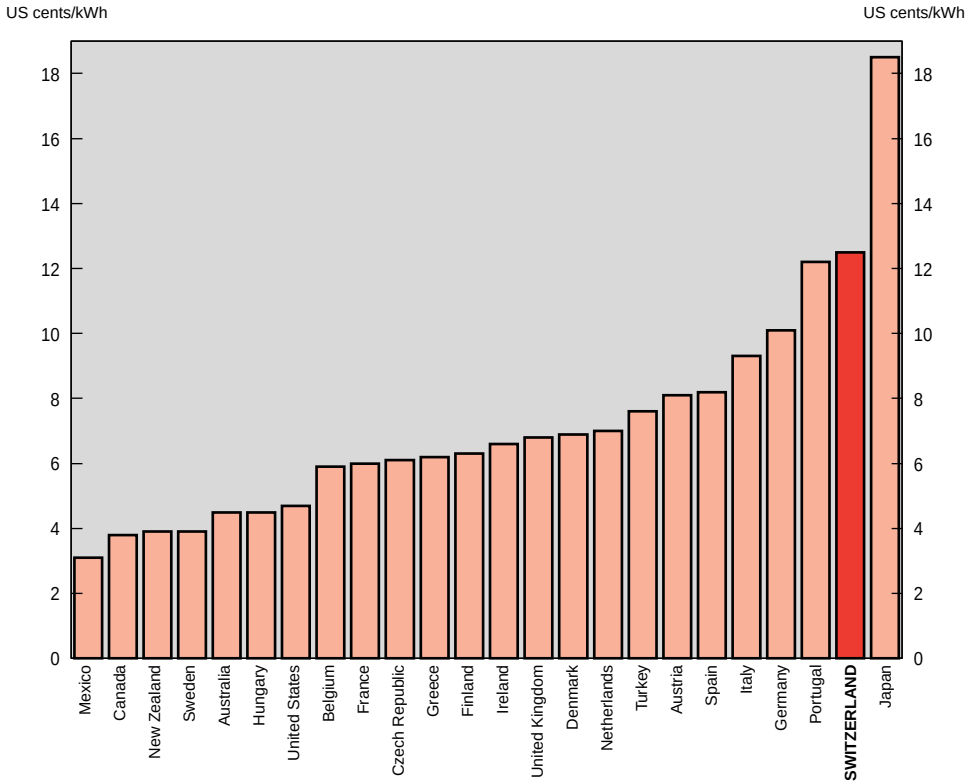
1. Data is from business tariff basket. The basket is expressed in the form of an index that makes 1990 = 100. The weighted average is calculated with 1992 telecommunication mainlines.
2. Excluding Luxembourg.

Source: OECD.

Energiewirtschaft, 1997). Based on this report, the Federal Government is currently drafting a proposal for a new law. In view of forthcoming reform, several large foreign producers have already acquired substantial participation in two Swiss electricity companies which control major transmission lines.

The recent package of measures of the Federal Council to stimulate the ailing economy also comprises an initiative to lift the restraint imposed by the so-called *Lex Friedrich* on purchases of real estate in Switzerland by foreigners who are not in possession of a permanent establishment permit.[69] The revised law completely liberalises foreign investment in business real estate. Foreign

Figure 17. **ELECTRICITY PRICES IN INDUSTRY, 1995**[1]



1. Average for the year; 1994 for Australia, Belgium and Canada.
Source: OECD, International Energy Agency.

residents with an annual work permit are now also allowed to buy a primary residence without authorisation, while the restrictions of the *Lex Friedrich* on the purchase of holiday homes and on financial investment in real estate other than business real estate will remain in place. The objective of the reform is to improve conditions for foreign productive investment in Switzerland.

Measures to reform business taxation

The Swiss business tax system is one of the most complex in the OECD, and its reform has repeatedly been advocated.[70] Current Federal Government proposals to reform the taxation of businesses – the draft reform of March 1997 – focuses on a select number of particularly pressing problems, but leaves a comprehensive reform of the system for the future.

Corporate tax reform

With corporate income tax revenues of 2.0 per cent of GDP in 1995, Switzerland's corporate tax burden in a narrow sense is among the lowest in the OECD (Table 14). The gap relative to other European OECD countries has indeed widened since the 1960s, even though the Swiss corporate tax to GDP ratio has increased during this period (Table 15). A broader definition, which takes account of other direct taxes related to business activities such as employers' social security contributions[71] corroborates that the level of Switzerland's aggregate business taxation compares favourably with that of most other OECD countries (Table 14, last column).

High compliance costs

However, the federal structure of the Swiss business tax system imposes high compliance costs on firms, such as filling out multiple non-uniform tax documents, dealing with complications arising from the dissimilarities between the federal and cantonal tax systems (*e.g.* biennial *versus* annual systems, differences in depreciation allowances, etc.) and keeping track of tax reforms. These costs become especially high when the activity of the firm is spread over several cantons.

The complexity of the Swiss corporate tax system goes well beyond this regional aspect. There are as many as eight different direct taxes levied on corporations in Switzerland.[72] While most of these taxes can be found individually in other countries, their coexistence within the same system is quite unique. Having many different taxes levied on the same entity not only increases compliance costs, but also reduces the transparency of the system. Moreover, compliance costs tend to weigh more heavily on small and medium-sized firms. Recent estimates of compliance costs of small- and medium-sized enterprises (SMEs)

Table 14. **The main direct taxes related to business activities, 1994**

As a percentage of GDP

	Corporate income tax	Social security contribution paid by employers	Taxes on payroll and workforce paid by enterprise	Taxes on property (by other than households)	Taxes on corporate net wealth	Other taxes paid solely by business	Total
Switzerland	**2.0**	**3.7**	**–**	**–**	**0.5**	**–**	**6.2**
Australia	4.2	–	2.0	–	–	–	6.2
Austria	1.3	7.3	2.8	0.2	–	–	11.6
Belgium	2.8	9.4	–	0.0	–	–	12.2
Canada	2.4	4.0	–	–	–	0.4	6.8
Czech Republic	6.4	12.1	0.1	–	–	–	18.6
Denmark	1.9	0.3	0.3	1.1	–	0.1	3.8
Finland	1.9	10.0	–	0.3	–	–	12.2
France	1.6	11.8	1.1	0.4	–	1.8	16.7
Germany	1.1	7.8	–	0.2	–	–	9.1
Greece	2.4	6.0	0.3	–	–	–	8.7
Hungary	1.7	8.3	0.1	–	–	–	10.1
Iceland	0.8	2.4	–	0.6	–	0.3	4.1
Ireland	3.3	3.2	0.6	–	–	–	7.1
Italy	3.7	8.7	0.1	–	0.3	–	12.8
Japan	4.1	5.0	–	–	–	0.1	9.2
Luxembourg	7.6	5.6	–	–	–	–	13.2
Poland	3.4	12.8	0.6	–	–	–	16.8
Netherlands	3.3	3.0	–	–	–	0.1	6.4
New Zealand	4.4	–	0.3	0.0	–	–	4.7
Norway	3.5	5.9	–	–	0.1	0.0	9.8
Portugal	2.3	5.0	–	–	–	0.0	7.3
Spain	1.7	9.6	–	–	–	–	11.3
Sweden	2.8	12.6	0.7	0.3	–	–	16.4
Turkey	1.3	1.9	–	–	–	–	5.3
United Kingdom	2.7	3.4	–	1.9	–	–	8.0
United States	2.5	3.7	–	1.7	–	–	7.9
Unweighted average:							
OECD total	2.9	5.8	0.3	0.3	0.0	0.2	9.4
OECD Europe	2.7	6.9	0.3	0.3	0.0	0.2	9.9
EU	2.7	6.9	0.4	0.4	0.0	0.1	9.7

Source: OECD, *Revenue Statistics of OECD Member Countries 1965-1995*, Paris, 1996.

Table 15. **Corporate tax burden, 1965-94**

Corporate taxes as a percentage of GDP

	1965	1970	1975	1980	1985	1990	1993	1994
Switzerland	**1.5**	**1.8**	**2.3**	**1.8**	**1.9**	**2.1**	**1.9**	**2.0**
OECD total	2.3	2.5	2.4	2.6	2.8	2.8	2.8	2.9
OECD Europe	1.8	1.9	2.0	2.3	2.7	2.6	2.7	2.7
EU-15	2.0	2.2	2.2	2.3	2.6	2.7	2.6	2.7

Source: OECD, *Revenue Statistics of OECD Member Countries 1965-95*, Paris, 1996.

which are caused by the tax system amount to an average of SF 14 000 per enterprise per year (Zarin-Nejadan, 1997). These represent more than 40 per cent of the estimated cost on SMEs from government regulations and amount to about 3 per cent of total machinery and equipment expenditures.

Progressive taxation of profits

A feature of the Swiss corporate tax system, which is unique among OECD countries, is the progressive taxation of profits at the federal level as well as in the majority of cantons, where the tax rate is often an increasing function (within a certain interval) of the so-called ''return intensity'', defined as before or after-tax profits divided by equity capital and reserves (Ammann, 1988). The return intensity of the firm is supposed to reflect its profitability and thereby its capacity to pay taxes. While the analytical soundness of this proxy can itself be questioned, the implicitly assumed relationship between the profitability of a firm and the total income of its shareholders is even less plausible. Hence, no justification on equity grounds can be invoked for such a system (Cornevin-Pfeiffer and Manzini, 1992).

The progressivity of the profit tax schedule as a function of return intensity favours capital-intensive over labour-intensive firms. This bias against labour use appears increasingly inappropriate given the unprecedented rise in Swiss unemployment since 1990, which calls for the abolition of obstacles to job creation. Besides, the progressive taxation of profits is not neutral with respect to different sources of financing. It favours investments financed by retained earnings and equity over those financed by debt, as the former increases the denominator of the return intensity. The system penalises risky new firms generating high profits, as such businesses tend to be undercapitalised and have little self-financing capacity. On the other hand, it privileges large well-established firms with considerable self-financing capacity, such as banks and insurance companies. This distortion can be particularly damaging as 99 per cent of Swiss firms are small and medium-sized (less than 250 employees) and employ about three-quarters of the labour force in the private sector[73] (Table 16).

Two further consequences of the current progressive taxation of profits are worth noting. One is the inevitable occurrence of ''jumps'' in the marginal tax rate given that the law usually specifies a bracket for the average tax burden. An illustration of these jumps is provided in Figure 18 for the federal direct tax on

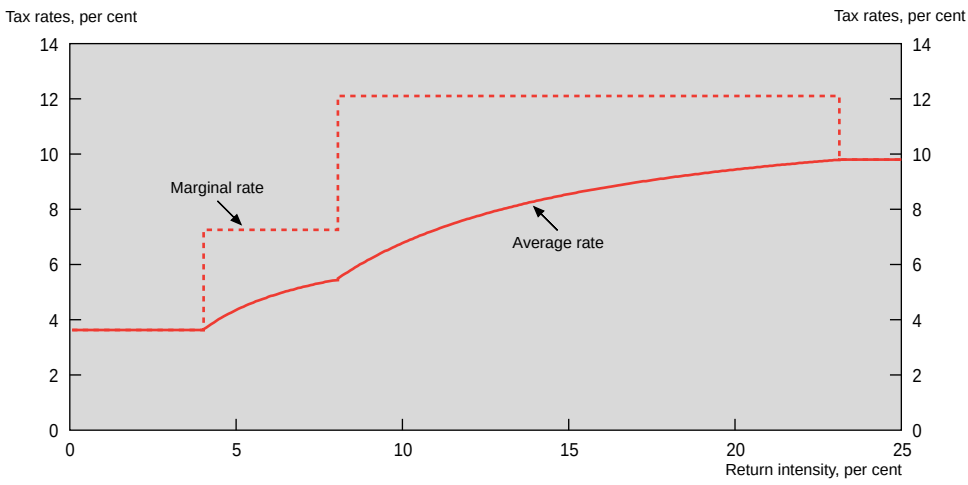
Table 16. **Employment in private firms**[1]

Size of firm (number of employees)	Employment according to the 1995 census		Change in employment	
			1985-1991	1991-1995
	Persons	Per cent	Per cent	Per cent
1 to 9	871 075	30.1	17.5	–0.7
10 to 49	694 594	24.0	14.6	–5.8
50 to 249	590 905	20.4	10.6	–10.0
250 and above	734 073	25.5	12.5	–13.5

1. Full- and part-time.
Source: Office fédéral de la statistique.

profits. Under this system a firm with a return intensity slightly below 23.1 per cent is taxed at a higher marginal rate than one with a return intensity above this threshold. Currently the progressive part of the direct federal tax schedule applies to about 70 per cent of all firms, which represent more than half of the pertinent tax base. Another odd consequence of the progressive schedule as a function of return intensity is the ‘‘bracket creep’’ phenomenon; that is the tax burden rises

Figure 18. **MARGINAL AND AVERAGE TAX RATES OF THE FEDERAL DIRECT TAX ON PROFITS**



Source: Amman, Y. (1988), “Marginale Steuersätze von Aktiengesellschaften”, Mitteilungsblatt für Konjunkturfragen, No. 4; OECD.

automatically as a result of inflation, which is unusual for business taxation. This is so because the return intensity increases with inflation as its numerator is likely to vary in proportion to the general price level while its denominator (including the nominal value of shares) lags behind.

Taxation of capital

Another salient feature of the Swiss corporate tax system is the taxation of capital at both the federal and cantonal levels. The capital tax is levied on the denominator of the return intensity. It is levied at a flat rate for the federal direct tax as well as in most cantons. In general, there is little justification for such a tax on businesses especially if a wealth tax is already levied at the personal level, except that it is a reliable source of tax revenue for the government. This critique is particularly valid for cantons, which all have wealth taxes on individuals, in contrast to the Confederation which does not levy such a tax. As real-estate taxes, the capital tax represents a fixed cost for business and can be particularly onerous to firms during economic downswings. In addition, the capital tax introduces a bias in favour of debt-financing and against projects financed by equity or by retained earnings, compensating to an unknown extent the reverse bias originating from the progressivity of the profit tax.

The Federal Government's recent reform proposal is for the replacement of the current progressive taxation of profits by a flat rate of 8.5 per cent, which would raise additional revenues from profit taxes of about SF 120 million per year. Further, the Federal Council proposes to abolish the corporate capital tax, which would entail a reduction in annual tax revenue by some SF 320 million (Table 17). Moreover, a variety of measures are planned to reduce the compliance costs for small- and medium-sized enterprises by various administrative reforms.

The taxation of holding companies

Another element of reform addresses the tax treatment of holding companies,[74] which in the past benefited from the low levels of taxation, making Switzerland an attractive place for establishing such companies. However, Switzerland imposes a withholding tax on interest on bank accounts and bonds, on dividends and on other cash distributions to shareholders, including the repatriation of profits to non-Swiss investors. As long as other countries applied

Table 17. **Revenue effects of business tax reform**

Proposed measures	Extra revenue (SF million)	Revenue reduction (SF million)
1. Tax exemption of capital and revaluation gains from new participations of holding companies as well as from holding companies' sales of participations		100
2. Flat rate federal profit tax of 8.5 per cent	120	
Abolition of federal tax on the firm's capital and reserves		320
3. Cut in the stamp duty on share issues from 2 to 1 per cent		120
4. More favourable tax treatment of firms' acquisitions of own stocks		n.a.
5. New stamp duty of 2.5 per cent on life insurance premia	250	
Total	370	540
Balance	–170	

Source: Administration fédérale des finances.

such taxes to all non-resident foreigners, Switzerland was able to attract holding companies on the basis of its low rates of profit tax. With the EU's parent-subsidiary directive of 1992, which liberalised cross-border payments from affiliates to parent companies within the EU, investors established in the EU receive the full amount of dividends from their affiliates located in the EU. If these dividends are instead paid to a Swiss holding company, then some countries levy a withholding tax. The holding company may claim a refund based on the double taxation treaty between Switzerland and the country where the controlled company is located. However, this refund may be paid out with a substantial delay and may remain partial. Another EU directive applies a neutral tax regime in the event of restructuring arising within the EU, while Switzerland taxes profits with respect to book values when such restructuring takes place or when a Swiss company sells an affiliate abroad. All this reduced the attractiveness of Switzerland as a location for holding companies and probably contributed heavily to the decline in foreign direct investment inflows after the EU directives came into effect. To restore (at least in part) Switzerland's attractiveness as a location for holdings, and thereby to safeguard the jobs created by holding companies, the Federal Government's tax reform proposal would align the Swiss

treatment of capital gains from restructuring of *new* acquisitions to that of the EU. The pertinent changes are estimated to entail a shortfall in federal tax revenues of some SF 100 million per year.

Stamp duties

A further tax policy measure is the reduction of the stamp duty on new share issues from 2 to 1 per cent, while maintaining the exemption limit of SF 250 000. This reform is estimated to lead to an annual revenue shortfall of SF 120 million. The recent amendment of the stamp duty law, in effect since January 1996, which cut the rate from 3 to 2 per cent, still implied a substantial extra burden on equity-financed projects compared with other OECD countries. Furthermore, the proposed more generous tax treatment of the acquisition of companies' own stocks aims at increasing entrepreneurial flexibility. But in order to recuperate part of the revenue losses from the measures outlined above, the Federal Council plans to re-introduce a stamp duty of 2.5 per cent on life insurance premiums which is deemed justifiable in view of the relatively privileged tax treatment this form of saving enjoys. If introduced, the new stamp duty is estimated to yield annual revenues of SF 250 million.

Assessment

Although the draft reform law remedies important anomalies of business taxation, the Swiss corporate tax system remains among the most complex in the world, imposing high compliance cost on firms. This is only partly a consequence of fiscal federalism. Hence, the task of a thorough reform of the corporate tax system remains at both the federal and the cantonal level. Further reform should include the abolition of the double taxation of dividends, although OECD estimates suggest that Switzerland's corporate ''tax wedge'', a measure of the total distortion caused by taxes at the corporate level, is significantly lower than the OECD average (OECD, 1991*b*).

IV. Education and training

Education and training make a vital contribution to Switzerland's prosperity. Human capital can be considered one of the major productive inputs underpinning the country's high level of labour productivity. Switzerland invests heavily in human capital. The state spends almost 6 per cent of GDP on education and training. In addition, the private sector invests substantially, mainly through foregone earnings while receiving instruction, through the costs to enterprises of providing on-the-job training (*e.g.* for apprentices) and through tuition fees for education services. As for physical capital, the contribution of human capital to overall labour productivity reflects the efficiency of this investment. There are many features of Switzerland's education and training system which contribute to efficient investment in human capital, most notably the strong emphasis on the relevance of education and training to labour-market requirements and the availability of educational pathways suited to students with a wide variety of abilities and interests. But there are other aspects, including the financing of tertiary education and the negligible role of competition from the private sector in the supply of educational services which may mitigate against an efficient use of resources.

Switzerland's enviable labour market performance also attests to the effectiveness of its education and training system. As in other countries with the dual apprenticeship system, the transition from education to work is relatively smooth. And the high proportion of the population which has completed at least upper secondary education contributes to keeping unemployment low by international comparison and to compressing the distribution of wage income. But, as in other countries, the information technology revolution and globalisation are creating new requirements in labour markets. Jobs for unskilled workers are becoming scarce and other workers need to be more skilled than in the past and more ready to continue learning throughout their working lives if they want to hold down

well-paying jobs. Adapting to these requirements is the major challenge now facing the education and training system in Switzerland. At the same time, Swiss governments are under pressure to enhance the efficiency with which they use resources, including for the provision of education and training, which represents a large proportion of government expenditure. What is required are institutional arrangements which enable individuals to invest efficiently in human capital throughout their working lives and which also ensure that equity objectives are achieved. This chapter reviews the efficiency and equity implications of institutional arrangements for the provision of education and training services in Switzerland and discusses actual and potential reforms to enhance the system's performance; an overview of the structure of the system is provided in Annex I.

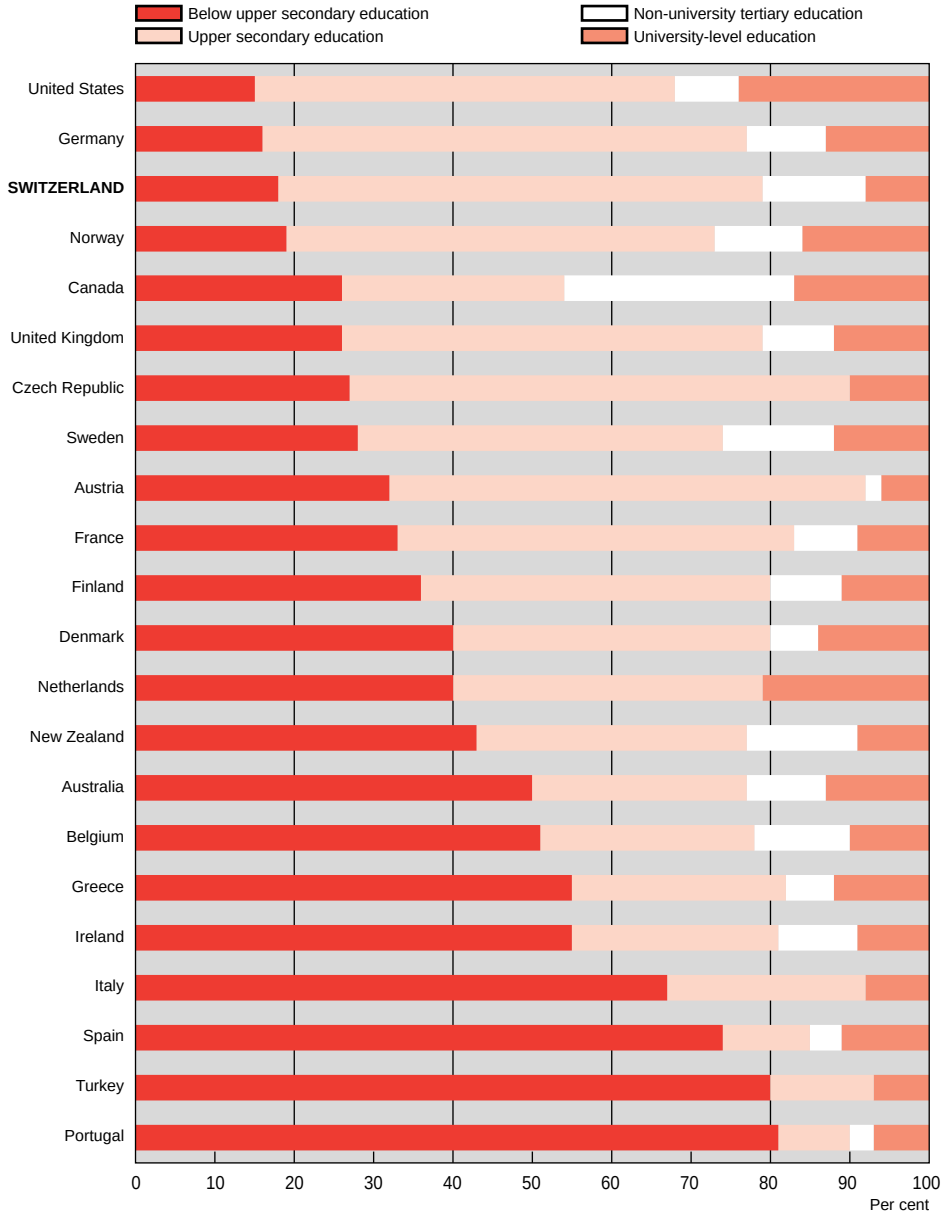
Attainment and structure of the formal education system

The proportion of the adult working-age population in Switzerland (defined as the population aged 25-64) which has completed at least upper secondary education is the third highest in the OECD (Figure 19). Only 18 per cent have a lower level of attainment, less than half the OECD average. A steadily rising proportion of each age-cohort has been completing at least upper secondary education, reaching 89 per cent for the population aged 25-34 (graduates of the 1980*s*). This is about 20 percentage points higher than the OECD average. Tertiary attainment, on the other hand, is similar to the OECD average (20 per cent). Attainment is above average at the non-university tertiary level (13 per cent) and below average at the university level (8 per cent). However, whereas tertiary attainment has been constant in Switzerland for each graduating age cohort since the 1960s, it has risen on average in OECD countries by about one-fifth since then.

There is a large gap between male- and female attainment in Switzerland (Table 18). In 1994, 69 per cent of the adult population which had not completed upper secondary education was female, well above the proportion in most other OECD countries (and as high as in Germany). Moreover, this ratio had only fallen to 65 per cent for the population aged 25 to 34, compared with an OECD average of 50 per cent (*i.e.* equality). Females comprise only a small proportion of the population having completed tertiary education, especially at the non-university level. Although considerable progress has been made in raising female

Figure 19. **EDUCATIONAL ATTAINMENT OF THE ADULT WORKING-AGE POPULATION, 1994**[1]

As per cent of total



1. Distribution of the population aged 25 to 64 by the highest completed level of education.
Source: OECD (1996*a*).

Table 18. **Percentage of women in the adult working-age population by level of attainment, 1994**[1]

	Early childhood, primary and lower secondary education	Upper secondary education	Non-university tertiary education	University-level education
Switzerland	**69**	**53**	**23**	**31**
Australia	60	34	47	47
Austria	61	44	59	36
Belgium	51	47	62	36
Canada	49	54	50	45
Czech Republic	66	46	–	40
Denmark	55	44	57	47
Finland	49	52	54	42
France	57	47	56	43
Germany	69	49	35	36
Greece	52	51	44	44
Ireland	47	58	53	42
Italy	52	49	–	44
Netherlands	56	46	–	42
New Zealand	57	38	66	43
Norway	50	50	48	47
Portugal	52	49	74	49
Spain	52	47	36	50
Sweden	46	50	55	48
Turkey	43	35	–	31
United Kingdom	61	46	58	37
United States	49	53	55	46
Country mean	55	47	44	42

1. Highest completed level of education. Population aged 25 to 64 years.
Source: OECD (1996*a*).

participation in tertiary education, particularly at university level, the ratio of female-to-male tertiary attainment remains much lower than in most other countries.

Many pathways to complete upper secondary education[75]

Vocational education

An important factor contributing to the high proportion of the population which has completed secondary education is the wide variety of vocational-education pathways available to reach this level of attainment. By far the most important form[76] of vocational education is apprenticeship (see Box 1 for a

Box 1. **The apprenticeship system**

Apprenticeships generally begin after nine years of compulsory schooling (primary and secondary I), when students are 15 or 16 years old, and last for two, three or four years. Apprentices receive on-the-job training under the supervision of a master (*Meister*)[1] and attend a vocational school for one to two days per week. In recent years, they also have been required to attend introductory courses which are intended to ensure that they have the basic aptitudes necessary to make good progress in their apprenticeships, regardless of the nature of their employer.[2] Programme content is largely determined by employers, ensuring that it remains relevant, and codified in federal regulations. Cantons supervise the development of apprentices and certify them only after an external examination has been passed. These certificates are recognised nationally. Cantons also pay for the costs of running vocational schools.

An apprenticeship contract must conform to the provisions of the Federal Law on Vocational Education 1978 (*la loi fédérale sur la formation professionnelle (LFPr*)). Such a contract is considered to be a training contract rather than an employment contract. An important implication is that apprentices expect to contribute to the costs of their apprenticeship training through accepting low wages – indeed, some apprentices receive no pay at all. Low apprentice wages are an especially important incentive for small firms to engage apprentices as these firms do not retain a high proportion of their apprentices once they are qualified. For such firms, the pay-back for investing in apprenticeships is in cheap labour, especially towards the end of the apprenticeship. For large companies, the main incentive to engage apprentices is to assure the future supply of skilled employees. Such companies tend to have high retention rates. This both enables them to amortise their investments in apprentices over a longer period and reduces the risk of hiring weak performers on the external labour market.

1. An apprentice master must be qualified to perform this function. To gain this qualification, it is necessary to undertake further studies both in the profession concerned and in pedagogy. In a *craft/small business* type of apprenticeship, the apprentice works in close collaboration with the master; in an *industrial* type of apprenticeship, the apprentice spends up to 45 per cent of his time in workshops working under the supervision of full-time instructors (masters). Apprenticeship regulations fix the minimum ratio of masters to apprentices in a workplace. This helps to ensure that adequate resources are available in the workplace to provide high quality training to apprentices.
2. These courses generally last several weeks and take place in training centres (they may also be held in schools or in employers' special workshop for apprentices). The courses are given by full-time instructors or specialists in the field who devote part of their time to teaching. The addition of these courses (since 1978) to apprenticeship programmes is the reason for which the system is now often referred to as a *tripartite* system.

description of the system). Apprentices combine on-the-job training under the supervision of a master (*Meister*) with regular attendance at a vocational school (one to two days per week).[77] This alternation between regular instruction and on-

the-job training is the reason for which the system is also known as the *dual system*. There is a wide variety (over 400) of apprenticeships available, although most apprenticeships are in a relatively small number of fields: office work/ administration, machine operation and para-medical care account for 55 per cent of new apprenticeships. The variety of apprenticeships available is such that students of most ability levels and interests can be accommodated within the system. A vocational education at the upper secondary level can be obtained alternatively by studying full-time in specialised vocational schools[78] or in public-school-workshops.[79] Approximately 16 per cent of students at the upper secondary level are enrolled in these schools. They are more popular in the latin parts of the country and among women.

Approximately two-thirds of the adult working-age population has completed at least a vocational education at the upper secondary level, with this proportion rising to three-quarters for the age group which completed its upper secondary education in the 1980s (Table 19). Vocational education remains more popular among men and in German-speaking regions. Differences by gender have been stable during the past two decades, but by linguistic region they have continued to narrow. Of the population which completed its upper secondary education in the 1980s, 77 per cent in German-speaking regions have at least a basic vocational qualification, compared with 66 per cent in the rest of the country.

The proportion of an age cohort entering a vocational programme after completing compulsory schooling rose steadily until the mid-1980s, but has since declined by 3 percentage points, falling to 54½ per cent in 1993 (Table 20).[80] This decline is entirely accounted for by students who had chosen the more advanced stream in lower secondary school.[81] An increasing proportion of these students have been opting for general education at the upper secondary level, especially in academic schools. The fact that there has been no decline in the proportion of students from the less demanding stream passing directly to vocational education undoubtedly reflects their more limited options in general education. But this also underlines the continuing importance of vocational education as an effective means of providing these students with attractive pathways to complete upper secondary education. The main counterpart of rising participation in general education has been a decline in the proportion of students dropping out of the education system after compulsory education; this proportion fell to 15 per

Table 19. **Vocational education by gender, linguistic region and age cohort**

Per cent of respective cohort

At least a basic vocational education	Age group (year of 16th birthday)						Total
	70 and older (before 1939)	60-69 (1940-49)	50-59 (1950-59)	40-49 (1960-69)	30-39 (1970-79)	25-29 (1980-89)	
Men	60	70	76	79	77	78	69
Women	42	44	54	63	68	69	57
German-speaking regions	53	61	70	74	77	77	69
Men	65	74	82	82	79	79	78
Women	45	48	58	66	73	73	60
French- and Italian-speaking regions	39	44	53	62	61	66	54
Men	46	59	59	69	69	73	63
Women	34	34	46	57	55	61	48
Total	49	55	65	71	72	74	65

Source: Office fédéral de la statistique (1996*a*).

Table 20. **Educational choices after compulsory schooling**

Per cent of students completing compulsory education

Academic year	Exit from education system after compulsory education[1]	Passage from education with				
		elementary requirements to a		advanced requirements to a		
		vocational education	general education	vocational education	diploma school or another school of general education	academic school[1]
1978	22.6	22.3	0.7	30.7	6.4	18.6
1985	19.1	24.5	1.0	33.0	5.3	16.5
1993	14.6	24.2	1.9	30.4	7.6	21.4

1. Preparing students for the *maturité*.

Source: Office fédéral de la statistique (1996*a*).

Table 21. **Secondary school drop-out rate**

As a per cent of students completing compulsory education

Academic year	Exit from education system after compulsory education[1]	Re-entry[2] into upper secondary education	Drop-outs
1978	22.6	12.8	9.8
1985	19.1	13.5	5.6
1993	14.6	7.2	7.4

1. Preparing students for the *maturité*.
2. Estimate.

Source: Office fédéral de la statistique (1996).

cent by 1993. Some of the drop-outs return to education – 14 per cent of all the beginners in upper secondary education have not attended school in the previous year. The percentage of the students dropping out of the educational system definitively after compulsory school has dropped from 10 per cent in 1980 to 7 per cent in 1993 (Table 21). In latin parts of the country, this proportion fell to only 3 per cent. In German-speaking parts of the country the drop-out rate for males was also low (6 per cent), but it reached 22 per cent for females.

Reflecting recent trends in participation in vocational education, the proportion of a cohort successfully completing a vocational programme of two or more years peaked in 1989 and subsequently fell to 72 per cent for men (–4 percentage points) and 58 per cent for women (–10 percentage points) in 1993. The more severe decline in female-graduation rates has reversed a large part of the convergence in rates achieved during the 1980s. Despite an increase in the proportion of the population successfully completing a general programme at upper secondary level, the overall graduation rate from upper secondary school has fallen from a peak of 89 per cent in 1989 to 83 per cent in 1993. This decline seems to be due mainly to the rising proportion of immigrants in this age group (following the war in Yugoslavia); they make up about 4 to 6 per cent of the age group in question and in most cases seek employment directly after completing compulsory schooling. Inadequate language skills seem to be an important barrier to their continued participation in the education system.

General education

The principal form of general education at the upper secondary level is that offered by the academic schools (*gymnase*) which prepare students for the

maturité examination (the equivalent of A levels or the *baccalauréat*), the qualification necessary to enter a university. In most cantons, entrance to these schools is selective (there is an entrance examination) and occurs at 15 or 16 years of age. By 1993, the proportion of students leaving compulsory education who went on to a *gymnase* reached 21 per cent, a significant increase from the mid-1980s (Table 20). This proportion is much the same for men and women but is significantly higher in latin regions than in the German-speaking regions. In 1993, 16 per cent of the relevant cohort gained a *maturité*, with the proportion being the same for men and women. The other main type of general education at the upper secondary level is offered by diploma schools (*école du degré diplôme*). The education offered by these schools is at a lower level than the *gymnase* and there is often an emphasis on languages. These schools are mainly attended by women preparing for a subsequent vocational education (such as teaching or medical care).

Diversity of tertiary educational options

The wide variety of educational options available to participants in the dual system continues at the tertiary level. Professional as opposed to academic training is offered by numerous non-university institutions: they teach the skills necessary to perform competently in a profession but, unlike universities, do not aim at preparing their students to undertake research. A wide range of subjects is taught by these institutions, including technology, agriculture, business administration, teacher-training, social work and arts and crafts. There are two levels of such institutions: Advanced Vocational Colleges (*les écoles professionnelles supérieures*), which offer diploma programmes lasting three years full-time or four-years part-time; and Technical Colleges (*les écoles techniques*), which offer diploma programmes lasting two years.[82] In most cases, students gain access to these institutions by having completed an apprenticeship of three or four years; although rare, it is also possible to gain access with a *maturité* certificate and at least one year of approved on-the-job training. The programmes offered by these institutions are considered to be of high quality (OECD, 1991*a*) and in many cases are equivalent to university programmes in other countries. As at the secondary level, the link between theory and practice in vocational education remains important. This is assured at the tertiary level by work experience requirements to obtain a diploma. In many cases this requirement is satisfied by

combining professional part-time work with part-time studies; in 1993/94, 55 per cent of students studied part time.[83] To gain a diploma from an Advanced College, it is also frequently required already to hold a diploma from a Technical College. Entries to colleges have risen substantially in recent years, rising from 22 per cent of a cohort in 1987 to 28 per cent 1993. Graduations from them amount to 25 per cent of the population at the typical age of graduation, almost twice the OECD average (Table 22). The graduation rate for males is almost twice that for females, reflecting the higher rates of male participation throughout the dual system.

Switzerland also has 10 university-level institutions, comprising eight cantonal universities and two Federal Institutes of Technology (FITs).[84, 85] University attendance rates have increased significantly in recent years, but nevertheless remain low. This is principally the counterpart of high participation in the dual system. Most students in the dual system do not possess a university entrance qualification: neither the federal proficiency certificate nor a diploma from a Technical College qualifies its holder for entrance to a university. Although a diploma from an Advanced Technical College can serve as a university entrance qualification,[86] this is generally not a very attractive option: there are few courses at universities which build on those offered at Advanced Technical Colleges, students tend to be in their late twenties or early thirties before gaining such a diploma, and the opportunity cost of prolonging their education would be high. Another factor contributing to low participation rates is long first-degree programmes. University graduation rates are much lower in countries with such programmes than in those with short programmes (Table 22). Compared with countries having long first-degree programmes, Swiss graduation rates are only a little below average. At the doctoral level, however, Swiss graduation rates are the second highest in the OECD.

University graduation rates for males are substantially higher than those for females, though less so than in non-university tertiary institutions (Table 22). This gender difference is large compared with other OECD countries, including those which also have a dual system (Germany and Austria). However, it should decline in the next few years reflecting the trend towards gender equality in entries to university: in 1996, 46 per cent of first-year students were women, compared with 36 per cent in 1985.

Table 22. **Tertiary graduation rates, 1994**

Tertiary graduates as a percentage of the population at the typical age of graduation

	Non-university tertiary programmes			Short first university degree programmes (*e.g.* US Bachelor's)			Long first university degree programmes (*e.g.* German *Diplom* or Italian *Laurea*)			Second university degree programmes (*e.g.* US Master's)			Ph.D. or equivalent		
	Total	Men	Women	Total	Men	Women	Total	Men	Women	Total	Men	Women	Total	Men	Women
Switzerland	**24.6**	**32.1**	**17.3**	**–**	**–**	**–**	**8.6**	**10.9**	**6.3**	**–**	**–**	**–**	**2.9**	**4.0**	**1.7**
Australia	–	–	–	31.5	25.4	37.9	–	–	–	11.5	10.4	12.6	0.7	0.9	0.4
Austria	4.9	2.9	6.9	–	–	–	9.1	10.0	8.2	–	–	–	1.1	1.6	0.7
Belgium[1]	22.9	19.0	26.9	–	–	–	14.2	15.2	13.1	4.0	4.8	3.1	0.5	0.8	0.3
Canada	–	–	–	30.2	25.5	35.1	–	–	–	4.8	4.9	4.7	0.7	1.0	0.5
Czech Republic	5.3	3.2	7.5	1.4	1.2	1.5	12.3	12.5	12.0	–	–	–	0.1	0.1	0.0
Denmark[1]	8.9	11.1	6.6	20.2	16.1	24.6	6.3	7.1	5.4	1.3	1.3	1.3	0.4	0.6	0.2
Finland	25.1	14.0	36.8	7.5	9.3	5.5	13.5	11.8	15.3	–	–	–	2.5	2.9	2.1
France[1]	25.4	22.1	28.8	–	–	–	13.7	12.9	14.5	–	–	–	5.3	5.6	5.1
Germany[1]	10.7	9.4	12.1	–	–	–	12.6	14.6	10.4	–	–	–	1.5	2.0	0.9
Greece[1]	4.6	4.0	5.1	–	–	–	11.7	9.5	13.9	0.2	0.2	0.2	0.3	0.5	0.2
Hungary[1]	–	–	–	5.4	4.6	6.3	11.0	9.9	12.2	1.6	2.0	1.2	–	–	–
Ireland[1]	13.9	14.5	13.2	22.8	22.3	23.4	–	–	–	4.1	4.3	3.8	0.7	0.9	0.4
Italy	9.0	6.6	11.5	0.8	0.5	1.0	10.6	9.8	11.5	–	–	–	1.8	2.1	1.6
Japan	28.3	17.2	40.0	23.4	31.2	15.3	–	–	–	1.9	3.1	0.7	0.5	0.8	0.1
Mexico	1.7	–	–	–	–	–	7.1	–	–	–	–	–	–	–	–
Netherlands[1]	–	–	–	17.4	17.2	17.6	9.3	10.0	8.6	–	–	–	1.7	2.2	1.1
New Zealand[1]	10.2	6.7	12.9	21.9	19.9	24.0	–	–	–	9.3	9.9	8.7	–	–	–
Norway	47.0	42.0	52.3	17.6	12.5	22.9	5.3	4.8	5.9	7.7	8.7	6.6	0.7	1.1	0.4
Portugal	6.1	4.0	8.1	1.8	1.2	2.4	12.1	9.2	15.0	0.9	0.8	0.9	–	–	–
Spain[1]	1.1	1.1	1.1	8.6	6.8	10.4	12.6	11.4	13.9	–	–	–	0.8	0.9	0.6
Sweden	11.6	6.4	17.1	6.9	4.8	9.1	6.6	7.0	6.1	2.7	1.8	3.6	1.6	2.2	0.9
Turkey[1]	2.0	2.2	1.8	7.1	8.8	5.4	–	–	–	0.4	0.5	0.3	0.3	0.4	0.2
United Kingdom	25.1	21.8	28.5	27.0	26.5	27.5	–	–	–	8.0	8.1	8.0	1.0	1.4	0.6
United States	21.9	17.9	26.0	31.8	28.4	35.2	–	–	–	11.9	11.2	12.6	1.1	1.4	0.9
Country mean	13.5	11.7	16.4	15.7	14.6	16.9	10.4	10.4	10.8	3.5	3.6	3.4	1.1	1.4	0.8

1. 1993 data.

Source: OECD (1996*a*).

Reforms to enhance the attractiveness of the dual system

Reforms have been introduced in recent years to create a pathway for students in the dual system through to a level equivalent to university studies (''*Hochschulen*''). These reforms have entailed the creation of a pathway (*maturité professionnelle* (MP)) at upper secondary level which prepares apprentices better for tertiary level studies and of new technical institutions called Colleges of Higher Education (*Hautes Écoles Specialisées* (HES), or – in German – *Fachhochschulen)*. This responds to the growing demand in Switzerland, as in other countries, for highly-skilled workers. It is also intended to make the dual system more attractive and hence to stem the flow in recent years towards the academic *gymnase*-university pathway. The new pathway has a much greater emphasis on general education[87] than other pathways in the dual system. This is intended to make graduates more flexible.

Apprentices who prepare for an MP receive considerably more general instruction than traditional apprentices.[88] Flexible arrangements are being made for apprentices receiving instruction in general education so that they do not regularly have to be away from the workplace for too long.[89] An MP gives its holder both the right to exercise the relevant profession and to enter a HES. Arrangements for the implementation of MP in different fields have been progressively introduced since 1992.[90] The MP has proved to be popular. Even though it is still in an introductory phase, 11½ per cent of apprentices are currently preparing an MP.

The HES are to be created by regrouping and upgrading the present Advanced Technical, Commercial and Arts-oriented Colleges.[91] The HES are intended to train highly-qualified professionals by offering a combination of practical and theoretical education in the framework of 3-4 year degree programmes. It is intended that these institutions integrate a wider range of activities than the Advanced Colleges they replace. These activities will include: applied research and development; knowledge and technology transfer; services to industry and to small and medium-sized enterprises (networking); continuing education for teachers and graduates; quality assurance; and building partnerships with other tertiary education institutions. Basic research will continue to be carried out at universities. Legislation was passed in 1995 creating the legal framework for the establishment of HES.[92] Due to delays related to the establishment of selec-

tion criteria and the creation of a counselling committee, the first HES degrees will not be delivered until after 2000.

Achievement

Swiss schools have a good record in imparting to students skills and knowledge in the core subjects of reading (comprehension), mathematics and science. Mastering these subjects is important if students are to be equipped to enter the labour market and to become lifelong learners who analyse, reason and communicate their ideas effectively. In recent international studies[93] of achievement of 14 year-olds, Swiss students obtained scores above the average of their counterparts in other OECD countries in reading and maths and an average score in science (Table 23). Only four of the 26 countries taking part in the mathematics study had significantly higher scores than Switzerland while in science this number rose to twelve. Overall, this is a good achievement in view of the higher age at which Swiss students start school.[94] Nevertheless, the weaker relative performance in science has raised concerns. Investigations into the reasons for this performance suggest that the main factors are that there are fewer lessons in natural science in the Swiss curriculum and that there is less emphasis on terminology than in other countries.[95] Another important dimension of these results is the difference between the scores of the highest-achieving and lowest-achieving groups of students, which can be interpreted as an indicator of the school system's contribution to equality of economic opportunity. The interquartile range (*i.e.* between the 75th and 25th percentiles) divided by the mean of scores is the same in Switzerland as the OECD average. In addition, despite large differences in financial resources per student from one canton to another, there are no significant differences in achievement if differences in social structure are taken into account.[96]

The Swiss experience that achievement is not dependent on expenditure per student (within the range of observations) is consistent with experience in other countries.[97] In an effort to identify factors which might contribute to good performance in mathematics at least, a recent study (Whitburn, 1997) examined teaching methods in Switzerland and Japan. These countries were selected as they differ in many important respects (including in expenditure per student (see below)) but have consistently good performance in mathematics. A number of

Table 23. **Student achievement**[1]

	Reading[2]		Mathematics[3]		Science[3]	
	Mean	Inter-quartile range[4] divided by mean	Mean	Inter-quartile range[4] divided by mean	Mean	Inter-quartile range[4] divided by mean
Switzerland	**103.0**	**0.2**	**105.6**	**0.2**	**99.8**	**0.2**
Australia	–	–	102.7	0.3	104.2	0.3
Austria	–	–	104.5	0.3	106.7	0.2
Belgium						
Flemish	–	–	109.5	0.2	105.2	0.2
French	89.2	0.2	101.9	0.2	90.1	0.3
Canada	98.8[5]	0.2[5]	102.1	0.2	101.5	0.2
Czech Republic	–	–	109.3	0.2	109.8	0.2
Denmark	100.0	0.2	97.3	0.2	91.4	0.3
Finland	109.0	0.2	–	–	–	–
France	106.2	0.2	104.3	0.2	95.2	0.2
Germany[6]	99.6	0.2	98.6	0.2	101.5	0.3
Greece	96.4	0.2	93.8	0.3	95.0	0.2
Hungary	–	–	104.1	0.2	105.9	0.2
Iceland	102.8	0.2	94.4	0.2	94.5	0.2
Ireland	96.8	0.2	102.1	0.3	102.9	0.3
Italy	96.6	0.2	–	–	–	–
Japan	–	–	117.2	0.2	109.2	0.2
Netherlands	97.2	0.2	104.8	0.2	107.1	0.2
New Zealand	105.6	0.3	98.4	0.3	100.6	0.3
Norway	97.8	0.2	97.5	0.2	100.8	0.2
Portugal	–	–	88.0	0.2	91.8	0.2
Spain	91.2	0.2	94.4	0.2	98.9	0.2
Sweden	105.8	0.2	100.6	0.2	102.3	0.2
United Kingdom						
England	–	–	98.1	0.3	105.5	0.3
Scotland	–	–	96.7	0.3	98.9	0.3
United States	102.8	0.2	96.9	0.3	102.1	0.3
Country mean	100.0	0.2	100.0	0.2	100.0	0.2

1. Test score achieved by children in the eighth grade. They are mostly 14 years old.
2. 1991.
3. 1994.
4. Difference between scores of the 75th and 25th percentile.
5. British Columbia.
6. Western Germany.

Source: OECD (1995*b*) and OECD (1996*a*).

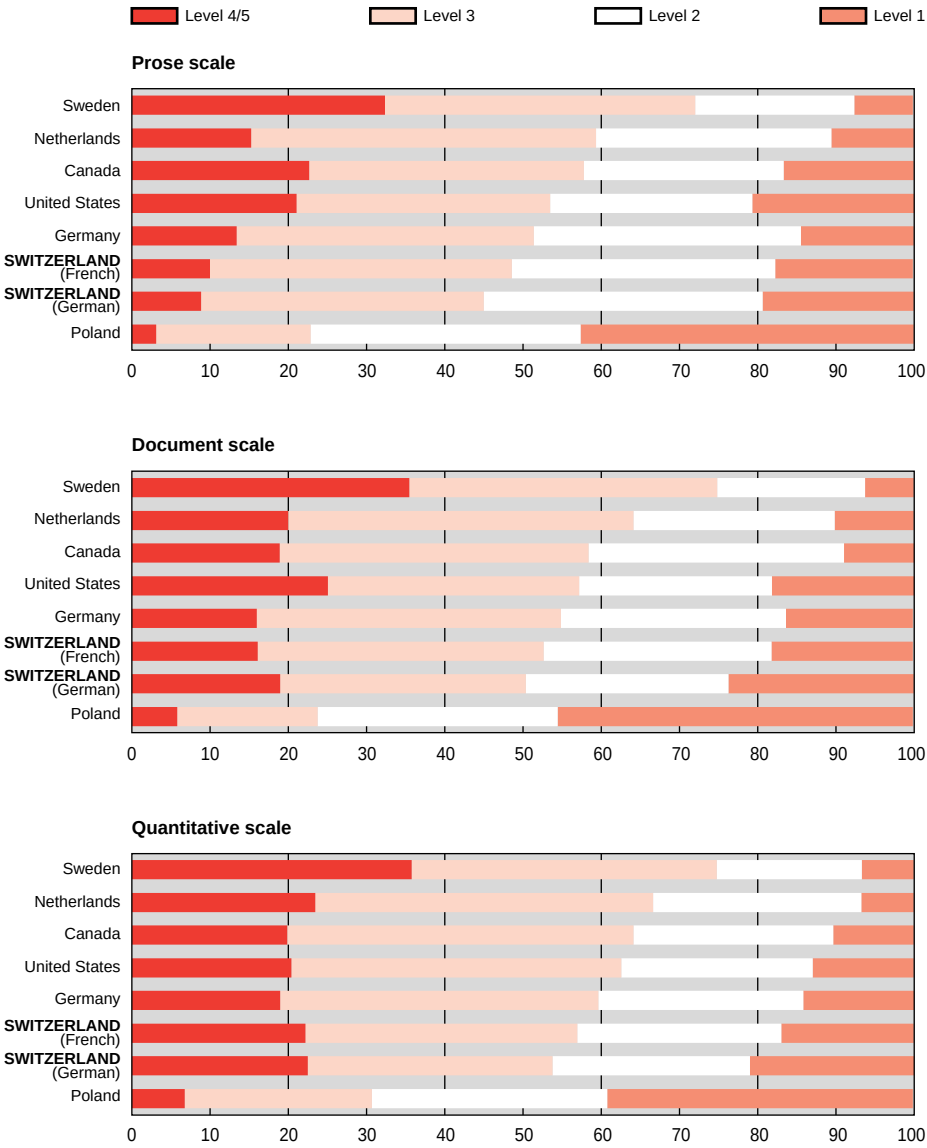
common features were found which may contribute to good performance, including: more time spent on the basics of arithmetic than on more general mathematical topics; pupils learning to do arithmetic operations in their heads before doing them on paper; widespread use of standardised manuals which have been subject

to pilot tests; whole-class interaction, ensuring that pupils are following the lesson; and great efforts to ensure that pupils do not fall behind, including providing extra coaching where necessary.

Adult literacy

Literacy is a key skill for acquiring other competences.[98] Today, adults in industrialised countries require higher levels of literacy than ever before to function well in a society which is becoming more complex and in which demand for low-skill labour seems to be falling (or is being truncated by high wage floors). In view of the growing importance of adult literacy, Switzerland and six other countries recently took part in a special survey (First International Adult Literacy Survey, 1994). Literacy was defined as a continuum, with proficiency levels indicating how well adults use written information to function in society. Three domains – prose, document and quantitative[99] – were tested, and a person's literacy ability was ranked on a scale from 1 to 5, with 1 indicating very poor skills and 3 regarded by experts as a suitable minimum for coping with the demands of everyday life and work in a modern society. Overall, almost half of Swiss adults surveyed had literacy levels corresponding to the lowest two categories and less than one-fifth scored in the top two categories; scores were better for the quantitative part and worse for the prose part (Figure 20). Allowing for the margins of error associated with these estimates, adult literacy in Switzerland is similar to that in the other participating countries except Sweden (where it is significantly better) and Poland (where it is significantly worse). As in the other countries, educational attainment is a major determinant of literacy. Nevertheless, a comparable level of attainment can produce very different literacy results across countries. For example, a German-speaking Swiss with tertiary qualifications has a 28 per cent chance of scoring at one of the top two literacy scales on the document scale, whereas a comparably qualified Swede has a 55 per cent chance. As educational attainment is higher amongst young people, they also tend to have higher literacy scores. In Switzerland, this difference is particularly marked. This is surprising in view of the small rise in attainment in Switzerland in recent years (as noted above, attainment has long been high in Switzerland). It suggests that in recent years the Swiss education system has become relatively

Figure 20. **ADULT LITERACY, 1994**[1, 2]



1. Percentage of adults (16-65) at each literacy level on the International Adult Literacy Survey scales.
2. Countries are ranked in descending order of the percentage of the population scoring at literacy level 3 and better.

Source: OECD (1996*a*).

Table 24. **Low literacy and migrant status**

Percentage of level 1 results[1]

	Swiss German		Swiss French	
	Immigrants	Non-immigrants	Immigrants	Non-immigrants
Prose	58	10	34	11
Document	57	9	32	10
Quantitative	48	6	29	

1. The lowest of the five scales in the International Adult Literacy Survey. Level 1 indicates very poor skills.
Source: OECD and Statistics Canada (1995).

more effective in imparting literacy skills and/or that the Swiss work and home environment is relatively ineffective at maintaining them.

A special factor weighing on literacy scores in Switzerland is the large immigrant population (comprising about 20 per cent of the population aged 16-65). Immigrants were over-represented in the groups with poor literacy levels (Table 24). This probably reflects the fact that their mother tongue[100] is neither German nor French and that most of them have low levels of educational attainment. Allowing for this, the education system performs better for Swiss nationals than is implied by the above results. However, immigration has little bearing on the scores for persons holding tertiary qualifications, which were well below those in the top performing countries (Sweden).

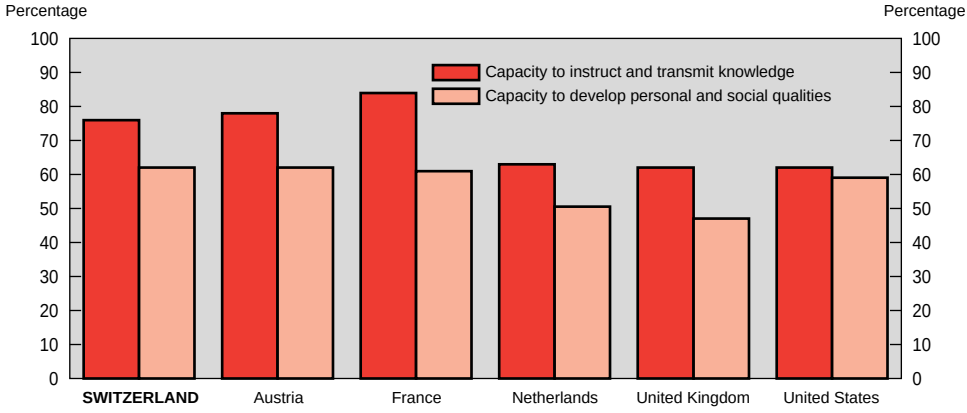
Responsiveness of outputs to client demands

The Swiss education system is highly decentralised. At the compulsory level, there are many bodies, with elected local officials playing an active role in managing them. The proximity of these managers to the communities which they serve facilitates feedback and adaptation of the system to clients' preferences.[101] Moreover, the fact that these schools are locally financed serves to heighten the community's interest, as citizens can see and influence how their taxes are being spent. On the other hand, strict zoning and the virtual absence of public subsidies for private schools[102] limit competition and hence the pressure to adapt to client demands: parents unhappy about the education their child is receiving in the local school can only switch schools at great expense (by moving or sending their

child to a private school). Nevertheless, the high degree of confidence which the Swiss have in their compulsory school system suggests that it is indeed responsive to community preferences (Figure 21).[103] Moreover, the low and stable proportion of students in private schools (4 per cent) is also indicative of satisfaction with the services being supplied by public schools: there are other countries with equally unfavourable financial conditions for attending private schools but with a higher proportion of students nevertheless attending them. Indeed, even the lower priority accorded to natural science (see above) in Swiss curricula seems to reflect public preferences, as does the increased emphasis on modern languages in the new *maturité* curriculum being put in place (Table 25).

Responsiveness to employer requirements is one of the major strengths of apprenticeship training. Employer associations largely determine the content of apprenticeship programmes, although they do consult with cantonal authorities and labour unions. Once agreement has been reached, regulations are made fixing the content of programmes. Although they are reviewed regularly, progress in updating them can be slow. For example, although the authorities consider that

Figure 21. **PUBLIC CONFIDENCE IN SCHOOLS**[1]
1993 Survey



1. Of the population who consider that the teaching of at least one subject or one quality is essential or very important, the percentage who are very confident or rather confident in the ability of schools to carry out these tasks well.

Source: Office fédéral de la statistique (1995*b*).

Table 25. **Public views on the relative importance of school subjects**[1]

Rank	**Switzerland**	Germany	France	Netherlands	United Kingdom	United States
1	Nat. lang. (84)	Nat. lang. (92)	Nat. lang. (97)	Nat. lang. (90)	Math. (93)	Math. (96)
2	Math. (82)	Math. (92)	Math. (88)	For. lang. (85)	Nat. lang. (88)	Nat. lang. (92)
3	For. lang. (77)	For. lang. (91)	For. lang. (87)	Comp. (75)	Comp. (72)	Comp. (86)
4	Comp. (71)	Comp. (79)	Comp. (69)	Math. (69)	Nat. Sc. (66)	Nat. Sc. (85)
5	Phy. Ed. (67)	Phy. Ed. (73)	Hist./Geo. (69)	Nat. Sc. (64)	For. lang. (56)	Hist./Geo. (80)
6	Civic Ed. (65)	Hist./Geo. (72)	Civic Ed. (67)	Hist./Geo. (47)	Hist./Geo. (50)	Civic Ed. (77)
7	Nat. Sc. (63)	Nat. Sc. (67)	Nat. Sc. (63)	Civic Ed. (41)	Phy. Ed. (41)	Phy. Ed. (62)
8	Hist./Geo. (62)	Civic Ed. (64)	Phy. Ed. (50)	Phy. Ed. (41)	Civic Ed. (36)	For. lang. (53)
9	Art Ed. (58)	Art Ed. (43)	Art Ed. (31)	Art Ed. (31)	Art Ed. (26)	Art Ed. (47)

Abbreviations: Nat. lang. = National language(s);
Math. = Mathematics;
For. lang. = Foreign languages;
Comp. = Computing;
Phy. Ed. = Physical education;
Civic Ed. = Civic education;
Nat. Sc. = Natural Science;
Hist./Geo. = History/Geography;
Art Ed. = Art education.

1. The numbers in brackets are the percentage of persons who consider the subject essential or very important.

Source: Office fédéral de la statistique (1995*b*).

there is too much specialisation in apprenticeships and too many of them, little progress has been made in increasing the weight of general education in all apprenticeships[104] and in reducing their number. Similarly, the introduction of new apprenticeships in the service sector has not kept pace with the growing importance of this sector.[105] The authorities' recognition of the importance of employer involvement in vocational education can also be seen in arrangements for the establishment of the HES. In addition to consulting with professional associations, a mixture of part-time and full-time teachers is being sought so as to enhance the flow of information from enterprises on course content.

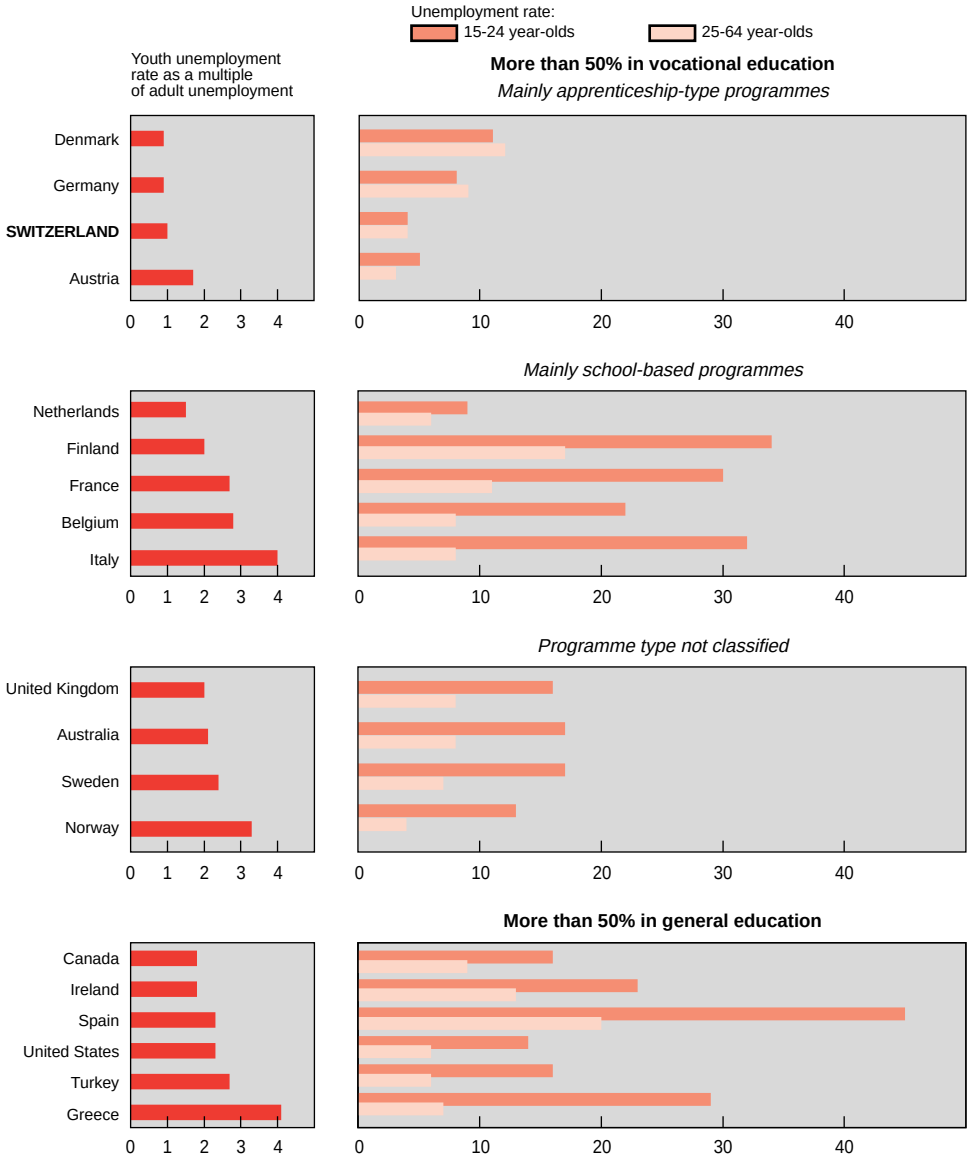
Universities also consult with professional associations on course content in vocational fields such as medicine and law. Nevertheless, reforms are planned which would significantly enhance responsiveness to clients. There is to be a shift in university management towards defining outputs and giving universities greater autonomy in deciding how to produce them within a budget constraint (New Public Management (OECD, 1995*a*). As part of this process, universities will be required to conduct regular surveys on client satisfaction. While these moves should increase responsiveness to client demands, they are unlikely to be as effective as would be competition from private universities; the latter are insignificant[106] in Switzerland owing to the high subsidies for public universities (see below) but the absence of subsidies for private universities.

The transition from education to work

The smoothness of the transition from education to work depends on many institutional features (especially labour-market regulation), including the interface between the education system and the labour market. In Switzerland, as in other (predominantly) German-speaking countries, the transition is gradual within a highly structured framework. Apprenticeship programmes give young people a chance to learn directly about the world of work while still being able to pursue more theoretical studies (OECD, 1996*b*). This tends to result in a smooth transition from education to work. The youth (15-24 years old) unemployment rate is the same as the adult (25-64 years old) rate, as in other countries in which apprenticeship-type programmes dominate at the upper secondary level (Figure 22).[107] By contrast, in countries where the apprenticeship system is not

Figure 22. **RELATIVE YOUTH UNEMPLOYMENT BY TYPE OF SCHOOL SYSTEM, 1994**[1]

Proportion of upper secondary level enrolments



1. Apprentices are considered to be part of the labour force.

Source: OECD (1996*b*).

Table 26. **The employment situation by regions[1] and age groups**

Per cent

	German-speaking cantons				French-speaking cantons			
	30-54		15-29		30-54		15-29	
	Rate of employment	Rate of unemployment	Rate of employment	Rate of unemployment	Rate of employment	Rate of unemployment	Rate of employment	Rate of unemployment
1992	85.9	1.9	63.0	3.0	81.6	3.5	56.0	7.9
1993	84.3	2.8	62.0	4.0	81.4	4.2	52.6	9.8
1994	84.3	3.0	60.3	3.8	80.4	4.7	52.6	10.4
1995	84.7	2.3	60.1	3.4	82.8	4.3	51.3	9.3
1996	85.6	2.4	59.4	4.0	81.6	4.8	52.8	11.2

1. The cantons of Fribourg (German and French-speaking) and Ticino (Italian-speaking) are excluded.

Source: Wolter *et al.* (1997).

dominant, youth unemployment rates are two to four times higher than adult rates.

Young people encounter greater difficulties in finding a job in the latin parts of Switzerland than in the German-speaking regions.[108] Youth unemployment rates are 2.3 times the adult rate in French-speaking cantons, compared with only 1.7 times adult rates in German-speaking regions[109] (Table 26). These ratios have remained fairly stable over the course of the recession, suggesting that the differences among linguistic regions are structural. In part, this difference reflects the greater role of the dual system in the German-speaking parts of the country. Nevertheless, this does not explain all of the difference.[110] Even for graduates of the dual system, unemployment is much higher in latin regions. For example, in 1995, one year after graduation from a non-university tertiary institution,[111] the unemployment rate was more than twice the adult rate in latin regions compared with less than 50 per cent higher in the rest of the country.[112] For university graduates, this ratio was also higher in latin regions than in the rest of the country, although the difference was less marked.[113] An important factor explaining these differences is that there is greater interaction between young people and the labour market in the German-speaking part of the country in both the dual- and general education systems.[114] This may take the form of part-time work, full-time work before commencing advanced studies,[115] the practical training of students or non-university training for graduates of teacher training schools. In addition, German-speaking university graduates demonstrate more flexibility in the kinds of jobs they are willing to accept to facilitate their entry into the labour market and in their initial wage demands.

Labour-market outcomes by level of educational attainment

Earnings premia and rates of return to education

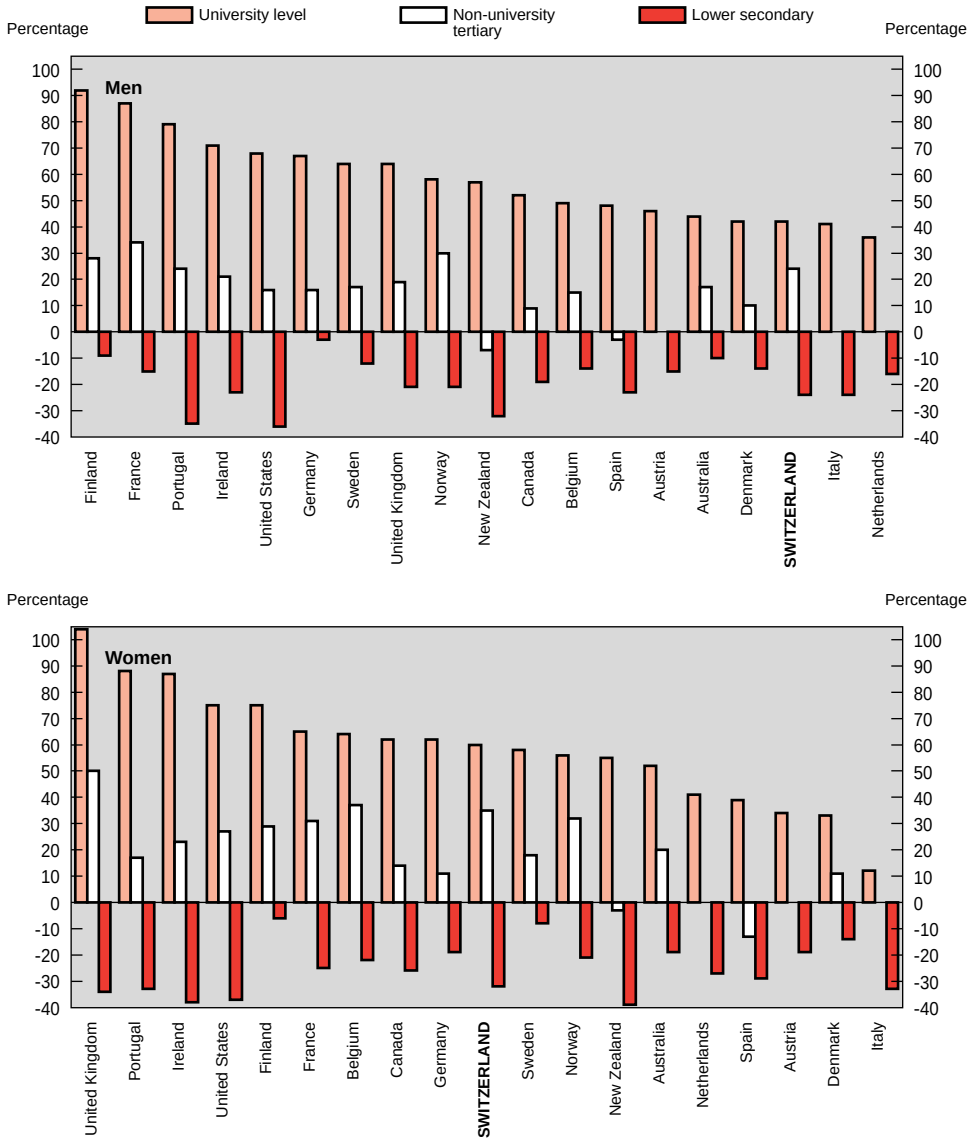
An important motivation for individuals to invest in education and training is that the acquired knowledge and skills tend to raise their productivity and hence, in a competitive market, their earnings potential.[116] How steeply wage rates rise with educational attainment depends on a variety of factors including the contribution which education and training make to raising productivity, the efficiency with which enterprises are able to use skilled labour, the relative scarcity of the skills in

question and the extent to which markets are competitive. Thus, earnings premia by level of educational attainment (relative to some basic level) may be a crude indicator of the effectiveness of the education and training system in raising productivity, although they undoubtedly reflect other factors as well.

Swiss workers who have completed upper secondary education earn a significant premium (24 per cent) over workers without any post-compulsory qualifications (Figure 23). This is clearly above the OECD average (19 per cent) and the fourth highest differential among the countries listed. The relatively low earnings for less-educated workers may be related to the absence of minimum-wage legislation and the abundant supply of workers at lower qualification levels as a result of the availability of foreign labour, especially seasonal workers. At the other end of the spectrum, the premium earned by university graduates over earnings for individuals with only upper-secondary qualifications is the third lowest in the OECD for males but about average for females. By contrast, the earnings premium for non-university tertiary graduates is one of the highest in the OECD, while the premium for university qualifications over non-university tertiary qualifications is one of the lowest in the OECD. The main factor underlying this pattern of earnings premia is undoubtedly that the border between university and non-university tertiary education in Switzerland differs from that in other countries: many of the professional programmes taught in Advanced Colleges in Switzerland lead to high-earning professions and would be taught in universities in other countries. Finally, female earnings in Switzerland rise more strongly with attainment than male earnings, increasing from 47 per cent of male earnings for persons with no post-compulsory qualifications to 60 per cent for university graduates. In contrast, the OECD averages of gender-based earnings differentials are independent of the level of educational attainment: female earnings average 63 per cent of male earnings. The relatively low earnings of females relative to males in Switzerland for persons without tertiary qualifications may reflect the greater supply of females in these categories (Switzerland has a large gender gap in attainment, especially below university level), a high incidence of part-time work amongst low attainment females and the more restricted choice of professions for females at this level of attainment.

A pilot survey conducted in 1991 on employers based in the canton of Geneva provides further evidence on the earnings profile of the work force as a function of the qualification level (Revaz and Rietschin, 1992). Its results confirm

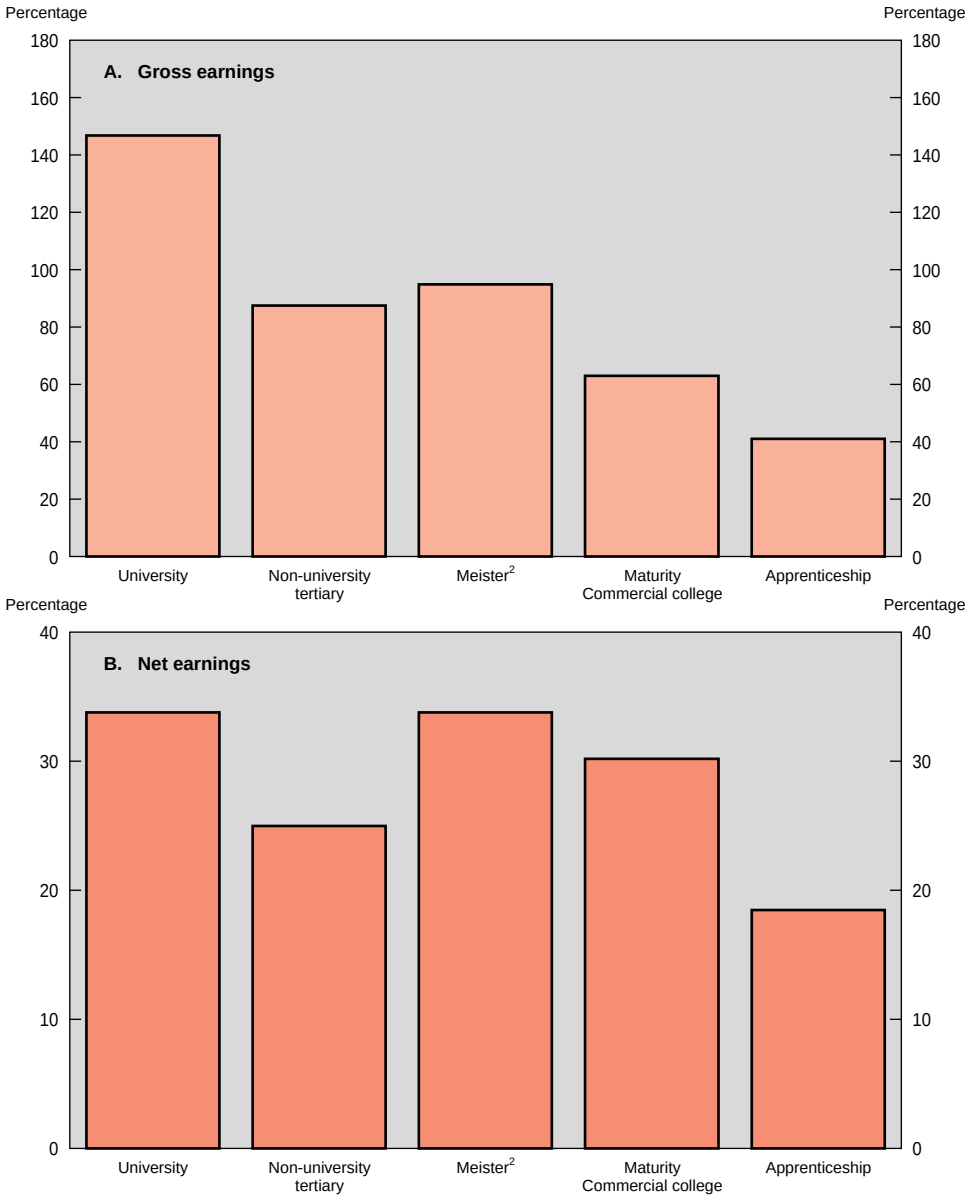
Figure 23. **MEAN ANNUAL EARNINGS BY LEVEL OF EDUCATIONAL ATTAINMENT AND GENDER**[1, 2]



1. Per cent deviation from mean annual earnings at the upper secondary level.
2. Persons 25 to 64 years of age. For Austria, 1991. For Belgium, 1992. For Australia, Denmark, Finland, Ireland, Italy, Netherlands, Norway, Portugal, Spain and Sweden, 1993. For other countries, 1994.

Source: OECD (1996*a*).

Figure 24. **EDUCATION-RELATED EARNINGS PREMIA, 1992**[1]



1. Compared with income earned by workers with no qualification beyond compulsory education.
2. Apprentices master.

Source: Wolter (1994).

that (gross) earnings rise with the level of qualification attained (Figure 24, Panel A). University graduates earn 2½ times as much as unqualified workers and 1½ times as much as upper-secondary graduates. In general, every additional year of investment in human capital beyond compulsory education increases annual earnings by 7.2 per cent.[117] A recent study based on these data reports a much less differentiated wage distribution once various sources of bias are eliminated (Wolter, 1994). By correcting income for bias due to differences in life-time earnings profiles, monetary- and non-monetary costs, risks of failure, tax rates (due to the progressive nature of tax schedules) and the age-difference for labour market entry, the (net) earnings differential between persons with university attainment and unqualified workers falls to 34 per cent (Figure 24, Panel B).

A recent study shows that overall rates of return, which adjust private returns for taxes and allow for public expenditures on education, are higher for upper secondary education than tertiary education in OECD countries (Table 27).[118] Women tend to have higher rates of return than men, particularly at

Table 27. **Overall rates of return to education**[1]

Per cent

	Men			Women		
	International Standard Classification of Education			International Standard Classification of Education		
	3[2]	5[3]	6/7[4]	3[2]	5[3]	6/7[4]
Switzerland	**12**	**9**	**7**	**14**	**9**	**7**
Australia	11	7	11	15	8	11
Canada	10	7	12	11	8	12
Denmark	11	4	10	12	4	8
Finland	10	10	14	8	12	14
France	9	19	18	11	18	15
Germany	2	11	12	7	10	9
Ireland	20	17	16	29	11	22
Italy	23	n.a.	12	54	n.a.	6
Netherlands	15	n.a.	10	21	n.a.	9
New Zealand	34	n.a.	9.8	32	n.a.	10
Sweden	8	7	10	7	6	8
United States	21	9	13	18	12	13
Average	14	10	12	18	10	11

1. Private rates of return adjusted for taxes and public expenditures on education.
2. Upper secondary level.
3. Non-university tertiary level.
4. University level.

Source: OECD (1997).

the upper secondary level. While these features also hold for Switzerland, rates of return are nevertheless lower than the OECD average regardless of educational level and gender, sometimes by a considerable margin. This might be related to the relatively high level of educational attainment of the Swiss population as compared to other OECD countries. Another interesting finding is that while on average university-level education results in a higher rate of return than non-university tertiary education in OECD countries, the opposite is true for Switzerland. This is a further indication of high relative earnings ensuing from non-university tertiary education and can be explained by the fact that advanced technical and commercial colleges in Switzerland provide skills and qualifications equivalent to university-level tertiary education in many other countries.

Labour-force participation and unemployment[119]

Although labour force participation rates vary considerably across countries, there is a clear relationship between the level of educational attainment and labour force participation in OECD countries. This general pattern also holds for Switzerland. In 1996, labour force participation rates were 73 per cent for persons having only completed compulsory and lower levels of education, 83 per cent for the upper-secondary level, 94 per cent for the non-university tertiary level and 90 per cent for the university level, giving an overall average participation rate of 83 per cent.[120] Swiss labour force participation rates are above the OECD average, the difference being particularly large for non-university tertiary and below upper secondary levels. In contrast to most other OECD countries, the participation rate is higher for the non-university tertiary level than for the university level. This might again reflect that the border between these two levels of education differs in Switzerland from that in other countries. It also reflects the fact that the graduates of the non-university tertiary level are predominantly male.

Unemployment rates in OECD countries tend to decrease as a function of the level of educational attainment. This relationship also applies broadly to Switzerland. In 1996, when the overall jobless rate stood at 3.6 per cent (Household Survey basis), unemployment rates were 6.5 per cent for those with attainment at the compulsory or lower levels, 3.1 per cent for the upper secondary level, 1.4 per cent for the non-university tertiary level[121] and 4.6 per cent for the university level. The relatively high unemployment rate for university graduates is in striking contrast to the situation in most other OECD countries, where this

group tends to have the lowest unemployment rate. Again, this probably reflects international differences in the criteria for classifying tertiary institutions as non-university or university: taken as a whole, tertiary graduates have the lowest unemployment rate in Switzerland, as in other countries. Finally, as in other countries, unemployment of persons with low educational attainment has risen in the 1990s relative to that for the overall labour force.

Costs of education and training[122]

Education services in Switzerland are mostly paid for by the public sector; the exact magnitude of private provision and financing is unknown but is believed to be quite small, except for the implicit cost to firms participating in the dual system. In 1994, education accounted for 18.7 per cent of total public expenditure. Although this proportion has declined slightly since the beginning of the 1990s, it remains the second highest after that for social security.[123] Compulsory schooling accounts for almost half of public expenditure on education and is paid for by communes and cantons (Table 28). Universities account for the next largest part (17 per cent) and are paid for almost equally by the Confederation and the cantons. Almost as much again is spent on vocational education, but this is mainly paid for by cantons. In all, the cantons account for a little over half of public expenditure on education, the communes for one-third and the Confederation for the remainder.

Swiss public expenditure on education was 5.6 per cent of GDP in 1993, a slightly larger share than the OECD average (5.4 per cent) (Figure 25). Switzerland's expenditure share exceeded the OECD average at both the combined primary and secondary levels and the tertiary level. The ratio of public educational expenditure to GDP soared in the early 1970s from 3.9 per cent in 1970 to 5.3 per cent in 1975, then remained relatively stable until the late 1980s, and since then has risen again. This trend increase contrasts with the evolution observed in most other OECD countries, where public expenditure on education has fallen back to a similar (or lower) proportion of GDP to that observed in the early 1970s. Public expenditure on education in Switzerland is particularly high when allowance is made for the country's demographic structure and full-time equivalent enrolment rate. If the proportion of the Swiss population aged 5 to 29 were as high as the OECD average and per capita service provision remained

Table 28. **Public expenditure on education, 1994**

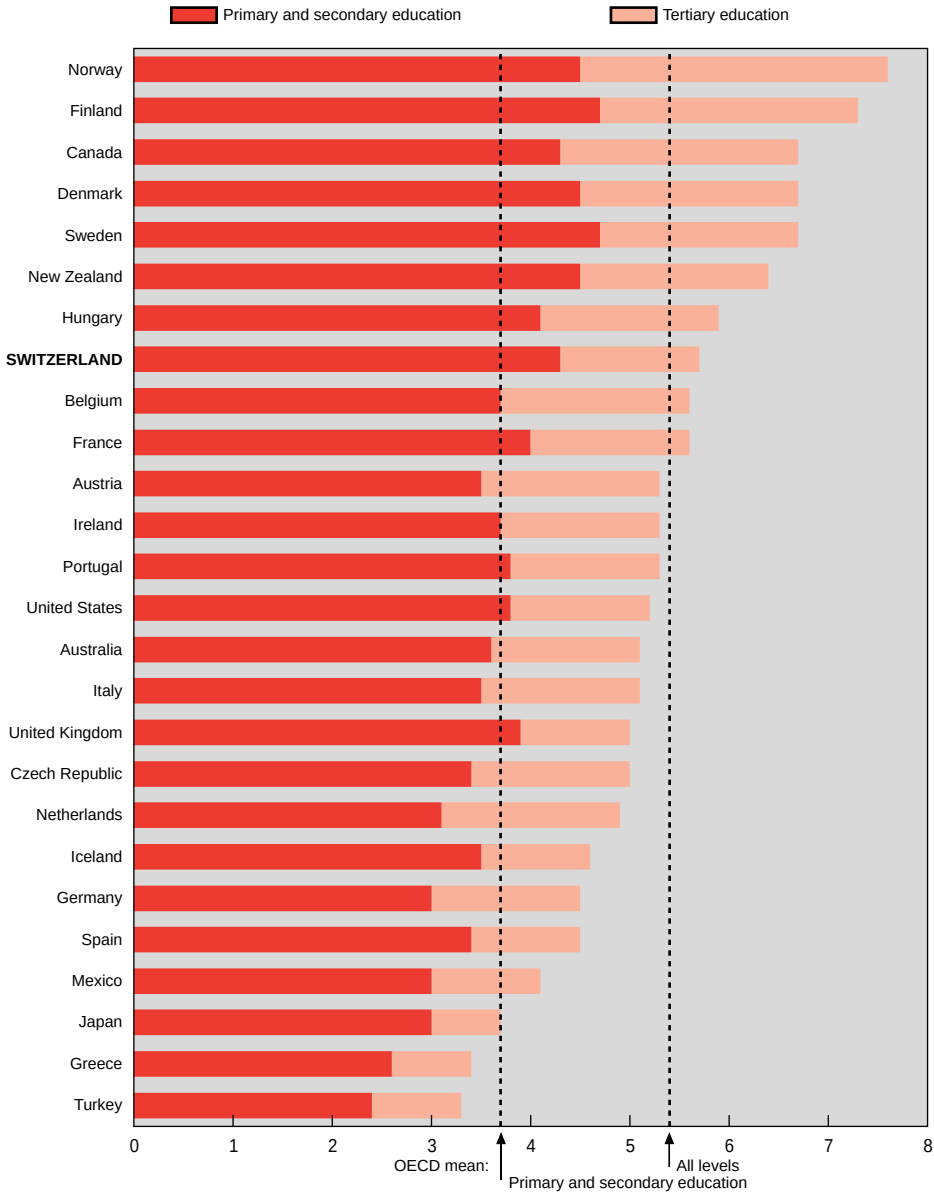
SF million

Type of education	Total		Level of government		
	Absolute amount	Percentage	Confederation	Cantons	Communes
Kindergarten	691.1	3.4	–	235.5	455.6
Primary and lower-secondary schools	9 417.1	46.6	21.5	3 812.2	5 583.4
Specialised schools	737.4	3.6	–	404.3	333.1
Professional training	2 976.5	14.7	547.6	1 914.6	514.3
Teacher training	403.5	2.0	–	390.5	13.0
Upper-secondary schools	1 557.0	7.7	11.8	1 489.5	55.7
Advanced professional training	544.4	2.7	106.1	420.3	18.0
Universities including FITs	3 513.9	17.4	1 773.7	1 738.6	1.6
Other	373.2	1.8	24.2	328.5	20.5
Total	20 214.1	100.0	2 484.9	10 734.0	6 995.2
Personnel costs	13 511.8	66.8	–	–	–
of which:					
Teaching staff	9 331.3	46.2	–	–	–
Administrative staff	2 184.4	10.8	–	–	–
Social security contributions	1 928.4	9.5	–	–	–

Source: Office fédéral de la statistique (1994).

Figure 25. **PUBLIC EXPENDITURE ON EDUCATION, 1993**[1]

As a percentage of GDP



1. Includes direct expenditure (*i.e.*, paid to educational institutions) and indirect expenditure (*i.e.*, education subsidies to households, excluding subsidies for living costs).

Source: OECD (1996*a*).

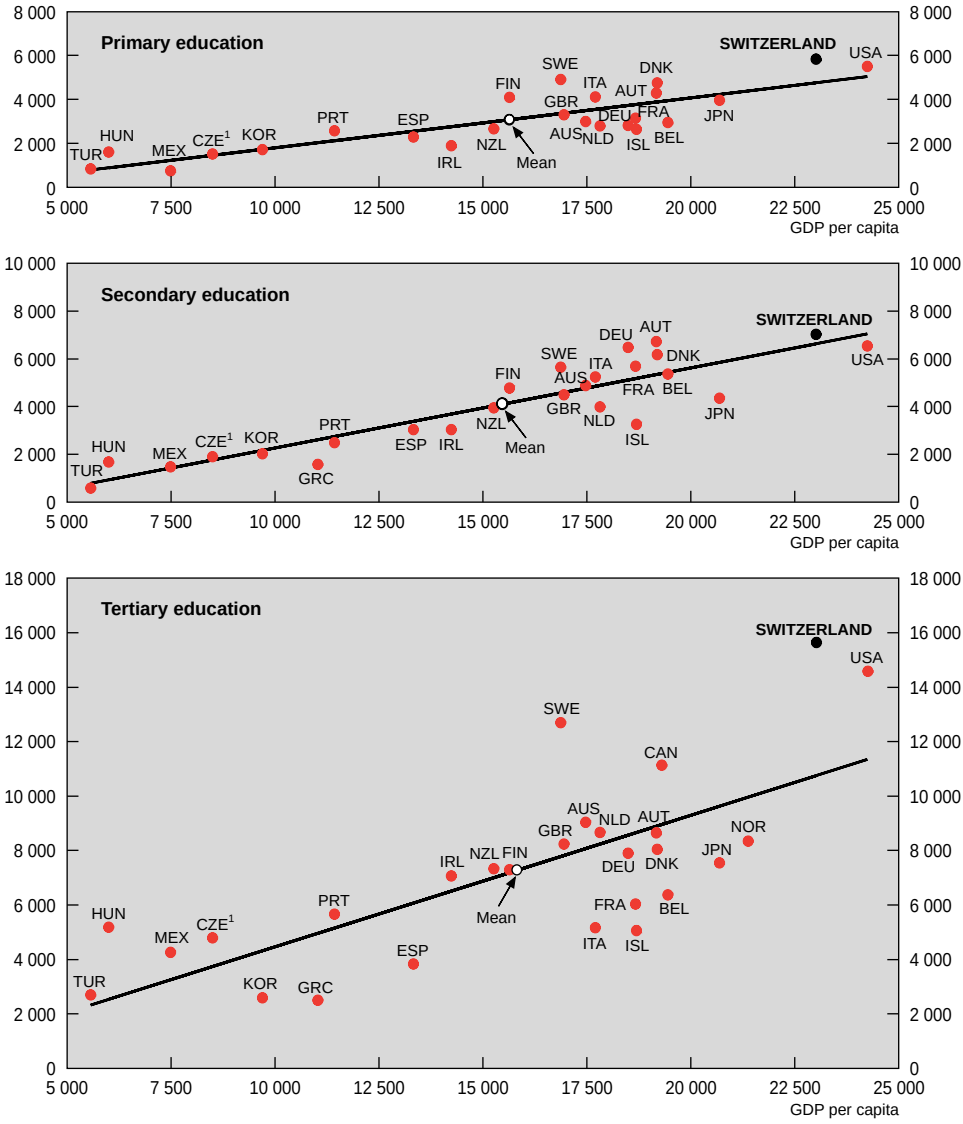
the same, the ratio of expenditure on education to GDP would be 0.7 percentage point higher (OECD, 1996*a*, Chart F1.2B). Similarly, a full-time equivalent enrolment rate as high as the OECD average would result in the expenditure ratio being 0.4 percentage point higher.[124]

Expenditure per student in Switzerland was US$7 011 (converted using Purchasing Power Parities (PPPs)) in 1993, far above the OECD average (US$4 293) and exceeded only in the United States. As the Swiss figure, unlike that for the United States, does not include private expenditure, total expenditure per student in Switzerland was in fact probably the highest in the OECD. As in other countries, expenditure per student rises sharply with the level of education, ranging from US$2 335 at the pre-primary level to US$15 731 at the tertiary level.[125] Expenditure per student was the highest in the OECD at every level of education except the pre-primary level, where it was well below the OECD average; the differential was particularly large for tertiary (especially university-level) education. To some extent, the high expenditure per student in Switzerland is related to its high level of GDP per capita (Figure 26). Even so, expenditure per student remains above what would be predicted on the basis of Switzerland's GDP per capita, especially for tertiary education. Indeed, tertiary education is even more costly in relative terms than indicated by these data as completion time in Switzerland is relatively long. Cumulative expenditure per student over the entire duration of tertiary education averages US$64 500 in Switzerland, far more than in any other OECD country. A factor contributing to the high cost of tertiary education is that Swiss educational institutions tend to be relatively small, limiting their ability to realise scale economies.

An unusually large proportion of education expenditure in Switzerland is on capital items. For example, at the combined primary and secondary level, capital expenditure accounts for 11.7 per cent of total outlays on educational institutions, far above the OECD average (7.6 per cent); as in most other countries, this share is higher still for tertiary education. The relatively large share of capital expenditure might be related to the fact that Swiss educational institutions enjoy ample space in a country where space is scarce and real estate prices are among the highest in the world.[126] Also, construction costs are relatively high in Switzerland, partly due to strict regulations and norms.[127] Safety regulations tend to be particularly severe for school buildings.

Figure 26. **EDUCATIONAL EXPENDITURE PER STUDENT IN RELATION TO GDP PER CAPITA**

In PPPs (US$), 1993



1. Expenditure in public and private institutions. For Mexico, Austria, Germany, Greece, Italy, Portugal, Czech Republic, Norway and Switzerland: public institutions only. For Belgium, Finland and United Kingdom: public and government-dependent private institutions.

Source: OECD (1996*a*).

Compensation of staff is also relatively high, accounting for 79 per cent in total current expenditure compared with an OECD average of 69 per cent. This mainly reflects relatively high teaching salaries in Switzerland. For example, the starting annual salary of a primary-level teacher in 1994 was US$30 571 (converted using PPPs) (1.3 times GDP per capita), compared with an OECD average of US$18 702 (1.1 times GDP per capita). At the lower secondary level, the differential was even greater (US$36 095 (1.5 times GDP per capita) compared with US$19 685 (1.1 times GDP per capita)).[128] These salaries are particularly high, given that Switzerland is one of the richest countries in the OECD: the ratio of teacher salaries to GDP per capita tends to be inversely related to GDP per capita because teachers represent a smaller proportion of the pool of well-educated workers in a rich country than in a poor country. At the tertiary level, a further possible explanation for the large cost differential is that teaching and research assistants in universities have full-employee status, in contrast to the situation in countries such as the United States where assistantships look more like scholarships to graduate students and therefore cost less in salaries and social-security contributions. The relatively low ratio of students to teaching staff at all levels of education except the pre-primary level is also a factor explaining high expenditure per student in Switzerland.

Tertiary education: financing and efficiency

Tertiary education in Switzerland is mainly funded by the state. At the non-university level, almost half of students attend public institutions, 30 per cent attend subsidised private institutions and the rest attend private institutions. Tuition fees are low in public institutions and the subsidies for private institutions are large in relation to tuition fees: hence, the public sector pays for most of the tuition costs in this sector. For universities, the public share of tuition costs is considerably higher: almost all students attend public universities and the fees charged are very low[129] (1.3 per cent of total university revenues).

Public funding of education can be rationalised on efficiency grounds insofar as education yields external benefits to society, such as greater ease of communication in a literate society and a shared sense of community; these benefits are particularly important in a multi-lingual country. As individuals do not take these benefits into account, they would tend to under-invest in education

in the absence of public subsidies. It is generally considered that the external benefits of education are greatest for primary education and lowest for tertiary education (Psacharopoulas, 1995). Hence, on efficiency grounds, the state's share of total expenditure on education should be highest for compulsory schooling and lowest for tertiary education. The state does indeed bear a higher proportion of expenditure on compulsory schooling than on non-university tertiary education but this is not so for university education. Moreover, it is not obvious why, in the tertiary sector, the state should be so much more generous with universities than with non-universities: it is implausible that the ratio of social- to private benefits of tertiary education should be so much higher for university education. While it is probable that efficient pricing of university education would entail students paying a larger share of the costs, it is not possible to indicate what that share should be as there is no agreement on the magnitude of external benefits. However, it is difficult to believe that the breakdown of benefits is anything like what is implied by current funding arrangements: 99 per cent external, 1 per cent private.

Imperfect capital markets provide another justification for state involvement in the financing of tertiary education. Under these conditions, individuals tend to under-invest in education as they are liquidity constrained and are unable to diversify adequately the risk of the investment.[130] Moreover, the under investment appears to be greatest among students from modest backgrounds. The traditional response to these concerns in many countries, including Switzerland, has been for the state to assume a high proportion of the costs of tertiary education. But this weakens the incentive for students to take into account the costs of their tuition when making decisions about which courses to take and how fast to complete their studies. It may also have the perverse effect of re-distributing wealth to individuals with higher lifetime earnings than the average taxpayer. Another solution has been for the state to underwrite student loans to pay tuition fees. But default rates on these loans have been high and risk-averse individuals are still discouraged from investing in tertiary education. A solution which gets around these problems is for the state to make repayments conditional on income exceeding a specified amount and to collect them through the income tax system. Australia introduced such a system in 1989 and a number of countries have since implemented similar schemes.

Even so, many students (especially those from poorer backgrounds) will still under-invest in education if they do not have access to finance to cover their living costs. Cantons provide grants for this purpose to students of limited means, but only 12.7 per cent of post-compulsory students (14.5 per cent for university and 18.0 per cent for non-university tertiary students) qualified in 1993.[131] Moreover, these grants cover only a small proportion of living costs.[132] In all, cantons' expenditure on grants in 1993 (SF 321 million) amounted to only 2.4 per cent of their total expenditure on education.[133] Cantons also provide student loans, but on a much smaller scale (SF 36 million in 1993). Consideration has recently been given to replacing grants by loans in the cantons of Zurich, City of Basle, Fribourg and Ticino as a result of worsening cantonal finances. In the event, the idea has been rejected so far on the grounds that administrative costs (including those related to default risk) would be too high to justify such a shift. For poorer cantons, which receive a substantial federal subsidy under the present grant system, such a shift is even less worthwhile (Scheuber, 1996). However, a loan system with income-contingent repayments collected through the income tax system would alleviate these problems and improve access to tertiary education for students from modest backgrounds (without transferring wealth to individuals with higher lifetime earnings than the average taxpayer).

Financing universities is becoming an increasing strain on cantons. One response of the cantons has been to make reforms aimed at enhancing university efficiency; another has been to re-negotiate the sharing of costs between university- and non-university cantons (Box 2). With a view to increasing efficiency, cantonal university laws recently have been amended to strengthen the autonomy and management structure of the universities. In future, academic institutions will receive global budgetary packages and will be free to allocate funds among different units according to self-determined priorities. Also, the universities will be run by stronger central governing bodies (*Rectorats*) capable of taking rapid decisions and implementing necessary measures in a constantly evolving scientific and economic environment. In exchange, universities are urged to specialise further by forming a limited number of relatively homogeneous units of education and research (*unités d'enseignement et de recherche*) in co-ordination with other universities. The development of these units will be given priority in the course of the universities' future growth. Finally, the universities' teaching, research and administrative staff will be subject to regular evaluation.[134] For the

Box 2. **University financing arrangements**

Cantons presently finance 60 per cent of total public expenditure on universities (including the two Federal Institutes of Technology) excluding research grants or 40 per cent including them, with the Confederation financing the remainder.[1] But the Confederation's contribution has not kept pace (basic grants actually fell in nominal terms between 1994 and 1995) with growing student numbers: the real amount of basic grants per diploma delivered has been declining steadily since the early 1980s, falling to 75 per cent of its 1980 level in 1995 (Table 29). In response to the growing financial burden of hosting a university, an inter-cantonal agreement[2] on financial contributions to cantonal universities was negotiated providing for payments from non-university cantons in proportion to the number of students they send to cantonal universities. The amount of the contribution increased progressively from SF 3 000 per student in 1981 to SF 8 500 in 1993. Since 1993, this amount has been indexed to inflation, reaching SF 8 913 in 1996, for a total transfer of SF 228 million. In exchange, university cantons guarantee students from other cantons the same rights as their own students.

Table 29. **Basic subsidies to cantonal universities**

	Subsidies (in SF million)	Number of degrees delivered	Basic subsidies per degree (in SF)
1980	189.2	4 258	44 431
1981	189.0	4 415	40 188
1982	193.1	4 735	36 240
1983	206.1	4 774	37 246
1984	231.6	4 987	38 941
1985	243.2	5 025	39 227
1986	256.1	5 148	40 018
1987	268.0	5 459	38 940
1988	280.4	5 599	38 983
1989	294.0	5 900	37 625
1990	307.3	6 101	36 084
1991	321.2	6 117	35 513
1992	350.1	6 348	35 879
1993	362.1	6 613	34 474
1994	370.5	6 598	35 037
1995	368.9	6 811	33 202

Source: Conférence universitaire suisse.

The per capita compensation provided for by the present agreement does not cover costs, irrespective of the field of study. Estimates for 1993 show that annual costs vary

(continued on next page)

(continued)

from SF 21 000 in human and social sciences to SF 156 000 in medical sciences, with an average of SF 45 000 per student. At about SF 9 000, the present lump-sum amount of the contribution therefore stands at a fifth of the true average cost per student. This agreement has been re-negotiated recently for the period after 1998. Under the new agreement, a differentiation has been introduced among 3 groups of subjects and the average contribution has been raised by about 50 per cent.[3] The non-university cantons have refused to increase their contributions further towards full compensation of costs on the grounds that the host cantons benefit from having a university on their territory (*Conférence universitaire suisse*, 1995). In particular, they gain highly qualified labour to the detriment of other cantons as a result of attracting students who tend to stay and work in the host cantons after finishing their studies (''brain drain''): with the exception of St. Gall, all host cantons receive a higher proportion of university graduates than their share of students entering university (Schneiter and Zarin-Nejadan, 1994). In return for an increase in their contributions, the non-university cantons have obtained from host universities some degree of participation in decision making. The issue of university financing has yet to be taken up in the framework of the revision of the law on fiscal equalisation.

1. According to the federal law of 1991 (replacing the original law of 1968 which came into effect in 1969), the federal subsidies consist of basic grants for operating costs, investment grants for buildings and equipment, and extraordinary grants to promote special projects of general interest for Swiss higher education for a limited time. The extraordinary grants have been effective in recent years in upgrading the computer equipment of universities, developing continuing education programmes, promoting student mobility and creating posts for junior university staff. In the context of the revision of the federal law on fiscal equalisation, it has been stated that the Confederation should in future concentrate its efforts on financing research and leave the financing of education to the cantons. This may not be compatible with the growing co-ordination role of the Confederation in higher education, especially in the context of European integration.
2. This agreement was first signed in 1981 for the period 1981-86. It has since been renewed twice for the periods 1987-92 and 1993-98.
3. The per student contributions have been fixed at SF 9 500 for human and social sciences, SF 23 000 for exact and natural sciences and SF 46 000 for medicine. For the two latter groups, the contributions will be phased in progressively from 1999 until 2002.

teaching staff, external referees will be used. Some of these measures have already come into effect. In some cases, the strengthening of the central power in the universities has given rise to stiff resistance by the lower bodies which have resented the restriction of their prerogatives. Staff evaluation procedures have so far remained quite soft without any serious consequence in case of a negative

result; Swiss universities have yet to introduce regular and transparent teacher evaluation by students.

The federal law on universities is also being revised to enhance efficiency. The new law aims to promote a more labour-market-oriented output and greater productive efficiency as well as to adapt to the increased autonomy granted by the cantons to their universities in the framework of new cantonal legislation. The new law should provide the central government with a better instrument for steering the national education and research policy in the years ahead. Federal subsidies will be used to finance services (*prestations*) rather than expenditure. For instance, by rendering basic subsidies proportional to the number of diplomas delivered instead of the number of enrolments, the new law seeks to encourage the cantonal universities to take steps to reduce the relatively high drop-out rates, especially during the early years of university studies. An obvious risk with this approach, however, is that universities may seek to achieve the objective by lowering standards.

Efficiency could also be raised through better co-ordination of university activities. Swiss universities are small, lacking critical mass in many areas. This problem also has been recognised at the non-university tertiary level, with the result that the federal law on the creation of the HES obliged colleges wishing to be upgraded first to amalgamate to form more efficient units. At the university level, the revision of the federal law on the financial assistance to universities is also expected to provide incentives for inter-cantonal co-ordination. So far, the scope of Swiss-wide co-ordination efforts has been limited to the mutual recognition of studies in order to promote student mobility. At the regional level, however, co-ordination has gone somewhat further, especially in the western part of Switzerland where it is most needed given the higher-than-average university density.[135] The most salient feature of this co-ordination has been the so-called BENEFRI framework convention signed in 1993 by the universities of Bern, Neuchâtel and Fribourg which allows students to attend courses and sit examinations in all three universities. It even requires professors to teach in different universities. In some cases, regional co-operation has crossed borders by including tertiary institutions in neighbouring countries.[136] A critical factor in improving co-ordination and rationalisation in many cases has been pressure from federal authorities, which has had to overcome fierce local resistance.[137] Although the reform of the university system goes in the right direction, it has been carried

out so far in isolation from the reform of non-university higher education leading to the creation of the HES. In many cases, the universities still do not know whether they should maintain certain training programmes or leave them to the HES and if so under which compensation arrangements.

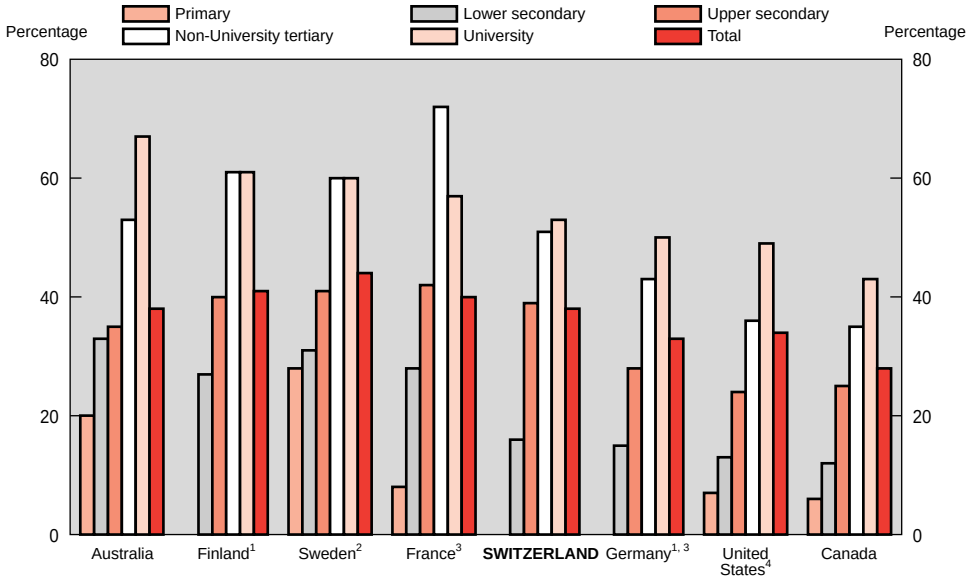
Continuing education

Knowledge and skills depend to a large extent on initial education (*i.e.*, that received before completing the transition from education to work). But they need regular maintenance and upgrading if workers are to remain productive. Continuing education and training gives individuals an opportunity to refresh and to complement previously-received education and training and therefore to maintain their stock of human capital. OECD Education Ministers agreed in January 1996 that ''lifelong learning will be essential for everyone as we move into the 21st century and has to be made accessible to all'' (OECD, 1996*c*). This consensus is motivated in particular by the recognition that the pace of innovation, both in products and in production methods, has accelerated markedly, requiring workers to have greater knowledge than in the past and continually to acquire new skills in order to remain productive. Against this background, job-related as well as more general continuing education and training are recognised as key elements in a lifelong learning strategy.

Job-related continuing education in Switzerland is mainly privately financed[138] and mostly provided by firms parallel to employment: there is no such thing as ''training leave'' in Switzerland. During the past decade, Swiss firms have increased their efforts in developing continuing education and training programmes, although these efforts have generally lagged behind those observed in other countries such as the United States, Germany and France (OECD, 1992). Nevertheless, participation in these programmes is broadly in line with that in other OECD countries (Figure 27). Also, as in other countries, participation rates are positively related to educational attainment: they range from 16 per cent for persons without post-compulsory attainment to 53 per cent for persons with university-level attainment. Once again, and in contrast to most other countries, there is a significant gender gap in participation: 42 per cent for males *versus* 34 per cent for females. On the other hand, unlike in other countries, participation

Figure 27. **PARTICIPATION IN JOB-RELATED CONTINUING EDUCATION AND TRAINING, 1993**

As a percentage of employed population aged 25 to 64



1. The data shown under lower secondary education for Finland and Germany apply both to those having attained primary and lower secondary education.
2. The reference period for Sweden is 6 months.
3. 1994.
4. 1995.

Source: OECD (1996*a*).

rates in general continuing education are higher for women than for men (*Office fédéral de la statistique*, 1995*a*).

Continuing education and training moved up the public policy agenda in Switzerland in the late 1980s. Pressed by complaints from the private sector over the chronic shortage of qualified labour, the Federal Council proposed to Parliament in 1989 a series of special measures, dubbed *continuing education offensive*, to promote job-related as well as more general continuing education. These measures came into effect in 1990 and were initially allocated a total budget of SF 399 million to be spent over six years. This amount was later reduced substantially as a result of fiscal consolidation efforts.[139] The programme included measures to promote vocational and university-level continuing educa-

tion as well as a specific element targeting the development of CIM (Computer Integrated Manufacturing) techniques. The latter was destined to help Swiss firms, especially SMEs which were lagging behind other industrialised countries in the use of computer-based production techniques, to upgrade their production technology and the pertinent qualifications of their work force. Seven regional CIM training centres were created to provide basic as well as continuing training and facilitate the transfer of technology to SMEs. CIM-oriented research projects have also been financed on a selective basis as a function of their innovation content. Although the federal programme appears to have been successful,[140] the degree of public-sector involvement in continuing education remains low in Switzerland.[141] This is likely to limit participation in continuing education to a sub-optimal level owing mainly to capital market imperfections (see above).

Summary and conclusions

Switzerland's education and training system performs well. Attainment up to at least the end of upper secondary education has been high since the 1950s. This success is largely attributable to the dual apprenticeship system, which provides educational pathways to students with a wide range of abilities and interests. Attainment at the tertiary level is around the OECD average, but is much higher at the non-university tertiary level. This reflects the dominance of the dual system: graduates from this system do not acquire a university entrance qualification. Swiss pupils do well in international studies of scholastic achievement, and the dispersion of achievement is relatively low. Adult literacy, however, is only average. This partly reflects the high proportion of immigrants in the Swiss population, most of whom have mother tongues other than German or French. However, the mediocre performance of tertiary graduates in the literacy survey suggests that other factors are important as well. The markedly better performance by international comparison of young adults relative to older adults in the literacy survey, despite a smaller rise in educational attainment than in most other countries, suggests that the Swiss education system has become relatively more effective in imparting literacy skills and/or that the Swiss work and home environment is relatively ineffective at maintaining them.

The Swiss education system is responsive to client requirements, especially in the dual system. Responsiveness in the school system comes from decentral-

isation of decision-making and financing. However, responsiveness is limited by strict zoning and the absence of public subsidies for private schools. In the dual system, employers largely determine course content, ensuring that it is relevant to labour-market requirements. Universities also consult professional bodies on course content. New Public Management reforms planned in the university sector should bring teaching further into line with client requirements. In response to concerns that the dual system may not be well adapted to meet the growing demand for workers who are flexible and/or possess advanced skills, a new pathway *(maturité professionnelle-hautes écoles spécialisées*) has been created which places more emphasis on general education and enables participants to continue their education to a level comparable to a university degree. This reform should also help to stem the flow of able students from the dual system. It is, however, unfortunate that the on-going reforms of university and non-university higher education have been carried out in isolation. To some extent, this is attributable to the federal structure and to direct democratic decision procedures, which favour isolated problem solving. It may also be related to the rather complex administrative structure of educational policy at the federal level leading to a multiplication of decision-making and co-ordination bodies. Some simplification of this structure could enhance the efficiency of the educational system by ensuring a better co-ordination of various reforms.

Another strength of the Swiss system is that the school-to-work transition is relatively smooth: unemployment among young adults (when apprentices are considered to be members of the labour force) is no higher than for the rest of the adult population of working age. This success is largely attributable to the apprenticeship system. As in other countries, earnings and employment prospects improve with educational attainment. But in Switzerland, the improvements appear to be particularly marked where attainment is through the dual system. Insofar as the tertiary sector is concerned, these differences partly reflect the fact that the boundaries between non-university tertiary institutions and universities in Switzerland differ from those in other countries. Rates of return on post-compulsory education in Switzerland are comparable across educational pathways.

The major drawback of the Swiss education and training system is that it is very expensive. This partly reflects factors specific to Switzerland, such as regional decentralisation, the multilingual context and a commitment to high

quality service. But there are also other factors. Much more is spent on capital equipment than in other countries, teachers salaries are high in relation to national income per capita, and the student-to-teacher ratio is low. The high capital expenditure seems to be related to the large amount of space which educational institutions occupy in a country with high real estate prices and to excessively strict building codes and norms. At the tertiary level, the full-time-employee status of teaching- and research assistants and the small size of institutions also contributes to high costs per student. Capital expenditure could be reduced by reviewing building codes and norms to bring them more into line with those in other countries. Educational authorities could reconsider whether teachers' salaries need to be so high to attract able individuals and whether there is scope to lower the teacher-student ratio. Decisions on these matters should be based on research into their effects on student achievement. Costs in the education system could also be reduced by exposing public providers to greater competition from private providers. For this to occur, public funding would need to follow student choices between public and private providers.

Another possible shortcoming concerns the financing of university education, which may be inefficient and inequitable. Almost all the costs of universities are paid for by the state and there is limited assistance with students' living costs. These arrangements will be inefficient if they do not correspond to the breakdown of the benefits of university education into private and external components and insofar as liquidity-constrained and/or risk-averse students may still under-invest in tertiary education owing to the limited assistance with living costs. Insofar as the latter factor is a greater barrier to students from poorer backgrounds, these arrangements are also a source of social inequity. Social equity may be further undermined by the public financing of universities, where this provides large subsidies to individuals with higher lifetime earnings than the average taxpayer. One arrangement which could be more efficient and fairer would be to provide loans for students to contribute towards tuition- and living costs with income-contingent repayments through the tax system.

Developments associated with globalisation and the information-technology revolution have underlined the need for more emphasis to be given to the maintenance and upgrading of skills throughout a person's working life. The pace of innovation, both in products and in production methods, has accelerated markedly requiring workers to have greater knowledge than in the past and

continually to acquire new skills in order to remain competitive. Participation in continuing education in Switzerland remains low, but is comparable to the situation in many other OECD countries. The state contributes little to the financing of continuing education. Reform of the financing of tertiary education could free the resources necessary to provide at least some increased public support for continuing education.

Another area in which there is scope for improvement is gender equality. Although equality has almost been attained on the academic educational pathway, substantial gender differences persist in the dual system. This reflects the dominance of apprenticeships in mainly male occupations. Gender equality in the dual system could be promoted by developing more apprenticeships in fields attractive to females and by encouraging greater social acceptance of female participation in traditionally ‘‘male’’ occupations. Equally, greater efforts may be required to assist immigrants from non-German or French speaking backgrounds to integrate them better into the education system and ultimately, the labour market.

Notes

1. The *Office fédéral de la statistique* (OFS) adjusted the Swiss National Accounts to EU norms in the Spring of 1997. This entailed a change of the price basis of constant price data from 1980 to 1990 and gave rise to a re-estimation of quarterly series of the GDP and its components. Since the full details of the revisions were not available at the time of finalisation (July 1997), the analysis of this Survey is based on the old National Accounts. However, the rate of change of the real GDP in 1996 is the same according to old and new National Accounts (–0.7 per cent).

2. However, the revised National Accounts suggest an increase in real GDP (at 1990 prices) by 0.8 per cent in 1995.

3. Switzerland suffered from a ten-year period of stagnation between 1929 and 1939.

4. Until recently Finland's growth performance had been worse than that of Switzerland, due *inter alia* to a collapse of its export markets in the former Soviet Union, which caused dramatic output losses during the 1991-93 period. But Finland has enjoyed a strong recovery since (OECD, 1997*a*).

5. According to the revised National Accounts, construction investment volumes fell by 6.6 per cent in 1996.

6. These figures are estimates of the Konjunkturforschungsstelle (KOF) of the Eidgenössische Technische Hochschule (ETH) in Zürich. An official breakdown of construction investment data is not available yet.

7. The average share of construction investment in Switzerland's nominal GDP is among the highest in the OECD, second only to Japan. It reached 20 per cent in 1973 before it contracted in the wake of the first oil price shock.

8. The revised OFS statistics report an increase in private consumption volumes by 0.8 per cent in 1996.

9. The new law on sickness insurance (*''Loi fédérale sur l'assurance maladie''*) introduced *inter alia* compulsory sickness insurance and complete freedom of movement between sickness insurance funds, in order to stimulate competition between funds. The new law also abandoned the subsidisation of sickness insurance funds and replaced it by direct transfers to low-income households. As an initial effect, sickness insurance contributions rose by some 25 per cent in 1996 and by another 12 per cent at the beginning of 1997, whereas only three-quarters of the funds allocated by the federal government to assist low-income households in financing their social security contributions were exhausted.

10. However, the revised National Accounts are for flat (0.2 per cent) machinery and equipment investment volumes in 1996. A contraction of the seasonally adjusted machinery and equipment investment (in real terms) is reported for the fourth quarter of 1996 and the first quarter of 1997.

11. Statistics of the *''population active occupée''* for 1996 are still provisional and may be subject to downward revision in the light of the latest results of the annual Labour Force Survey of the *Office fédéral de la statistique* (OFS).

12. These categories are frontier workers, seasonal workers, workers with annual permits and those with permanent work permits.

13. Statistics of hours worked are available only with a long delay.

14. Roughly 30 000 persons were subject to the termination of their unemployment benefits in 1996. About two-thirds of them remained registered as unemployed. The others either found a job, took up training, continued to search for a job without the assistance of the labour office, left the country, or withdrew from the labour force.

15. The OFS calculates an alternative measure of the unemployment rate from the results of the annual Labour Force Survey. This measure is derived in accordance with recommendations of the ILO and is therefore internationally comparable. From 1991 to 1993, it was higher than the rate of registered unemployment. In 1996, the Survey measure of unemployment was 3.7 per cent of the labour force, up from 3.2 per cent in 1995.

16. No systematic statistical analysis of this bias has been made for Switzerland so far. But drawing on work done in other countries, Fischer and Zurlinden (1994) arrive at a bias of about 1 per cent a year for Switzerland.

17. To a large extent rent increases are linked to rises in mortgage rates by federal decree, the magnitude of the increase depending on the actual level of the mortgage rate.

18. The weakening of merchandise exports in May 1997 seems to be the result of fewer work days than normally.

19. Import growth was also boosted by additional expenditures (+SF 460 million) for aircraft and related equipment in 1996.

20. Simulations with the OECD Secretariat's INTERLINK model suggest that with no change in the effective Swiss franc exchange rate from its 1992 level, the economy might have stagnated instead of fallen in 1993 and then have grown faster than potential output in the period from 1994 to 1996.

21. The seasonally-adjusted monetary base comprises notes in circulation and sight deposits held with the National Bank. In addition to its seasonal adjustment, the series is corrected for the biennial peak in note circulation at the end of even years, when property and income taxes are assessed, because of evidence that taxpayers try to lower their tax base by converting parts of their financial assets into currency.

22. The SNB's ''1 per cent money supply rule'' is based on the assumption of average potential output growth of about 2 per cent, an inflation objective of 1 per cent and an assumed trend increase in the income velocity of base money by 2 per cent per year; the latter is expected to be a consequence of the growing use of cashless means of payment. During the 1990s, the

average growth of currency in circulation – which accounts for around 90 per cent of base money – was indeed about 1 per cent.

23. This was the result of the implementation of the Swiss Interbank Clearing system (SIC) and the introduction of new prudential liquidity regulations in 1988, discussed in OECD (1990), Chapter II.

24. This reshuffling was driven by efforts of some bigger banks to further optimise their cash management procedures.

25. A study by Peytrignet (1996) suggests unstable demand for currency in circulation in recent years. Given that the share of currency in the monetary base amounts to about 90 per cent this also raises doubts about the stability of the demand for base money.

26. The Hodrick-Prescott filter decomposes changes in the GDP deflator into a smooth non-stationary component – used as an approximation of expected inflation – and a stationary cyclical component.

27. See ''Teuer bezahlter Leichtsinn der Banken'', *Neue Zürcher Zeitung* of 26/27 April 1997, p. 9.

28. The Bank of Canada's MCI uses the weight of one-third on the real exchange rate. This is based on the observation that on average in Canada a change in the real short-term interest rate by 1 percentage point exerts three times the effect on real output growth as a change in the real exchange rate by 1 per cent. The study by Lengwiler (1997) estimates weights of one-half to one-third for Switzerland before 1990. However, it suggests to use the weight of one-fifth for 1990s, which implies a very high interest elasticity of final demand. Simulations with the OECD INTERLINK model would tend to give exchange rate changes a much higher relative weight than one-fifth, so that the Bank of Canada's choice of one-third also appears plausible for Switzerland. Note that the weight has primarily an effect on the level of the MCI and in most circumstances lesser impact on its movement.

29. The MCI is defined as $MCI = -(r - r^*) - (e/e^* - 1)/w$, where r is the real (CPI deflated) three-month Euro-Swiss franc interest rate, r^* the long-term average of r, e the real effective exchange rate (CPI based), e^* the long-term average of e, and 1/w the weight on the exchange rate deviation from its average. Since the real exchange rate in Switzerland is characterised by a significant upward trend, the MCI calculated for Switzerland uses the deviation of the real exchange rate from its (linear) trend rather than from its long-term average. A positive MCI implies easy monetary conditions (relative to long-term averages).

30. The normative short-term interest rate r^* (''Taylor rule'') is defined as $r^* = \phi r + p + (p - p^*)/2 + GAP/2$. This makes r^* the sum of the long-term (ten years) average real short-term interest rate ϕr, the actual inflation rate p, one-half of the difference between actual inflation p and the inflation objective p^* and one-half of the output gap, defined as the difference between actual and potential output as a percentage of the latter.

31. The SNB pursues the objective of price stability, which at present it defines as measured CPI inflation of 1 per cent, in view of the usual upward bias in the consumer price index. However, on several occasions members of the Board of the National Bank referred in public statements to a less stringent target, considering CPI inflation of 1 to 2 per cent as tolerable.

32. The performance of the Taylor rule over the period 1990 to 1992 is also less plausible than in the recent past, *cf.* Figure 14, lower panel.

33. This is in accordance with recommendations of the Kommission für Konjunkturfragen (KfK) that the National Bank should make more explicit in its statements that the conduct of monetary policy takes exchange rate movements into account. The SNB's announcement of its policy intentions in 1997 is in line with this recommendation. In the case of persistently strong upward pressures on the Swiss franc, the KfK's recommendations include as a last resort a temporary pegging of the franc to the Deutschemark or the Euro. See ''Die Schweiz und die Europäische Wirtschafts- und Währungsunion'', Mitteilungen der KfK, August 1996, where the KfK discusses the possible effects of European monetary integration and Switzerland's options.

34. The deficits referred to in this subsection are for general government and are measured using the definitions in the *Statistique Financière Révisée (SFR*).

35. This includes those parts of the social security system for which payments are compulsory for all employed persons. Hence, medical insurance and pension funds are excluded.

36. The cyclically-adjusted budget balances discussed in this section were provided by the *Office fédéral des questions conjoncturelles*. The methodology used to calculate them is described in Amman (1995).

37. The data discussed in this subsection are on a financial accounts basis unless otherwise stated.

38. The organisation which operates Switzerland's air-traffic control system.

39. This concept excludes the surplus (SF 1 billion) from the federal civil servants' pension fund (*Caisse fédéral des pensions (CFP)*).

40. The deficits referred to in this subsection are measured on a SFR basis.

41. Cantons finance half of the deficit of the unemployment insurance system, with the Confederation financing the other half.

42. The data discussed in this subsection are on a financial accounts basis unless otherwise stated.

43. These loans are included in the budget because they are unlikely to be repaid.

44. The exceptions are debt interest, cantons' share of the Confederation's receipts, subsidies for social security and compulsory contributions to international organisations.

45. The fall in personnel expenditures reflects both cuts in employment and the transfer of posts to enterprises gaining financial autonomy.

46. The deficits referred to in this subsection are measured on an SFR basis.

47. Their finances are sensitive, however, to developments in inflation and in interest rates. No great divergences from budget assumptions are expected in these areas.

48. Cantons' 1997 budgets were based on the projections made in November 1996 that unemployment would average 170 000 in 1997.

49. The need to verify that expenditures are indeed additional to what had already been planned was an important factor in the decision to centre the programme on the public sector. This verification can be made by referring back to published budgets.

50. The data discussed in this subsection are on a financial accounts basis unless otherwise stated.

51. This long lag is a result of the biennial method of imposing and collecting the federal income tax.

52. An increase in the unemployment insurance contribution rate from 2 to 3 per cent of the gross wage – split between employer and employee – combined with a higher (indexed) wage ceiling had already been introduced in 1995 by emergency decree (''*arrêté urgent*'').

53. Although subsidising short-time work – if rigorously controlled – can contribute to lower unemployment in the short run, such schemes can be used by firms to retard structural adjustment. Indeed, a recent study shows that subsidies for short-time work which were introduced to bridge periods of cyclical weakness have to a large measure been used to cushion the effects of structural change (Hollenstein, 1997).

54. Since premiums are not experience rated, this scheme acts as a subsidy to sectors with an above average incidence of these benefits (mainly construction).

55. After extensive negotiations between the cantons and the Federal Government it was agreed that the cantons would increase the number of ALMP slots nationwide from about 10 000 in 1996 to 25 000 in early 1997 (the original target proposed by the Federal government was much higher) and to contribute to their financing through a lump sum contribution of SF 3 000 per slot and year. The minimum number of ALMP slots allocated to each canton depends on its unemployment rate and population. If additional slots are offered, the cantons do not have to contribute to their financing. However, in the event that a canton fails to meet its quota, it will have to pay 20 per cent of 80 additional passive daily unemployment benefits per unit shortfall.

56. The limits are respectively 250 and 400 days for those aged between 50 and 60 years and those 60 years and over.

57. In the event that no ALMP places are available, the unemployed person continues to be entitled to passive benefit payments for up to a further 370 days, hence a total of 520 days.

58. This restriction is controversial insofar as income received from ALMP employment is treated as any other income, *i.e.* it is subject to social security contributions, including contributions to the unemployment insurance scheme.

59. See Müller and Viredaz (1993); according to this study the most typical unskilled workers are male foreigners aged 25 to 49.

60. When preparing the reform in 1995, unemployment was lower than at present, and the National Council had estimated at the time that 66 000 slots would be needed, an estimate which was subsequently considered excessive.

61. Estimates of the *Forschungsstelle für Arbeitsmarkt- und Industrieökonomik* (FAI) at the University of Basel, put the increase at SF 3 billion (reported in Sheldon, 1997, p. 10).

62. For empirical evidence of the unemployment-increasing effect of longer duration of unemployment benefit eligibility see Scarpetta (1996). Empirical evidence of a strong effect of unemployment benefit duration on the incidence of long-term unemployment, and thereby on structural unemployment, is also reported in Sheldon, *op. cit.*

63. The current law only allows work from 6 a.m. to 8 p.m. and this makes it hard to schedule two-shift working. Companies can work longer hours but need official authorisation to do so.

64. However, the liberalisation envisaged by the reform proposal would have been comparatively modest, allowing employers in the retail sector to employ personnel six Sundays per year. Retailers could have taken advantage of this revision of the federal labour law only if the cantonal law also permitted shops to open on Sundays.

65. This has been analysed in some detail in OECD (1992), Chapter III.

66. Article 9 of the Act stipulates that ''the Competition Commission must be notified of concentrations of enterprises before they are carried out when, in the last accounting period prior to the concentration: *a*) the enterprises concerned reported joint turnover of at least 2 billion Swiss francs or turnover in Switzerland of at least 500 million Swiss francs; and *b*) at least two of the enterprises concerned reported individual turnover in Switzerland of at least 100 million Swiss francs''.

67. However, it should be noted that notification of a merger is mandatory ''when, on termination of a procedure initiated pursuant to the present law, a legally enforceable decision establishes that a participating enterprise occupies a dominant position in a market in Switzerland, and when the concentration concerns either that market or an adjacent market or a market upstream or downstream''.

68. For a brief review of the Swiss electricity industry see OECD (1995*a*), Chapter III.

69. This legislation grew out of concerns about the rapid increase in foreign acquisitions of Swiss property which began in the 1950s.

70. Tax reform issues were discussed in detail in OECD (1990), Chapter IV.

71. These are mandatory payments which increase the cost of labour for firms and, therefore, tend to be perceived as taxes.

72. These are the profit tax, the capital tax, the withholding tax (*impôt anticipé*), the stamp duty, the capital-gains tax, the real-estate tax, the parish tax and the minimum tax (a substitute for profit and capital taxes). In the canton of Geneva, a special trade tax (''*taxe professionnelle*'') is levied by the municipalities depending on sales, rent and personnel.

73. Despite this handicap, firms with less than 50 employees seem to have been particularly dynamic in recent years: in the period from 1985 to 1991 their employment grew faster than that of bigger firms, and during the 1991-95 period they reduced their workforce by much less than bigger firms.

74. A holding company is defined as a corporation which permanently holds substantial investments in the equity capital of other companies. To qualify as a holding company, a certain percentage of a company's total assets must consist of such holdings.

75. This section draws extensively on *Office Fédéral de la Statistique* (1996).

76. In 1993/94, 82 per cent of young people acquiring a vocational certificate had completed an apprenticeship. The others had attended a vocational school without on-the-job training, 14 per cent as full-time students.

77. Almost half (43 per cent) of apprentices attend school for two days a week, with the remainder attending 1-1½ days. Since the Law on Professional Training (LFPr) was revised

in 1978, there also has been the option for students who fulfil the entrance requirements to attend school an additional half day per week. However, few apprentices have chosen this option as employers tend not to appreciate the longer absences from the workplace and the resulting certificate confers few advantages over a regular proficiency certificate (*certificat de capacité*).

78. The most important of these are the schools of commerce. There are also schools which prepare their students for careers in the arts, paramedics or pedagogy as well as parts of the teaching profession and musical education.

79. These resemble those found in France, Belgium and the German Land of Baden-Wurtemberg. They are popular in watch-making and metallurgy and in the clothing industry.

80. Total entry rates to vocational education are higher because some students complete a general (usually non-academic) education before entering a vocational programme. Thus, in 1993 total entries to vocational- and general-education programmes respectively amounted to 80 per cent and 32 per cent of the population at the theoretical age to commence them. Often, students undertake a non-academic general education at the upper secondary level before beginning a basic vocational programme because the minimum age of entry to the vocational programme is higher than that at which compulsory schooling is completed (15 or 16). This is the case, for example, in vocational training in medical care. Total entry rates to vocational education also have fallen in recent years, from a peak of 89 per cent in 1985 to 80 per cent in 1993.

81. About two-thirds of students enrolled in this stream in lower secondary school.

82. In all, there are more than a hundred non-university tertiary institutions. The advanced vocational colleges comprise 28 Advanced Technical Colleges (*les Écoles Techniques Supérieures*), which offer programmes in engineering, architecture and chemistry (including agriculture), 14 Advanced Commercial and Management Colleges (*les Écoles Supérieures de Cadres pour l'Économie et l'Administration*), which offer programmes in commerce and business administration, and 7 Colleges of Advanced Arts (*les Écoles Supérieures d'Arts Appliqués*). The Advanced Technical Colleges play a key role in higher technical training, every year awarding three times as many diplomas as the two (university level) Federal Institutes of Technology combined.

83. However, only 27 per cent of students in Advanced Colleges studied part time. The professional experience requirements (generally, a certain number of years) to qualify in these cases tend to be fulfilled by full-time work.

84. There are also a few small university-level tertiary institutions with limited curricula such as the University College of Lucerne and the newly-created University of Italian-speaking Switzerland in Ticino.

85. Four of the cantonal universities (Zurich, Basle, Bern and St. Gall) and one of the FITs (Zurich) are located in the German-speaking part of Switzerland, which comprises 19 cantons and 71.2 per cent of the population; the other four cantonal universities (Geneva, Lausanne, Neuchâtel and Fribourg, the latter being bilingual) and the second FIT (Lausanne) are situated in the French-speaking part, which includes 6 cantons and 24.5 per cent of the population. The Italian-speaking part of Switzerland (mainly the canton of Ticino with

300 000 inhabitants) has recently created its own university. A detailed presentation of the Swiss higher education system can be found in *Office central universitaire suisse* (1996).

86. The most common university entrance qualification is the *maturité*; two-thirds of entries are on this basis. The other possible entrance qualifications are a degree from a foreign university or passing an entrance examination.

87. The new programme in general education at the upper secondary level focuses on the mother tongue, foreign languages, mathematics and natural sciences. Depending on the profession, apprentices also receive additional instruction in computing, physics, ecology or economics.

88. For example, during a four-year apprenticeship, an apprentice on the new pathway must spend 4 500 hours in on-the-job training and do 1 440 lessons in vocational subjects and as much again in general subjects. This is about 600 to 1 000 lessons more than a traditional apprentice.

89. The new arrangements even make it possible to do the required number of general education lessons in one year at the end of an apprenticeship of three or four years.

90. Arrangements for a MP were completed in the domain of technical agriculture in 1992, of technical subjects in 1993 and of commercial and artisanal subjects in 1995. Presently, arrangements are being made for the introduction of a MP in applied arts. This differentiation of MP is necessary to take into account the diversity of training requirements and the different orientations of the HES.

91. Other types of non-university higher education offered by teacher-training colleges, music academies as well as colleges for social workers, interpreters and paramedics will be transformed later into Colleges of Higher Education under agreements between cantons or by special federal laws.

92. Official recognition as a HES entitles an institution to federal subsidies, amounting to one-third of total expenditure.

93. These studies are the International Association for the Evaluation of Educational Achievement (IEA) Reading Literacy Study, conducted in 1991, and the IEA Third International Mathematics and Science Study, conducted in 1995. Students were set tasks in these studies which were selected to be compatible with the curricula in various countries: questions that might discriminate against students of some countries due to particularities of their curricula were avoided. A scale was constructed on which tasks of varying difficulty were placed and the ability of students was then expressed by a score on this scale (OECD, 1996*b*, p. 25).

94. The children examined in Switzerland were in their 6th or 7th school year, whereas in other countries they were in their 7th or 8th school year.

95. There is more emphasis in the Swiss curriculum on understanding and applying knowledge. Swiss students did well in questions which tested their understanding of the subject but poorly in those which tested their knowledge of terminology. A report (Moser *et al*., 1997) is to be available in Spring 1997 on the reasons for the weaker relative performance in natural science.

96. Financial resources per student within a canton, on the other hand, are equalised through intra-cantonal transfers.

97. OECD (1996*b*), p. 26; *The Economist*, March 29th 1997, p. 22. The remainder of this paragraph is based on this latter source.

98. This paragraph draws heavily on OECD (1996*b*), pp. 30-39.

99. *Prose literacy* is defined as the knowledge and skills needed to understand and use information from texts including editorials, news stories, poems and fiction; *document literacy* is defined as the knowledge and skills required to locate and use information contained in various formats, including job applications, transport schedules, maps, tables and charts; and *quantitative literacy* is defined as the knowledge and skills required to apply arithmetic operations, either alone or sequentially, to numbers embedded in printed materials, such as balancing a cheque-book, figuring out a tip, completing an order form or determining the effective interest on an advertised loan.

100. It has been suggested that a factor weighing on results in the German-speaking part of the country is that High German is not spoken in the home. The similarity between the German- and French speaking parts of the country in the proportion of the non-immigrant population scoring at the lowest literacy level (Table 23) suggests that this factor is unlikely to be very important.

101. However, direct democracy also tends to slow down the adoption of new ideas: the local public can at times be slower to appreciate the need for changes than professionals.

102. 3 per cent of students in compulsory schooling attend private schools which receive no public subsidies and a further 1 per cent attend private schools which are subsidised.

103. As in other countries, there is more confidence in the system's capacity to instruct and transmit knowledge than to nurture personal and social qualities in students.

104. While the *new maturité professionnelle* places greater emphasis on general education, only a minority of apprentices enrol in this programme.

105. A notable exception is the introduction of a new apprenticeship in computing.

106. Private supply of higher education is limited to a few graduate schools specialising mainly in management science.

107. If apprentices are considered as students rather than members of the labour force, the youth unenmployment rate is twice the adult rate.

108. This paragraph draws heavily on Wolter *et al*. (1997).

109. For the purposes of these statistics, apprentices are not counted as part of the labour force.

110. Participation in the dual system is still quite high (50 per cent) in the latin parts of the country. This compares with 70 per cent in the German-speaking regions.

111. As the Confederation controls Advanced Colleges, course content and final examinations are standardised across the country. Hence, differences in the labour market experiences of graduates from these institutions cannot be attributed to differences in the type of instruction they have received.

112. The Job-seekers rate was 10.1 per cent in latin regions and 3.5 per cent in German-speaking regions.

113. Compared with the adult rate, it was 2.1 times higher in latin regions compared with 1.8 times higher in the rest of the country.

114. The unemployment rate is lower for tertiary graduates who have exercised an activity in relation to their academic fields for more than a year before or during their studies. For such university graduates, the unemployment rate in 1995 one year after graduating was 5.9 per cent, compared with 8.4 per cent for graduates with less work experience (Diem, 1996).

115. For example, in German-speaking regions, the majority of graduates of Higher Business and Administration Colleges have worked for three years or more between the end of their apprenticeship and the beginning of their advanced studies, while in the French-speaking region, the corresponding proportion is just over 20 per cent.

116. This view is based on human capital theory. An alternative view is that education and training primarily play a screening role, identifying workers with attributes (such as intelligence, tenacity and reliability) desired by employers.

117. 7.3 per cent for men and 7.0 per cent for women. Ferro-Luzzi and Flückiger (1996). A similar study based on the Swiss Labour Force Survey of 1991 shows that an extra year of education increases annual earnings by 7.7 per cent on average, but it also moves workers to industries with 1.8 per cent higher wages. So almost a quarter of the return to education is due to ''sorting'' (Ferro-Luzzi, 1994).

118. These are rates of return with respect to the next lower level of educational attainment. They fall short of ''social'' rates of return since the latter should in principle incorporate externality effects and the macro-economic impacts of education.

119. The data discussed in this subsection can be found in OECD (1996*a*), Tables C11.1, C11.2, and R21(A).

120. The same general pattern is observed for men and women taken separately, resulting in overall labour force participation rates of 94 per cent for males and 71 per cent for females.

121. This figure is based on a small number of observations and should therefore be interpreted with care.

122. Unless otherwise indicated, the data discussed in this section in the context of international comparisons come from OECD (1996*a*).

123. Before 1993, education was the largest component of public expenditure in Switzerland. Since then, it has been surpassed by social security due to the rise of unemployment-insurance benefits.

124. OECD (1996*a*), Chart F1.2C. Full-time equivalent enrolments are relatively low because apprentices only attend vocational schools for one or two days per week.

125. Note that there is a wide variation in student costs within the Swiss tertiary sector. At US$17 807, the cost for a university student is almost double that for a non-university tertiary student (US$8 947).

126. For instance, the average surface of 14.4 m^2 per student occupied by universities in 1994 can be considered as quite generous by European standards (*Conférence universitaire suisse*, 1996*a*).

127. High construction costs in Switzerland are discussed in OECD (1994).

128. A primary teacher with 15 years experience earns about 30 per cent more than the median of gross average earnings for all employees. This margin doubles for a lower secondary teacher.

129. These fees vary from SF 500 to SF 688 per semester for students who are Swiss residents (including foreign immigrants). Non-resident foreigners pay higher fees, albeit less than double those charged to residents. These fees are low by international standards, especially in PPP terms.

130. These risks include not completing the programme and subsequently not earning as much as expected.

131. This proportion has been declining in recent years: in 1980 it was 15.2 per cent.

132. For example, the average grant in Neuchâtel in 1993 was SF 2 769, which represented about 20 per cent of the estimated living costs there for a nine-month academic year. Grants were higher in other cantons, but so were living costs. The highest grants were in the City of Basle (SF 8 221) and the cross-canton mean was SF 5 809. Tertiary students received the highest grants (SF 7 546 for university and SF 7 118 for non-university students).

133. Of this expenditure on grants, SF 99 million was for university students and SF 74 million for those studying at the non-university tertiary level (*Office fédéral de la statistique*, 1995*b*). Note that these grants are subsidised by the Confederation at a rate of 20, 40 or 60 per cent depending on the financial capacity of the canton.

134. In Geneva and Fribourg, universities have been subject to auditing by private firms.

135. French-speaking Switzerland has a long tradition of inter-university co-ordination in the framework of *Conférence universitaire romande*. This co-ordination has recently been extended to the entire western part of Switzerland in the framework of *Conférence universitaire de Suisse occidentale.*

136. An example is the so-called CLUSE network which includes Swiss and French universities.

137. A recent example is the merger between the schools of architecture of the University of Geneva and the Federal Institute of Technology at Lausanne.

138. In 1993, 61 per cent of job-related continuing education and training courses were totally or partially financed by firms, 29 per cent by the employees themselves and the rest by the public sector or other means (*Office fédéral de la statistique*, 1995*a*).

139. Initially, the amounts allocated to different components of the programme were the following: SF 162 million for professional continuing education, SF 135 million for university-level continuing education and SF 102 million for the *CIM Action Programme*. These amounts were later reduced to SF 101 million, 77.5 million and 90 million, respectively.

140. Among the three components of the programme, so far only university-level continuing education has been subject to an official evaluation (*Conférence universitaire suisse*, 1996*b*). The evaluation shows that as a result of the federal programme, all Swiss universities now host specialised units providing an extensive array of continuing education courses. In 1995, the Parliament accepted the extension of the federal programme by allocating an additional amount of SF 11.5 million to cantonal universities for the period 1996-99. From 1996 onwards, university-based continuing education should in principle become self-financing.

141. In 1993, only 7 per cent of continuing education courses were financed by the Confederation, the cantons and the communes, leaving as much as 54 per cent to the users themselves, 36 per cent to firms (total or partial financing) and 4 per cent to other sources (*Office fédéral de la statistique*, 1995*a*).

Bibliography

Ammann, Y. (1988), ''Marginale Steuersätze von Aktiengesellschaften'', *Mitteilungsblatt für Konjunkturfragen*, No. 4.

Ammann, Y. (1995), *Le budget de plein emploi – un réexamen*, Office fédéral des questions conjoncturelles, étude No. 20, Bern.

Banque nationale suisse (1996), ''La politique monétaire en 1997'', Monnaie et conjoncture, Bulletin trimestriel de la Banque nationale suisse, December.

Banque nationale suisse (1997), *89e rapport de gestion*, 1996.

Brunner, O. (1996), ''Der Arbeitsmarkt in Gegenwind'', *Die Volkswirtschaft*, April.

Bundesamt für energiewirtschaft (1997), *Marktöffnung im Elektrizitätsbereich*, BEW-Schriftenreihe, Studie Nr. 59, January.

Conférence universitaire suisse (1995), *Accord intercantonal sur le financement des universités – Examen des critères de financement sous l'angle économique*, Bern.

Conférence universitaire suisse (1996*a*), ''Inventaire des constructions universitaires'', *INFO CUS*, No. 5.

Conférence universitaire suisse (1996*b*), *Mesures spéciales de la Confédération en faveur de la formation continue au niveau universitaire – Rapport final*, Bern.

Cornevin-Pfeiffer, K. and A. Manzini (1992), ''Le financement de l'État'', in: L. Weber (ed.), *Les finances publiques d'un État fédératif – La Suisse*, Paris, Economica.

Diem, M. (1996), *La situation des diplômés universitaires sur le marché de l'emploi en 1995*, Office fédéral de l'éducation et de la science, Bern.

Ferro-Luzzi, G. (1994), ''Inter-Industry Wage Differentials in Switzerland'', *Swiss Journal of Economics and Statistics*, Vol. 130, No. 3.

Ferro-Luzzi, G. and Y. Flückiger (1996), *Analyse des inégalités entre les femmes et les hommes sur le marché du travail à Genève*, Observatoire universitaire de l'emploi, Geneva.

Fischer, A.M. and M. Zurlinden (1994), ''Geldpolitik mit formellen Inflationszielen: eine Übersicht'', *Monnaie et conjoncture*, Bulletin trimestriel de la Banque nationale suisse, March.

Hollenstein, H. (1997), ''Auswirkungen der Kurzarbeitsregelungen auf das Beschäftigungsverhalten der Schweizer Industrie'', *Die Volkswirtschaft*, January.

Kommission für konjunkturfragen (1996), ''Die Schweiz und die Europäische Wirtschafts- und Währungsunion'', *Mitteilungen der Kommission für Konjunkturfragen*, August.

Lengwiler, Y. (1997), ''Der 'Monetary conditions Index' für die Schweiz'', *Monnaie et conjoncture*, Bulletin trimestriel de la Banque nationale suisse, March.

Moser, U. *et al*. (1997), *Schools under Examination* (in German), Rüegger Verlag, Berne.

Müller, R.A. and Y.-D. Viredaz (1993), ''Les chômeurs sans qualification'', *La vie économique*, April.

Neue Zürcher Zeitung (1997), ''Teuer bezahlter Leichtsinn der Banken'', April 26/27.

OECD (1990), *Economic Surveys, Switzerland*.

OECD (1992), *Economic Surveys, Switzerland.*

OECD (1991*a*), *Reviews of National Policies for Education, Switzerland*.

OECD (1991*b*), *Taxing profits in a global economy – Domestic and international issues*.

OECD (1992), *Les relations entre l'enseignement supérieur et l'emploi – Les développements récents de la formation professionnelle continue – Contribution nationale – Suisse*.

OECD (1994), *Economic Surveys, Switzerland.*

OECD (1995*a*), *Economic Surveys, Switzerland.*

OECD (1995*b*), *Education at a Glance.*

OECD (1996), *Economic Surveys, Switzerland*.

OECD (1996*a*), *Education at a Glance*.

OECD (1996*b*), *Education at a Glance, Analysis*.

OECD (1996*c*), *Lifelong Learning for All*.

OECD (1997*a*), *Economic Outlook*, No. 61, forthcoming.

OECD (1997*b*), *The Role of Comparative International Indicators of Human Capital Investment in Public Policy*, OECD Secretariat Working Paper Objective 1.

OECD and Statistics Canada (1995), *Literacy, Economy and Society*, Ottawa.

Office central universitaire suisse (1996), *A Guide to the Swiss Higher Education System and its Qualifications*, Bern.

Office fédéral de l'éducation et de la science, Office fédéral de l'industrie, des arts et métiers et du travail, Office fédéral de la statistique et Secrétariat de la CDIP (1995), *Structures of the Education and Initial Training Systems in Switzerland*.

Office fédéral de l'énergie (1995), *Cattin group report*, Bern.

Office fédéral de la statistique (1994), *Annuaire statistique de la Suisse*, Bern.

Office fédéral de la statistique (1995*a*), *La formation continue en Suisse*, Bern.

Office fédéral de la statistique (1995*b*), *Les indicateurs de l'enseignement en Suisse*, Bern.

Office fédéral de la statistique (1996), *La formation professionnelle en Suisse, Bern.*

Office fédéral des questions conjoncturelles, études n° 20, Bern.

Office fédéral du personnel (1994), *Hautes écoles spécialisées : la politique de la formation suisse face à un nouveau défi*, Proceedings of a seminar organised in Sion 13-15 April 1994, Bern.

Peytrignet, M. (1996), ''Stabilité économétrique des agrégats monétaires suisses'', *Monnaie et conjoncture*, Bulletin trimestriel de la Banque nationale suisse, September.

Psacharopoulas, G. (1994), ''Returns to Education: A Global Update'', *World Development*, Vol. 22, No. 9.

Revaz, F. and R. Rietschin (1992), ''Enquête pilote sur les salaires dans le canton de Genève – Premiers résultats et brefs commentaires'', *La vie économique*, December.

Scheuber, A. (1996), *Stipendienpolitik in der Schweiz : Die Bedeutung der Stipendien für unser Land*, Chavannes-près-Renens, IDHEAP.

Schneiter and Zarin-Nejadan (1994), ''Impact de l'Université de Neuchatel sur l'économie cantonale'', *Cahiers de recherche en économie politique*, No 94-01, University of Neuchatel.

Secrétariat général de la Conférence suisse des directeurs cantonaux de l'instruction publique (1996), Développement de l'éducation : *Rapport national de la Suisse, Report prepared for the International Conference on Education*, Geneva, September.

Scarpetta, S. (1996), ''Assessing the Role of Labour Market Policies and Institutional Settings on Unemployment: A Cross-country Study'', *OECD Economic Studies*, No. 26.

Sheldon, G. (1997), ''Verstecktes Impulsprogramm für die Schweiz?'', *Neue Zürcher Zeitung*, February 26.

Straubaar, Th. and M. Winz (1992), *Reform des Bildungswesens*, Bern, Haupt.

Taylor, J.B. (1993), ''Discretion *versus* policy rules in practice'', *Carnegie-Rochester Conference Series on Public Policy*, No. 39.

Whitburn, J. (1997), ''When being top is not seen as best'', *The Times Educational Supplement*, January 10.

Wolter, S. (1994), ''Lohnt sich Bildung überhaupt?'', *Economic Focus*, Union Bank of Switzerland, September.

Wolter *et al*. (1997), *Different School Systems – Different Labour Market Results*, Swiss Federal Office for Industry and Labour (BIGA).

Zarin-Nejadan, M. (1997), ''Die Besteuerung der KMU in der Schweiz'', *Die Volkswirtschaft*, February.

Annex I

Institutional arrangements in the education and training sector*

General context

Constitutionally, education is a cantonal responsibility, although the federal government plays a role in funding universities and in planning, implementing and funding vocational education (see Box A1). Since the cantons have constitutional responsibility for education, there is no national education system as such, although all cantonal systems have the same basic structure. Variations occur mainly in the offer of programmes at the post-compulsory level and in the configuration of compulsory schooling. The communes also play a role in funding and administering compulsory education, although the definition of this role and its importance vary among cantons. In short, there is considerable diversity in the education system.

The education system has been described as highly selective, streaming students into programmes with either a basic curriculum or a broader-based one as early as after 5 or 6 years of primary school. The selection into different programmes in upper secondary education is also rather rigid. All upper secondary and tertiary education is optional – young people are under no legal obligation to continue their education after age 16, and public authorities are not required to assure educational opportunities for all.

Educational structure and enrolments

Almost all children (98 per cent) attend kindergarten for an average of 1.8 years before commencing primary school. Primary school starts at 6 or 7 years of age and lasts for four, five or six years depending on the canton. There are about 440 000 (1994) primary school students. Lower secondary education commences at 11 or 12 years of age and lasts for three to five years depending on the canton. This makes a total of nine years of compulsory education in all cantons. There are about 280 000 (1994) lower secondary students. Some cantons have organised their lower secondary level in a comprehensive form; in others, the different types of programmes can be classified into two groups:

* With the exception of Box A1, this annex consists of extracts from the description of Switzerland's education and training system in OECD (1996*a*).

Box A1. **Areas of responsibility and administration***

Distribution of responsibility

As the constitution only accords a few specific tasks in education to the Confederation, the cantons retain power with regard to the basic organisation of the education system. Legislation varies considerably from canton to canton, with the result that there are in fact 26 different education systems. Certain important areas, however, are covered by federal provisions in order to establish some degree of homogeneity. However, there is no federal or national Ministry of Education.

The tasks of the Confederation include the following:

- it is responsible for legislation concerning vocational training in certain sectors;
- it controls the Federal Institutes of Technology in Zurich and Lausanne;
- it regulates admission to medical studies;
- it subsidies the cantonal universities and scientific research.

In areas where legislative powers are held by the Confederation, it passes the necessary laws and delegates the corresponding executive powers to the cantons or, in rare cases, to private bodies. The cantons are therefore also largely responsible for an education system over which they do not have direct legislative powers. Their position is strengthened by the fact that the Confederation grants them the right of consultation.

Tasks in vocational training are shared between the Confederation, the cantons and professional organisations. The Confederation is constitutionally empowered to pass legislation in the field of vocational training for industry, trade, commerce, agriculture and domestic service. All other areas of vocational training – teaching, health care, social services, the arts – are the responsibility of the cantons. In general, the cantons are responsible for providing vocational training, including that covered by federal regulations. The private sector, through professional associations, not only takes care of the practical side of vocational training in most cases, but also runs schools or helps to finance certain technical colleges.

Cantonal universities are run by the cantonal department of education and enjoy a large degree of academic autonomy.

Intercantonal collaboration and co-ordination of the education system

In view of the diversity of characteristics and the different levels of responsibility, the main problem arising is coherence within the system as a whole. In order to enhance the co-ordination of education policy, the Swiss Conference of Cantonal Directors of Education (EDK/CDIP) was restructured in the 1960s and provided with a secretariat, and four regional organisations were set up. Co-ordination has also been promoted through the ''Agreement on the Co-ordination of Education''. This was drawn up by the

* This box is based on Office fédéral de l'éducation et de la science *et al.*, 1995 and on Secrétariat général de la Conférence suisse des directeurs cantonaux de l'instruction publique, 1996.

(continued on next page)

(continued)

cantons in 1970 ''in order to develop education and harmonise cantonal legislation on education''. The Agreement is intended first and foremost to be an instrument for voluntary collaboration and harmonisation (including on reforms, recognition of qualifications, co-operation in planning, research and statistics). It also includes compulsory provisions (concerning matters such as the age at which children start school, the duration of compulsory school attendance, the duration of schooling up to the school-leaving certificate, and the start of the school year). It has taken 15 years to apply the few compulsory principles contained in the Agreement owing to the difficulties encountered through direct democracy. The Agreement accords executive powers to the CDIP and its agents.

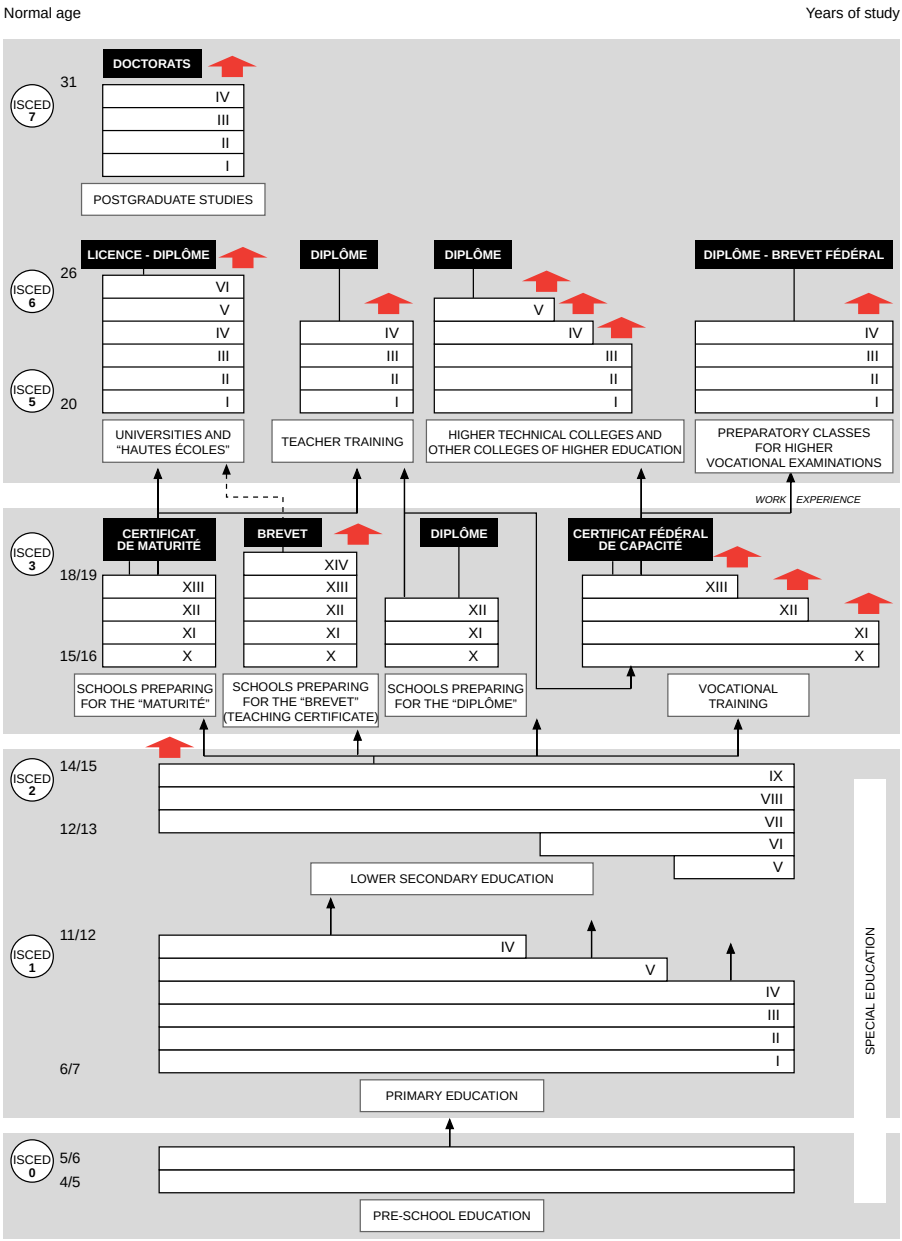
Administration

The Confederation's educational policy is administered by two ministries. The Ministry of Interior is responsible for Federal Institutes of Technology, subsidies to cantonal universities, scholarships, research, physical education and recognition of the cantonal *maturité*. The Ministry of Public Economy supervises professional training. Within each ministry, various entities are responsible for educational policy in overlapping domains. The Ministry of Interior embodies *Groupement pour la science et la recherche* (GSR), *Office fédéral de l'éducation et de la science* (OFES) as well as two co-ordination bodies, namely *Conseil suisse de la science* (CSS) and *Conférence universitaire suisse* (CUS). The relevant entities operating within the Ministry of Public Economy are *Office fédéral de l'industrie, des arts et métiers et du travail* (OFIAMT) which oversees general professional training and *Office fédéral de l'agriculture* which is responsible for agricultural professional training. At the cantonal level, the educational system is administered by the cantonal Ministry of Public Education. Some cantons also have consultative bodies called Educational Councils (*Erziehungsrat*). At the communal level, political authorities share the administration of the school system with the so-called School Commissions (*Commissions scolaires*) under supervision by the canton.

those schools with a basic required curriculum and those with an extended required curriculum. In 1994, 31 per cent of students attended the first type and 69 per cent either the second type or a comprehensive form.

Even though upper secondary education is not compulsory, in 1994 an estimated 93 per cent of the relevant cohort entered education at this level, and 85 per cent graduated from a programme of more than two years' duration. Nearly 20 per cent of the cohort attempted more than one programme in upper secondary education, with the most frequent case being attendance at a short (one or two years' duration) general programme followed by entering vocational education. Three-quarters of the cohort enter vocational

Figure A1. **THE SWISS EDUCATION SYSTEM**



Note: = Recognised exit point of the education system.
Source: OECD Secretariat.

education and two thirds finish a programme of two or more years' duration. 17 per cent of the cohort graduate from a programme giving access to university education.

Tertiary education is provided by ten universities, several different types of colleges of higher vocational education and many part-time vocational education courses. The latter usually lead to an examination approved by a public or professional body. All universities are public, while many of the programmes in the higher vocational education sector are private. 13 per cent of young adults enter one of the universities and 27 per cent a higher vocational education course/programme. Some of the higher vocational colleges will be promoted to *Fachhochschulen* from 1997 on, thus diversifying the tertiary sector.

In 1992/93 almost two million adults (40 per cent) attended at least one continuing education course. Only 19 per cent of these courses were offered by the public sector. 31 per cent were organised by employers and 50 per cent by other private entities, with most of the private organisations being market-oriented providers of continuing education. Educational policy in the fields of higher vocational and continuing education supports this pattern as there is a strong reliance on market forces to arrive at a pattern of efficient qualification of the workforce.

Curriculum and assessment

With the exception of vocational education, there is no nation-wide common school curriculum. However, for compulsory education, the conference of cantonal education ministers has established a series of targets defining the expected competencies in several key subject areas such as mathematics, the mother tongue and the first foreign language. For upper secondary general education, the targets have been set for the university entrance exam and the intermediate diploma. All of these inter-cantonal agreements pertaining to the expected level of competence allow for cantonal variation in curricula.

In contrast, the national curricula for vocational education in the form of apprenticeships are very detailed. They are compulsory for both the school-part and the enterprise-part and are established after intensive consultation with representatives of industry and trade unions.

There is an entrance requirement or exam for all post-compulsory programmes and for streams with a broader based curriculum in lower secondary education (with the notable exception of the universities which have up to now admitted all persons with a *maturité*). Assessment within a programme is usually school-based, with promotion between year levels not being automatic. In compulsory education, for example, one in six students repeats at least one grade. Diplomas at the end of post-compulsory programmes are usually granted after passing a government board approved school-based exam. Again, relatively high failure rates are not uncommon.

Teachers

The education of kindergarten, primary and lower secondary (basic demands) teachers is the responsibility of the cantons and is still far from uniform across the country. Nevertheless, some broad common features can be identified. All upper secondary teachers are educated in university-level institutions. The teachers of lower secondary programmes which provide the broader based curriculum are also educated at the university. In some cantons, the teachers in programmes providing the basic curriculum are taught in specialised teacher training colleges, usually at the tertiary level. Some primary teachers still attend 5-year colleges beginning after compulsory school. The others are educated at the tertiary level in teacher colleges. The policy established in 1995 plans to split all teacher education into an upper secondary part equivalent to the *maturité* and a tertiary part of at least three years' duration devoted to the acquisition of professional knowledge and skills.

Statistics on Swiss teachers are as yet incomplete, but will be completed by the year 2001. In compulsory schools some 62 200 teachers occupied 51 000 full-time posts, with 46 per cent of all teachers having part-time status. The proportion of women teachers in primary and lower secondary is 68 and 36 per cent, respectively, one of the lowest proportions among OECD countries.

Swiss teachers work longer hours than teachers in other OECD countries and enjoy salaries well above the average earnings. Salaries are higher at higher levels of education, and there is considerable variation of teachers' salaries among cantons for the same category of teachers.

Annex II

Calendar of main economic events

1996

January

The first of two steps of unemployment insurance reform comes into force, which *inter alia*:

- introduces waiting periods before the start of unemployment benefit payments;
- reduces the duration of compensation for income losses from short-time work and from bad weather;
- makes the definition of ‘‘suitable’’ work more stringent;
- raises penalties for unco-operative behaviour of job-seekers; and
- introduces intermittent pay supplements for low-paid jobs.

New procedures are introduced in the federal public service in order to make individual pay setting more flexible.

March

The Parliament approves a revision of federal labour law (*loi fédéral sur le travail*) which is intended to increase the flexibility of working time, *inter alia* by an extension of the definition of ‘‘normal’’ working hours and the abolition of the ban on night work for women, while reinforcing the protection of night workers.

The Parliament approves the Federal Council’s proposal to introduce a special VAT rate of 3 per cent (instead of 6.5 per cent) for the hotel trade.

June

The Federal Council submits to Parliament a plan for the construction and financing of transport infrastructure until the year 2017. It foresees, *inter alia* to raise funds through

an increase in the petrol tax by SF 0.10 per litre and a special levy on heavy road transport.

July

The Federal Parliament adopts a law on technical barriers to trade which aligns Swiss legislation to that of major trading partner countries.

The Parliament approves an amendment of the domestic market law which enhances the professional mobility of Swiss citizens and of foreign workers with permanent work permits.

The new Cartels Act comes into force, which prohibits the elimination of effective competition and restricts its limitation; the Act also includes provisions for merger control.

September

The Federal Council submits to Parliament a proposal to freeze federal expenditures in 1997 at their levels of the 1996 Budget and to leave them constant in real terms until the year 2000.

The National Bank lowers the discount rate by ½ percentage point to 1 per cent.

December

The Swiss people reject the proposed revision of the federal labour law in a referendum.

In the bilateral negotiations with the EU the Federal Government agrees on a timetable for scrapping its system of work-permit quotas for EU citizens and for instituting the free movement of persons. Quotas will be eliminated within six years of the treaty coming in force, but can be reintroduced subsequently in case of strong inflows of immigrants.

The Parliament adopts the federal budget for 1997, which provides for an increase in the deficit/GDP ratio by ½ percentage point to SF 6.3 billion (1.7 per cent of GDP). The increase largely reflects changes in accounting practices; abstracting from these changes, the budget provides for a small increase in the deficit compared with the 1996 Budget, reflecting a decline in revenues.

The National Bank announces to continue its easier monetary policy and to allow the monetary base to exceed its medium-term target of 1 per cent annual growth in 1997.

1997

January

The second step of unemployment insurance reform comes into force, introducing *inter alia* a shift away from passive income support to active labour market programmes (the ''activation principle''). The move implies a prolongation of the duration of benefit eligibility from previously 170, 250 or 400 workdays – depending on the unemployment contributions paid – to uniformly 520 workdays. To make the new system work, a network of 150 regional placement offices and 1 850 posts of job counsellors are to be created during 1997.

A decree comes into force which allows during the next two years to subsidise through the unemployment insurance fund up to half the previous wage of an employee retiring early, to make way for the recruitment of an unemployed person, who was out of work for at least six months.

The tenth revision of the old-age and survivors' insurance scheme (AVS) comes into force which, among other things, raises the retirement age for women from 62 to 63 in 2001 and from 63 to 64 in 2004 and grants pension credits to women who stop work to care for children or elderly relatives.

Switzerland and the United States agree to open their public procurement markets further to each other's companies. The agreement goes beyond the World Trade Organisation accord on public procurement to which both countries are signatories.

March

The Federal Council submits to Parliament a policy package to stimulate economic activity by bringing forward infrastructure maintenance expenditures of SF 561 million, most of which will be spent in 1998. In addition, the package introduces a flat-rate profit tax of 8.5 per cent (replacing the variable profit tax rate dependent on profitability), the abolition of the corporate tax on capital, the alleviation of the taxation of holding companies, and the re–introduction of the stamp duty on life insurance premiums. The estimated net effect of the tax measures on the Confederation's budget is a revenue reduction of SF 170 million per year. Moreover, the Federal government proposes to moderate the restraint on foreign investment in business real estate imposed by the *Lex Friedrich*.

April

The Parliament approves the federal expenditure programme for infrastructure investment but defers the decision on the proposed tax measures.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	Average 1987-96	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
A. Percentage change from previous year at constant 1980 prices											
Private consumption	1.1	2.1	2.1	2.2	1.5	1.5	–0.2	–0.6	0.9	0.7	0.3
Gross fixed capital formation	2.4	7.4	6.9	5.8	2.6	–2.5	–5.0	–2.5	7.2	2.3	1.3
Construction	1.1	5.4	6.4	6.9	1.9	–3.1	–2.3	–1.8	5.2	–4.3	–3.5
Machinery and equipment	4.6	11.3	7.9	3.9	3.7	–1.2	–9.6	–3.7	11.0	14.1	8.4
GDP	1.0	2.0	2.9	3.9	2.3	0.0	–0.3	–0.8	1.0	0.1	–0.7
GDP price deflator	3.0	2.6	2.4	4.2	5.7	5.5	2.6	2.0	1.9	2.5	0.1
Industrial production	1.7	1.2	7.8	1.5	2.7	0.0	–0.8	–1.9	4.4	1.9	–0.1
Employment	1.0	2.5	2.6	2.7	3.2	1.2	–1.6	–0.6	–0.2	0.2	0.2
Compensation of employees (current prices)	4.3	4.9	5.9	6.9	8.9	7.3	3.4	1.0	1.1	2.4	0.9
Productivity (real GDP/employment)	0.0	–0.4	0.3	1.1	–0.8	–1.2	1.3	–0.2	1.2	–0.1	–0.8
Unit labour cost (compensation/real GDP)	3.2	2.8	2.9	2.9	6.4	7.4	3.7	1.9	0.1	2.3	1.5
B. Percentage ratios											
Gross fixed capital formation as per cent of GDP at current prices	24.6	25.3	26.6	27.5	26.9	25.6	23.7	22.5	23.1	22.7	22.3
Stockbuilding as per cent of GDP at current prices	0.7	2.0	1.2	2.2	2.3	1.4	–0.3	–1.2	–0.7	0.5	–0.4
Foreign balance as per cent of GDP at current prices	2.4	0.8	0.4	–0.2	0.5	1.4	3.5	5.0	4.7	3.9	3.9
Compensation of employees as per cent of GDP at current prices	61.1	60.4	60.7	60.0	60.4	61.5	62.2	62.1	61.0	60.9	61.7
Direct taxes as per cent of household disposable income	18.0	17.9	18.1	17.5	18.0	17.2	17.7	17.6	18.8	17.8	19.2
Household saving as per cent of disposable income	10.9	8.4	9.9	11.0	12.2	13.0	12.7	11.5	10.1	10.5	9.2
Unemployment as per cent of total labour force	2.4	0.7	0.6	0.5	0.5	1.1	2.5	4.5	4.7	4.2	4.7
C. Other indicator											
Current balance ($ billion)	14.0	8.7	10.4	7.0	8.6	10.6	15.1	19.4	18.4	21.5	20.3

Source: OECD, *National Accounts*.

Table B. **Gross national product**

Million Swiss francs, current prices

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Private consumption	150 210	156 970	166 150	177 650	190 490	198 070	203 005	206 970	212 135	215 190
Public consumption	33 025	35 405	38 485	42 850	46 640	49 320	49 705	50 625	51 350	51 715
Gross fixed asset formation	64 370	71 480	79 860	84 545	84 810	80 375	77 020	81 445	82 340	80 340
Change in stocks	4 975	3 355	6 435	7 310	4 545	–980	–4 175	–2 535	1 900	–1 270
Domestic demand	252 580	267 210	290 930	312 355	326 485	326 785	325 555	336 505	347 725	345 975
Exports of goods and services	90 525	97 990	110 510	115 050	116 720	122 170	125 300	127 430	127 545	130 610
Imports of goods and services	88 420	96 790	111 080	113 415	112 130	110 190	108 005	111 015	113 265	116 490
Gross domestic product at market prices	254 685	268 410	290 360	313 990	331 075	338 765	342 850	352 920	362 005	360 095
Factor income from abroad	20 900	24 185	28 010	28 130	29 285	27 715	30 685	30 660	30 905	. .
less: Factor income paid abroad	9 495	9 645	13 200	14 535	14 970	14 285	15 130	17 945	15 350	. .
Gross national product at market prices	266 090	282 950	305 170	327 585	345 390	352 195	358 405	365 635	377 560	. .

Source: Office fédéral de la statistique.

Table C. **Gross national product**

Million Swiss francs, 1980 prices

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
Private consumption	119 290	121 845	124 560	126 430	128 285	128 065	127 330	128 485	129 400	129 735
Public consumption	26 115	27 235	28 365	29 695	30 150	30 105	29 745	30 125	30 080	30 020
Gross fixed asset formation	53 620	57 340	60 650	62 210	60 685	57 640	56 210	60 275	61 650	62 430
Change in stocks	4 335	2 715	4 105	4 335	2 455	–1 455	–2 825	–1 045	810	–2 360
Domestic demand	203 360	209 135	217 680	222 670	221 575	214 355	210 460	217 840	221 940	219 825
Exports of goods and services	75 695	80 090	84 090	86 630	86 005	88 890	90 340	93 430	96 210	98 520
Imports of goods and services	87 530	92 145	97 080	99 905	98 245	94 545	93 755	102 145	108 845	110 425
Gross domestic product at market prices	191 525	197 080	204 690	209 395	209 335	208 700	207 045	209 125	209 305	207 920
Factor income from abroad	17 415	18 810	19 520	20 570	19 855	17 675	18 565	19 120	20 370	. .
less: Factor income paid abroad	9 475	9 180	12 200	12 620	11 985	10 775	11 150	13 450	10 980	. .
Gross national product at market prices	199 465	206 710	212 010	217 345	217 205	215 600	214 460	214 795	218 695	. .

Source: Office fédéral de la statistique.

Table D. **Gross value added by main area of activity**

	1975	1985	1990	1991	1975	1985	1990	1991
	SF million				As a percentage of GDP			
Enterprises	124 685	203 930	281 094	296 530	89.0	89.5	89.5	89.6
Non-financial enterprises	116 282	180 834	251 724	296 530	83.0	79.3	80.2	89.6
Agriculture, forestry	6 487	8 180	9 664	9 939	4.6	3.6	3.1	3.0
Energy, metal ore mining	2 808	5 023	6 011	6 375	2.0	2.2	1.9	1.9
Industrial arts and crafts	41 687	58 794	76 720	77 842	29.7	25.8	24.4	23.5
Building and civil engineering	11 210	17 325	26 224	26 660	8.0	7.6	8.4	8.1
Distributive trades, hotels and catering, repairs	29 430	41 823	60 796	57 107	21.0	18.3	19.4	17.2
Transport and communications	9 140	14 552	18 556	20 096	6.5	6.4	5.9	6.1
Real estate and consultancy services	5 946	15 100	26 324	. .	4.2	6.6	8.4	. .
Rental and dwellings	5 290	9 878	13 949	. .	3.8	4.3	4.4	. .
Other services	4 284	7 293	10 887	. .	3.1	3.2	3.5	. .
Banking and insurance	8 403	23 096	29 370	. .	6.0	10.1	9.4	. .
Government (including social insurance)	14 452	26 065	36 412	40 022	10.3	11.4	11.6	12.1
Household and private non-profit institutions	2 453	4 655	6 783	7 431	1.8	2.0	2.2	2.2
Total (unadjusted)	141 590	234 650	324 289	343 983	101.0	102.9	103.3	103.9
Adjustments								
Imputed value of bank service charge	–4 565	–10 400	–14 566	–17 335	–3.3	–4.6	–4.6	–5.2
Import duties	3 130	3 700	4 267	4 427	2.2	1.6	1.4	1.3
Adjusted total = gross domestic product	140 155	227 950	313 990	331 075	100.0	100.0	100.0	100.0

Source: Office fédéral de la statistique, *National Accounts*.

Table E. **Producer and import prices**

May 1993 = 100

	1993[1]	1994	1995	1996	1996				1997
					Q1	Q2	Q3	Q4	Q1
Producer price index									
Total	99.9	99.5	99.4	97.6	98.3	97.8	97.3	97.0	96.9
Agricultural products	101.9	101.5	95.2	89.4	90.4	89.5	88.8	89.1	88.2
Manufacturing	99.7	99.1	99.4	97.9	98.6	98.1	97.6	97.2	97.2
Food and tobacco	99.9	100.1	99	98.2	98.2	98.2	98.5	97.8	97.9
Textiles and clothing	100	100.5	100.9	99.3	99.9	99.3	99.1	99.1	99.4
Paper and paper products	98.9	97.8	101.4	98.3	101.6	99.1	96.7	95.9	96.1
Petroleum products	95.8	90.4	87.8	96.6	91.4	94.2	94.4	106.3	106.9
Chemicals	99.2	95.5	92.2	89.1	90.3	89.4	88.7	87.9	86.6
Machinery	99.9	99.6	100	100.3	100.3	100.4	100.3	100.2	100.2
Transport equipment	100.4	101.8	102.8	102.3	102.8	102.7	102.7	101.1	101.1
Furniture and other articles	101.9	103.1	104.5	102.9	103.2	103.3	102.9	102.1	102.7
Electricity and gas	100.2	101.2	103	103.5	104.1	103.7	103.4	103.0	103.1
Domestic use	100	99.9	100	98.1	98.8	98.3	97.9	97.5	97.5
Exports	99.7	98.8	98.4	96.8	97.5	97.0	96.5	96.1	95.9
Raw materials	103.3	104.7	95.1	90.4	89.0	92.6	90.1	90.0	89.5
Semi-finished goods	99.6	98.9	99.4	96.9	98.1	97.1	96.5	96.0	95.9
Consumer goods	99.8	99.8	100.2	99.6	100.0	99.5	99.4	99.3	99.3
Capital goods	99.9	99.5	99.5	99.2	99.5	99.2	99.2	99.0	99.0
Import price index, goods	99.8	99.8	100	96.3	97.6	96.9	95.3	95.4	96.7
Raw materials	104.8	121.5	121.1	109.9	109.3	113.6	109.0	107.8	121.3
Semi-finished goods	99.6	100.3	105.7	99.8	102.9	100.1	97.7	98.6	100.2
Consumer goods	99.9	98.6	97.3	97.3	97.1	97.5	97.0	97.8	98.0
Capital goods	98.4	95.4	91.3	87	88.6	87.9	86.4	85.0	84.2
Total supply price index	99.9	99.6	99.6	97.2	98.1	97.5	96.7	96.5	96.8
Raw materials	103.8	111	104.7	97.7	96.5	100.4	97.1	96.6	101.3
Semi-finished goods	99.6	99.2	100.6	97.5	99.0	97.7	96.8	96.5	96.8
Consumer goods	99.9	99.3	99	98.7	98.8	98.7	98.4	98.7	98.7
Capital goods	99.3	97.9	96.4	94.5	95.3	94.9	94.3	93.6	93.4

1. As from May 1993, the wholesale price index has been replaced by the total supply price index, which is the weighted average of the producer price and the import price indices. Therefore indices for 1993 are the averages for May to December.

Source: Office fédéral de la statistique, *Indice des prix à la production et à l'importation*.

Table F. **Money supply**

Million Swiss francs, yearly average

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
1985 Definition										
Monetary base	33 403	31 521	29 910	28 934	29 247	29 056	29 498	30 070	30 129	31 185
Money supply										
M1	71 351	81 480	77 032	73 775	74 687	74 626	81 597	86 436	90 510	101 439
M2	138 503	149 098	179 037	202 338	208 883	209 995	193 321	189 727	187 811	189 549
M3	303 394	333 089	353 721	362 145	373 670	384 063	402 291	425 432	437 987	471 494
1995 Definition										
Monetary base	33 403	31 521	29 910	28 934	29 247	29 056	29 498	30 070	30 129	31 185
Money supply										
M1	107 906	122 558	117 535	111 570	113 744	116 011	128 212	135 419	144 651	161 736
M2	233 704	262 078	247 180	227 520	231 370	237 861	276 200	304 417	319 383	358 129
M3	300 855	329 695	349 184	356 083	365 565	373 230	387 924	407 706	416 683	446 185

Source: Banque nationale suisse, *Bulletin mensuel*.

Table G. **Interest rates and capital markets**

Million Swiss francs and percentages

	1991	1992	1993	1994	1995	1996	1995				1996				1997
							Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
Interest rates (average for the period)															
Discount rate (end of period)	7.0	6.0	4.0	3.5	1.5	1.0	3.0	3.0	2.0	1.5	1.5	1.5	1.0	1.0	1.0
Three-month deposits (Zurich)	6.2	6.4	4.6	5.0	4.5	4.0	5.2	4.7	4.4	3.8	4.1	4.2	4.0	3.7	3.5
Government bond yield	7.6	7.2	4.3	3.5	2.4	1.4	3.2	2.7	2.1	1.4	1.1	1.8	1.5	1.3	1.2
Savings deposits of cantonal banks	5.0	5.1	4.4	3.4	3.1	2.4	3.3	3.3	3.1	2.9	2.5	2.4	2.4	2.3	2.1
Memorandum items:															
Euro-dollar, three-month	5.9	3.7	3.2	4.6	5.9	5.4	6.2	6.0	5.8	5.7	5.2	5.4	5.5	5.4	5.4
Euro-bond yields (dollars)	8.2	6.8	6.1	7.0	6.8	6.3	7.9	7.0	6.3	6.0	5.8	6.5	6.6	6.2	6.3
Capital market															
Foreign bonds	31 931	25 196	45 121	36 872	37 630	36 039	8 450	6 239	10 973	11 967	9 777	10 619	7 999	7 644	9 526
Domestic bonds	17 631	25 793	34 909	25 157	24 102	23 785	6 417	5 540	5 700	6 445	5 676	6 488	6 174	5 447	
Public market issues[1]	16 202	24 558	32 757	24 793	23 781	23 285	6 362	5 393	5 654	6 372	5 452	6 399	6 083	5 351	6 275
of which:															
Government	4 332	12 286	15 793	11 166	8 574	9 681	3 029	2 425	1 513	1 607	2 344	2 232	2 435	2 670	3 194
Financial institutions[2]	9 663	8 624	11 786	10 530	10 805	10 320	2 741	1 882	2 924	3 259	2 277	3 584	2 267	2 193	2 025
Other private	2 207	3 648	5 178	3 097	4 402	3 283	593	1 086	1 218	1 505	831	583	1 382	487	1 056
Shares															
Public market issues	1 899	2 598	3 286	2 406	1 829	1 514	709	154	839	127	93	895	363	163	95
Redemptions	11 419	18 474	30 205	32 743	28 948	41 951	4 825	5 061	7 867	11 196	10 732	13 319	8 551	9 349	8 976

1. According to date of payment.
2. Including holding companies.

Source: Banque nationale suisse, *Bulletin mensuel*, and OECD, *Financial Statistics.*

Table H. **Foreign trade by area**

Million US dollars

	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
	Exports, f.o.b.									
OECD countries	36 639	41 091	42 020	52 197	50 579	53 840	50 575	56 652	66 138	64 670
EU	28 456	31 873	32 526	40 768	39 683	42 372	39 426	43 496	50 737	48 445
of which: Germany	9 871	10 916	10 718	14 046	14 580	15 357	14 483	16 408	19 377	18 070
Other OECD	8 183	9 218	9 494	11 429	10 896	11 468	11 149	13 155	15 401	16 225
of which: USA	4 005	4 344	4 589	5 080	5 038	5 593	5 636	6 428	7 051	7 492
Non-OECD countries	8 611	9 523	9 491	11 336	10 751	11 692	12 560	13 413	15 248	15 094
CEEC	860	933	972	1 140	758	743	571	407	581	536
OPEC	1 888	1989	1 592	1979	2 078	2 365	2 443	2 102	2 523	2 227
Other	5 863	6 601	6 927	8 217	7 915	8 584	9 546	10 904	12 144	12 330
Total	45 250	50 614	51 511	63 533	61 330	65 531	63 135	70 065	81 386	79 764
	Imports, c.i.f.									
OECD countries	45 923	51 181	52 914	63 254	59 962	60 039	55 510	61 982	73 304	71 418
EU	39 714	43 923	45 099	54 388	50 738	51 555	47 923	53 766	63 764	61 805
of which: Germany	17 383	19 263	19 589	23 492	21 711	21 962	19 794	22 232	26 939	24 555
Other OECD	6 209	7 258	7 815	8 866	9 224	8 484	7 587	8 216	9 540	9 614
of which: USA	2 687	3 125	3 723	4 269	4 867	4 191	3 921	4 479	5 114	5 611
Non-OECD countries	4 487	5 129	5 285	6 293	6 309	5 626	5 283	5 729	6 599	6 788
CEEC	411	351	409	433	688	429	115	155	253	252
OPEC	715	604	328	851	912	656	720	834	963	1 168
Other	3 361	4 174	4 549	5 008	4 708	4 542	4 447	4 740	5 383	5 369
Total	50 410	56 310	58 199	69 546	66 271	65 665	60 794	67 711	79 903	78 207

Source: OECD, *Foreign Trade Statistics*, Series A.

Table I. **Foreign trade by commodity group**

Million US dollars

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
	Exports, f.o.b.										
0. Food and live animals	731	982	1 154	1 145	1 143	1 380	1 394	1 495	1 499	1 618	1 866
1. Beverages and tobacco	138	188	228	258	255	352	377	416	396	513	545
2. Crude materials, inedible, except fuels	383	457	526	680	693	741	685	721	677	795	983
3. Mineral fuels, lubricants and related materials	91	65	62	34	22	45	92	71	86	92	82
4. Animal and vegetable oils and fats	18	17	19	18	20	23	23	22	20	18	20
5. Chemicals	5 879	8 137	9 976	11 016	11 228	13 769	13 890	15 770	16 096	17 951	21 236
6. Manufactured goods, classified chiefly by material	5 572	7 370	8 732	10 025	10 136	12 068	11 332	11 617	11 102	11 654	12 823
7. Machinery and transport equipment	8 546	12 249	15 045	15 904	15 643	19 652	19 062	20 058	18 759	20 807	25 673
8. Miscellaneous manufactured articles	5 892	8 095	9 548	10 763	11 828	15 149	14 318	15 447	15 703	16 738	18 112
9. Commodities and transactions, not classified according to kind	35	36	72	101	108	98	61	63	204	89	301
Total	27 284	37 595	45 361	49 945	51 076	63 276	61 234	65 678	64 543	70 276	81 641
	Imports, c.i.f.										
0. Food and live animals	1 949	2 536	2 960	3 045	2 867	3 353	3 393	3 464	3 328	3 681	3 985
1. Beverages and tobacco	392	531	651	715	726	891	884	833	785	864	1 007
2. Crude materials, inedible, except fuels	1 039	1 273	1 484	1 776	1 833	2 019	1 750	1 739	1 567	1 768	2 069
3. Mineral fuels, lubricants and related materials	3 074	2 421	2 252	2 066	2 299	3 155	3 080	2 813	2 423	2 280	2 317
4. Animal and vegetable oils and fats	64	66	56	56	54	69	63	74	78	99	111
5. Chemicals	3 665	4 798	5 721	6 727	6 875	8 108	7 900	8 629	8 644	9 766	11 721
6. Manufactured goods, classified chiefly by material	6 476	8 993	10 994	11 980	12 920	14 773	13 458	13 002	12 593	13 451	15 270
7. Machinery and transport equipment	7 974	12 225	15 975	18 201	18 264	21 988	21 139	20 229	18 335	20 899	26 795
8. Miscellaneous manufactured articles	5 822	8 275	10 415	11 710	12 265	15 037	14 584	14 791	14 066	14 885	16 676
9. Commodities and transactions, not classified according to kind	179	80	50	61	86	260	230	119	182	221	199
Total	30 632	41 197	50 557	56 335	58 189	69 651	66 480	65 692	62 000	67 915	80 152

Source: OECD, *Foreign Trade Statistics*, Series C.

Table J. **Balance of payments**

Billion Swiss francs

	1987	1988	1989	1990	1991	1992	1993	1994	1995[1]	1996[1]
Current account balance	**11.3**	**13.2**	**11.5**	**12.0**	**15.2**	**21.3**	**28.8**	**24.4**	**25.0**	**25.1**
Merchandise trade balance	–8.9	–9.3	–12.1	–10.0	–8.5	–1.4	2.5	2.2	1.0	1.0
Non-factor services balance	12.3	12.1	13.1	13.0	14.7	15.1	16.8	16.1	14.9	15.0
Factor services balance	10.1	13.0	13.2	12.2	12.7	11.7	13.5	10.8	14.0	14.0
Net unrequited transfers	–2.2	–2.5	–2.8	–3.2	–3.7	–4.2	–4.1	–4.7	–4.9	–4.9
Capital account balance	**–14.0**	**–21.8**	**–10.8**	**–18.2**	**–17.1**	**–22.8**	**–30.5**	**–22.5**	**–21.1**	
Foreign direct investment	1.1	–12.7	–9.2	–2.7	–5.6	–7.4	–13.1	–10.2	–11.7	
Abroad	1.9	12.7	12.8	8.8	9.4	8.0	12.9	14.8	14.0	
Into Switzerland	3.0	0.1	3.7	6.2	3.8	0.6	–0.1	4.6	2.3	
Porfolio investment	–2.6	–10.9	–4.9	–1.6	–17.2	–8.6	–26.4	–24.8	–5.2	
Abroad	9.5	20.2	15.4	0.8	25.2	13.6	44.8	26.1	11.0	
Into Switzerland	6.9	9.4	10.4	–0.8	8.0	5.0	18.5	1.2	5.9	
Banking sector	–15.0	–3.5	2.5	–10.4	8.0	–8.7	14.1	14.8	–1.1	
of which:										
Increase in credit claims	25.6	0.4	–11.2	8.3	–2.6	6.8	3.6	26.3	–11.2	
Increase in credit liabilities	20.0	4.9	13.6	16.7	2.7	–1.0	6.0	41.4	1.9	
Net increase in fiduciary funds	–7.5	–7.7	–24.1	–17.3	2.8	0.8	14.7	–0.5	7.6	
Enterprises	–2.9	0.1	–3.5	–1.5	–0.8	0.1	1.0	–2.0	1.7	
Increase in claims	–3.3	–2.7	–5.4	–4.3	0.3	–4.3	0.8	–11.9	0.8	
Increase in liabilities	0.3	2.9	1.9	2.8	–1.1	4.4	0.2	9.8	0.9	
Other private sector	4.9	5.5	4.8	–1.8	–1.1	2.1	–6.0	–0.1	–5.1	
Other public sector	0.5	–0.4	–0.6	–0.3	–0.4	–0.2	–0.1	–0.1	0.2	
Changes in national bank reserves[2]	–1.1	1.5	–3.8	2.4	–3.1	–6.9	–1.4	0.9	3.7	
Revaluation of national bank reserves[2]	–3.5	2.0	1.4	–4.0	1.7	0.7	0.8	–2.3	–3.5	
Errors and omissions	7.4	5.1	1.8	7.9	3.3	7.8	2.5	–0.3	–3.9	

1. Provisional data.
2. – = increase.

Source: Banque nationale suisse.

Table K. **Labour market: structural and institutional characteristics**

	1975	1980	1985	1990	1991	1992	1993	1994	1995	1996
Labour force (thousand)	3 540	3 629	3 721	3 839	3 905	3 896	3 946	3 947	3 937	3 958
of which: Foreigners,[1] end-August	754	706	756	955	989	977	950	947	939	911
Participation rate[2] (per cent)										
Total	. .	. .	60.4	64.2	64.7	64.2	64.3	63.8	63.6	63.9
Male	. .	. .	78.5	79.7	79.3	78.4	78.4	77.4	77.1	76.9
Female	. .	. .	43.6	49.7	51.1	51.0	51.2	51.2	51.0	51.8
Employment/population aged 15-64	74.5	74.3	74.8	83.2	83.1	82.0	81.2	80.2	80.4	. .
Employment by sector (per cent of total)										
Agriculture	7.6	6.9	6.1	4.2	4.0	3.9	4.0	3.8	4.1	4.5
Industry	42.2	38.1	35.6	32.2	31.0	29.7	28.9	28.9	28.8	27.8
Services	50.2	55.0	58.3	63.6	65.0	66.5	67.1	67.2	67.0	67.7
Unemployed (thousand)	10.2	6.3	30.3	18.1	39.2	92.3	163.1	171.0	153.3	168.6
Unemployed foreigners (thousand)	. .	. .	. .	7.6	16.9	36.7	63.5	68.3	64.7	75.9
Unemployment rate	0.3	0.2	1.0	0.5	1.1	2.5	4.5	4.7	4.2	4.7
Vacancies: Full-time[3]	2 813	12 312	7 875	16 711	10 145	7 741	5 370	4 776	4 898	5 040
Part-time	. .	. .	647	550	404	425	373	449	503	566
Number of days not worked (lock-out and strikes)	1 733	5 178	662	4 090	51	673	–	14 380	351	. .

1. Includes established foreign workers, workers on a one-year contract and seasonal and border workers.
2. Labour force as a percentage of the corresponding population aged 15 years old and over.
3. The decomposition between full and part-time begins in 1983, so for 1975 and 1980 it is total vacancies.

Source: Office fédéral de la statistique, *Annuaire statistique de la Suisse*, 1995; Département fédéral de l'économie publique, *La vie économique;* and OECD, *Labour Force Statistics.*

Table L. **The structure of taxation**

Per cent of GDP

	1980	1985	1990	1991	1992	1993	1994	1995	1996[1]
Tax receipts	30.8	32.0	31.5	31.2	31.9	33.2	33.6	33.9	35.1
Personal income tax	11.0	11.2	10.8	10.7	11.1	10.9	11.5	10.7	11.3
Corporate tax	1.8	1.9	2.1	2.0	2.0	1.9	2.0	1.9	2.0
Social security tax	9.5	10.3	10.4	10.7	11.0	12.4	12.2	12.7	13.0
Property taxes	2.2	2.6	2.5	2.2	2.3	2.4	2.5	2.4	2.4
Tax on goods and services	6.3	6.1	5.8	5.6	5.4	5.5	5.5	6.3	6.3
Memorandum:									
Income tax as a per cent of total tax	41.4	40.9	41.0	40.7	41.1	38.6	39.9	37.2	38.0

1. Provisional data.

Source: OECD, Revenue Statistics of OECD Member Countries.

Table M. **Interest rate margins in banking**[1]

Per cent of average balance sheet total

	1981	1988	1989	1990	1991	1992	1993	1994	1995
Large banks	1.13	1.37	1.37	1.25	1.53	1.68	1.92	1.44	1.22
Cantonal banks	1.03	1.12	1.13	1.22	1.34	1.35	1.48	1.20	1.22
Regional and saving banks	1.25	1.21	1.21	1.32	1.48	1.51	1.51	1.20	1.44
Loans associations and agricultural credit co-operatives	0.98	0.95	0.96	1.04	1.13	1.14	1.19	1.06	1.16
Other Swiss and foreign banks	1.90	1.99	2.05	2.01	2.11	2.13	2.49	1.78	1.90

1. Interest received less interest paid.

Source: OECD, *Bank Profitability (Statistical Supplement), Financial Statements of Banks*, 1997.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Greece
Population											
Total	Thousands	1994	17 840	8 031	10 124	29 251	5 206	5 088	57 960	81 407	10 430
Inhabitants per sq. km	Number	1994	2	96	332	3	121	15	106	228	79
Net average annual increase over previous 10 years	%	1994	1.4	0.6	0.3	1.6	0.2	0.4	0.5	2.9	0.5
Employment											
Total civilian employment (TCE)[2]	Thousands	1994	7 680 (93)	3 737	3 724 (92)	13 292	2 508	2 015	21 781 (93)	35 894	3 790
of which: Agriculture	% of TCE	1994	5.3 (93)	7.2	2.6 (92)	4.1	5.1	8.3	5.1 (93)	3.3	20.8
Industry	% of TCE	1994	23.7 (93)	33.2	27.7 (92)	22.6	26.8	26.8	27.7 (93)	37.6	23.6
Services	% of TCE	1994	71 (93)	59.6	69.7 (92)	73.3	68.1	64.9	67.2 (93)	59.1	55.5
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1994	331.6	198.1	227.9	544	146.7	97.2	1 328.5	1 832.3	73.1 (93)
Per capita	US$	1994	18 588	24 670	22 515	18 598	28 181	19 106	22 944	27 826	7 051 (93)
At current prices using current PPP's[3]	Bill. US$	1994	327.9	162.3	204.2	596.7	107	82.5	1 111.8	1 601.7	118
Per capita	US$	1994	18 382	20 210	20 166	20 401	20 546	16 208	19 201	24 325	1 450
Average annual volume growth over previous 5 years	%	1994	2.2	2.5	1.6	1.1	1.9	–1.6	1.1	2.6	1.4 (93)
Gross fixed capital formation (GFCF)	% of GDP	1994	21.4	24.8	17.4	18.7	14.8	14.3	18.1	18.5	17.4 (93)
of which: Machinery and equipment	% of GDP	1993	9.8	8.7	7.8	6.2	7.2	5.90	8.10	7.5	7.8
Residential construction	% of GDP	1993	5.2	6.3	4.6	6.1	2.9	3.7	4.9	6.5	3.6
Average annual volume growth over previous 5 years	%	1994	0.8	3.7	0.4	–0.1	–2.8	–12.9	–1	0.8	2.7 (93)
Gross saving ratio[4]	% of GDP	1994	16.8	25.3	22	16	17	16.6	19	21	15.5 (93)
General government											
Current expenditure on goods and services	% of GDP	1994	17.5	18.8	15	20.2	25.3	22.4	19.6	17.7	19.1 (93)
Current disbursements[5]	% of GDP	1993	36.9	48.4	55.3	49	61.1	58.9	51.5	45.6	51.2
Current receipts	% of GDP	1993	33.5	48.6	50.1	43	58.3	52.5	46.8	45.7	40.2
Net official development assistance	% of GNP	1993	0.34	0.31	0.41	0.46	1.03	0.76	0.66	0.44	..
Indicators of living standards											
Private consumption per capita using current PPP's[3]	US$	1993	10 803	10 546	12 090	11 863	10 042	8 814	11 395	10 733	6 367
Passenger cars, per 1 000 inhabitants	Number	1990	430	382	387	469	311	386	413	480[8]	169
Telephones, per 1 000 inhabitants	Number	1991	464	432	410	586	577	544	511	420[8]	413
Television sets, per 1 000 inhabitants	Number	1991	480	478	451	639	536	501	407	556[8]	197
Doctors, per 1 000 inhabitants	Number	1993	2.2 (91)	2.3	3.7	2.2	2.8 (92)	2.6 (92)	2.8	3.2 (92)	3.8 (92)
Infant mortality per 1 000 live births	Number		6.1	6.5	8	6.8	5.4	4.4	6.5	5.8	8.5
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1994	3	5.5	3.7	3.3	3.5	4.8	3.5	5.2	14.6
Consumer prices	%	1994	3	3.4	2.8	2.8	2.1	3.3	2.5	3.3	16.2
Foreign trade											
Exports of goods, fob*	Mill. US$	1994	47 363	44 881	137 259[7]	165 358	41 850	29 514	235 337	422 243	8 958
As % of GDP	%	1994	14.3	22.7	60.2	30.4	28.5	30.4	17.7	23	11.5 (93)
Average annual increase over previous 5 years	%	1994	5	6.7	6.5	7.1	8.3	4.9	5.6	4.4	3.4
Imports of goods, cif*	Mill. US$	1994	49 731	55 071	126 006[7]	148 297	35 932	23 091	220 508	376 566	21 111
As % of GDP	%	1994	15	27.8	55.3	27.3	24.5	23.8	16.6	20.6	30.1 (93)
Average annual increase over previous 5 years	%	1994	4	7.2	5	5.4	6.1	–1.3	3.5	6.9	5.4
Total official reserves[6]	Mill. SDRs	1994	7 730	11 523	9 505[7]	8 416	6 203	7 304	17 986	52 994	9 924
As ratio of average monthly imports of goods	Ratio	1994	1.9	2.5	0.9	0.7	2.1	3.8	1	1.7	5.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Iceland	Ireland	Italy	Japan	Luxembourg	Mexico	Netherlands	New Zealand
Population										
Total	Thousands	1994	267	3 571	57 190	124 960	398	93 010	15 382	3 526
Inhabitants per sq. km	Number	1994	3	51	190	331	153	47	377	13
Net average annual increase over previous 10 years	%	1994	1.1	0.1	0	0.4	0.8	2	0.6	0.8
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	138	1 168 (93)	20 152 (93)	64 530	162 (91)	32 439	6 631	1 560
of which: Agriculture	% of TCE	1994	9.4	12.7 (93)	7.5 (93)	5.8	3.7 (91)	25.8	4	10.4
Industry	% of TCE	1994	26.1	27.7 (93)	33 (93)	34	31.5 (91)	22.2	23	25
Services	% of TCE	1994	65.2	59.7 (93)	59.6 (93)	60.2	64.8 (91)	52.1	73	64.6
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	6.2	52	1 017.8	4 590	10.6 (92)	371.2	334.3	51.2
Per capita	US$	1994	23 199	14 550	17 796	36 732	27 073 (92)	3 991	21 733	14 513
At current prices using current PPP's[3]	Bill. US$	1994	5.1	54.3	1 068.4	2 593.7	11.7	673.3	285.9	57.3
Per capita	US$	1994	19 271	15 212	18 681	20 756	29 454	7 239	18 589	16 248
Average annual volume growth over previous 5 years	%	1994	0.6	4.7	1	2.1	4.1 (92)	3	2.3	2.5
Gross fixed capital formation (GFCF)	% of GDP	1994	15.2	15.1	16.4	28.6	20.4 (93)	20.7	19.3	20
of which: Machinery and equipment	% of GDP	1993	3.9	6.3	7.4	11.5	..	9.4	8.6	9.3
Residential construction	% of GDP	1993	4.4	4.1	5.3	5.4	..	4.9	5.1	4.9
Average annual volume growth over previous 5 years	%	1994	–4	1	–2.3	1.4	6.5 (92)	7.7	0.4	5.8
Gross saving ratio[4]	% of GDP	1994	16.9	19.5	18.8	31.2	60.2 (92)	15.1	24.4	20.7
General government										
Current expenditure on goods and services	% of GDP	1994	20.6	16	17.1	9.8	17.1 (92)	11.8[9]	14.2	14.7
Current disbursements[5]	% of GDP	1993	34.9	..	53.2	26.9	..	..	55.4	..
Current receipts	% of GDP	1993	35.9	..	47.1	32.9	..	..	54.5	..
Net official development assistance	% of GNP	1993	..	0.15	0.42	0.27	0.34 (92)	..	0.88	0.22
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	11 546	7 750	11 029	11 791	15 545	4 853	10 726	9 266
Passenger cars, per 1 000 inhabitants	Number	1990	464	228	478	282	470	85	356	440
Telephones, per 1 000 inhabitants	Number	1991	527	300	400	454	511	70	477	436
Television sets, per 1 000 inhabitants	Number	1991	319	276	421	613	267	148	485	443
Doctors, per 1 000 inhabitants	Number	1993	3	1.7 (92)	1.7 (91)	1.7 (92)	2.1 (92)	1	2.5 (90)	2
Infant mortality per 1 000 live births	Number		4.8	5.9	7.3	4.5 (92)	8.5 (92)	18	6.3	7.3
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	..	4.6	5.9	2.4	..	5.3	3.2	2.1
Consumer prices	%	1994	6.3	2.7	5.2	2	3.1	16.1	2.8	2.5
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	1 628	34 125	189 802	396 149	..	60 882	155 084	12 169
As % of GDP	%	1994	26.3	65.7	18.6	8.6	..	16.4	46.4	23.8
Average annual increase over previous 5 years	%	1994	2.7	10.5	6.2	7.6	..	21.7	7.6	6.5
Imports of goods, cif*	Mill. US$	1994	1 464	25 812	167 690	274 916	..	79 346	139 800	11 859
As % of GDP	%	1994	23.6	49.7	16.5	6	..	21.4	41.8	23.2
Average annual increase over previous 5 years	%	1994	0.7	8.1	1.9	5.5	..	25.5	6.1	6.1
Total official reserves[6]	Mill. SDRs	1994	201	4 189	22 102	86 214	..	4 301	23 655	2 540
As ratio of average monthly imports of goods	Ratio	1994	1.6	1.9	1.6	3.8	..	0.7	2	2.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population										
Total	Thousands	1994	4 337	9 900	39 150	8 781	6 994	60 573	58 375	260 651
Inhabitants per sq. km	Number	1994	13	107	78	20	169	78	238	28
Net average annual increase over previous 10 years	%	1994	0.5	–0.1	0.2	0.5	0.7	2.1	0.3	1
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	1970 (93)	4 372	11 760	3 926	3 772	19 664	25 044 (93)	123 060
of which: Agriculture	% of TCE	1994	5.6 (93)	11.5	9.8	3.4	4	44.8	2.2 (93)	2.9
Industry	% of TCE	1994	23.1 (93)	32.8	30.1	25	28.8	22.2	26.2 (93)	24
Services	% of TCE	1994	71.3 (93)	55.7	60.2	71.6	67.2	33	71.6 (93)	73.1
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	103.4 (93)	87	482.4	196.6	257.3	130.7	1 019.5	6 649.8
Per capita	US$	1994	23 984 (93)	8 792	12 321	22 389	36 790	2 157	17 468	25 512
At current prices using current PPP's[3]	Bill. US$	1994	95.3	122	531.7	153	167.4	319.3	1 030.2	6 649.8
Per capita	US$	1994	21 968	12 335	13 581	17 422	23 942	5 271	17 650	25 512
Average annual volume growth over previous 5 years	%	1994	2.1 (93)	1.4	1.5	–0.3	0.5	3.6	0.8	2.1
Gross fixed capital formation (GFCF)	% of GDP	1994	22 (93)	25.7	19.8	13.7	22.8	24.5	15	17.2
of which: Machinery and equipment	% of GDP	1993	..	..	5.7	5.7	7.5	10.3	7.3	7.7
Residential construction	% of GDP	1993	..	..	4.1	4	14.9[10]	9.1	3.1	4
Average annual volume growth over previous 5 years	%	1994	–3.93	2.7	–1.2	–7.6	–0.4	5.1	–2.1	4.6
Gross saving ratio[4]	% of GDP	1994	21.9 (93)	24.2	18.8	13.7	29.3	22.5	13.5	16.2
General government										
Current expenditure on goods and services	% of GDP	1994	22.1 (93)	17.2	16.9	27.3	14.1	11.7	21.6	16.4
Current disbursements[5]	% of GDP	1993	..	..	43.7	67.3	36.7	..	42.7	35.8
Current disbursements[5]	% of GDP	1993	..	..	40.1	59	36	..	36.8	31.7
Current receipts	% of GDP	1993	1.23	0.36	0.32	1.33	0.49	..	0.34	0.19
Net official development assistance	% of GNP	1993	9 826	7 780	8 412	9 240	13 730	3 617	10 942	16 444
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	378	260	307	418	441	29	361	568
Passenger cars, per 1 000 inhabitants	Number	1990	515	273	340	687	603	143	445	553
Telephones, per 1 000 inhabitants	Number	1991	423	187	400	468	406	175	434	814
Television sets, per 1 000 inhabitants	Number	1991	3.2 (92)	2.9	4.1	3	3	0.9	1.5 (92)	2.3 (92)
Doctors, per 1 000 inhabitants	Number	1993	5	8.7	7.6	4.8	5.6	52.6	6.6	8.5 (92)
Infant mortality per 1 000 live births	Number		4	..	7.2	5.4	..	..	6.7	2.8
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	2.7	9	5.6	5.7	3.9	73	4.6	3.6
Consumer prices	%	1994	34 645	17 072	73 129	61 122	70 467	18 456	205 170	512 627
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	30.9 (93)	19.6	15.2	31.1	27.4	14.1	20.1	7.7
As % of GDP	%	1994	5	6.1	10.5	3.4	6.4	9.5	6.1	7.1
Average annual increase over previous 5 years	%	1994	27 345	25 967	92 182	51 730	68 126	22 976	227 026	663 256
Imports of goods, cif*	Mill. US$	1994	23.3 (93)	29.9	19.1	26.3	26.5	17.6	22.3	10
As % of GDP	%	1994	2.9	6.6	5.2	1	3.2	37.9	2.8	7
Average annual increase over previous 5 years	%	1994	13 033	10 627	28 475	15 929	23 790	4 911	28 094	43 350
Total official reserves[6]	Mill. SDRs	1994	5.7	4.9	3.7	3.7	4.2	2.6	1.5	0.8

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

MAIN SALES OUTLETS OF OECD PUBLICATIONS
PRINCIPAUX POINTS DE VENTE DES PUBLICATIONS DE L'OCDE

AUSTRALIA – AUSTRALIE
D.A. Information Services
648 Whitehorse Road, P.O.B 163
Mitcham, Victoria 3132 Tel. (03) 9210.7777
Fax: (03) 9210.7788

AUSTRIA – AUTRICHE
Gerold & Co.
Graben 31
Wien I Tel. (0222) 533.50.14
Fax: (0222) 512.47.31.29

BELGIUM – BELGIQUE
Jean De Lannoy
Avenue du Roi, Koningslaan 202
B-1060 Bruxelles Tel. (02) 538.51.69/538.08.41
Fax: (02) 538.08.41

CANADA
Renouf Publishing Company Ltd.
5369 Canotek Road
Unit 1
Ottawa, Ont. K1J 9J3 Tel. (613) 745.2665
Fax: (613) 745.7660

Stores:
71 1/2 Sparks Street
Ottawa, Ont. K1P 5R1 Tel. (613) 238.8985
Fax: (613) 238.6041

12 Adelaide Street West
Toronto, QN M5H 1L6 Tel. (416) 363.3171
Fax: (416) 363.5963

Les Éditions La Liberté Inc.
3020 Chemin Sainte-Foy
Sainte-Foy, PQ G1X 3V6 Tel. (418) 658.3763
Fax: (418) 658.3763

Federal Publications Inc.
165 University Avenue, Suite 701
Toronto, ON M5H 3B8 Tel. (416) 860.1611
Fax: (416) 860.1608

Les Publications Fédérales
1185 Université
Montréal, QC H3B 3A7 Tel. (514) 954.1633
Fax: (514) 954.1635

CHINA – CHINE
Book Dept., China National Publications
Import and Export Corporation (CNPIEC)
16 Gongti E. Road, Chaoyang District
Beijing 100020 Tel. (10) 6506-6688 Ext. 8402
(10) 6506-3101

CHINESE TAIPEI – TAIPEI CHINOIS
Good Faith Worldwide Int'l. Co. Ltd.
9th Floor, No. 118, Sec. 2
Chung Hsiao E. Road
Taipei Tel. (02) 391.7396/391.7397
Fax: (02) 394.9176

CZECH REPUBLIC – RÉPUBLIQUE TCHÈQUE
National Information Centre
NIS – prodejna
Konviktská 5
Praha 1 – 113 57 Tel. (02) 24.23.09.07
Fax: (02) 24.22.94.33
E-mail: nkposp@dec.niz.cz
Internet: http://www.nis.cz

DENMARK – DANEMARK
Munksgaard Book and Subscription Service
35, Nørre Søgade, P.O. Box 2148
DK-1016 København K Tel. (33) 12.85.70
Fax: (33) 12.93.87

J. H. Schultz Information A/S,
Herstedvang 12,
DK – 2620 Albertslung Tel. 43 63 23 00
Fax: 43 63 19 69
Internet: s-info@inet.uni-c.dk

EGYPT – ÉGYPTE
The Middle East Observer
41 Sherif Street
Cairo Tel. (2) 392.6919
Fax: (2) 360.6804

FINLAND – FINLANDE
Akateeminen Kirjakauppa
Keskuskatu 1, P.O. Box 128
00100 Helsinki

Subscription Services/Agence d'abonnements :
P.O. Box 23
00100 Helsinki Tel. (358) 9.121.4403
Fax: (358) 9.121.4450

***FRANCE**
OECD/OCDE
Mail Orders/Commandes par correspondance :
2, rue André-Pascal
75775 Paris Cedex 16 Tel. 33 (0)1.45.24.82.00
Fax: 33 (0)1.49.10.42.76
Telex: 640048 OCDE
Internet: Compte.PUBSINQ@oecd.org

Orders via Minitel, France only/
Commandes par Minitel, France
exclusivement : 36 15 OCDE

OECD Bookshop/Librairie de l'OCDE :
33, rue Octave-Feuillet
75016 Paris Tel. 33 (0)1.45.24.81.81
33 (0)1.45.24.81.67

Dawson
B.P. 40
91121 Palaiseau Cedex Tel. 01.89.10.47.00
Fax: 01.64.54.83.26

Documentation Française
29, quai Voltaire
75007 Paris Tel. 01.40.15.70.00

Economica
49, rue Héricart
75015 Paris Tel. 01.45.78.12.92
Fax: 01.45.75.05.67

Gibert Jeune (Droit-Économie)
6, place Saint-Michel
75006 Paris Tel. 01.43.25.91.19

Librairie du Commerce International
10, avenue d'Iéna
75016 Paris Tel. 01.40.73.34.60

Librairie Dunod
Université Paris-Dauphine
Place du Maréchal-de-Lattre-de-Tassigny
75016 Paris Tel. 01.44.05.40.13

Librairie Lavoisier
11, rue Lavoisier
75008 Paris Tel. 01.42.65.39.95

Librairie des Sciences Politiques
30, rue Saint-Guillaume
75007 Paris Tel. 01.45.48.36.02

P.U.F.
49, boulevard Saint-Michel
75005 Paris Tel. 01.43.25.83.40

Librairie de l'Université
12*a*, rue Nazareth
13100 Aix-en-Provence Tel. 04.42.26.18.08

Documentation Française
165, rue Garibaldi
69003 Lyon Tel. 04.78.63.32.23

Librairie Decitre
29, place Bellecour
69002 Lyon Tel. 04.72.40.54.54

Librairie Sauramps
Le Triangle
34967 Montpellier Cedex 2 Tel. 04.67.58.85.15
Fax: 04.67.58.27.36

A la Sorbonne Actual
23, rue de l'Hôtel-des-Postes
06000 Nice Tel. 04.93.13.77.75
Fax: 04.93.80.75.69

GERMANY – ALLEMAGNE
OECD Bonn Centre
August-Bebel-Allee 6
D-53175 Bonn Tel. (0228) 959.120
Fax: (0228) 959.12.17

GREECE – GRÈCE
Librairie Kauffmann
Stadiou 28
10564 Athens Tel. (01) 32.55.321
Fax: (01) 32.30.320

HONG-KONG
Swindon Book Co. Ltd.
Astoria Bldg. 3F
34 Ashley Road, Tsimshatsui
Kowloon, Hong Kong Tel. 2376.2062
Fax: 2376.0685

HUNGARY – HONGRIE
Euro Info Service
Margitsziget, Európa Ház
1138 Budapest Tel. (1) 111.60.61
Fax: (1) 302.50.35
E-mail: euroinfo@mail.matav.hu
Internet: http://www.euroinfo.hu//index.html

ICELAND – ISLANDE
Mál og Menning
Laugavegi 18, Pósthólf 392
121 Reykjavik Tel. (1) 552.4240
Fax: (1) 562.3523

INDIA – INDE
Oxford Book and Stationery Co.
Scindia House
New Delhi 110001 Tel. (11) 331.5896/5308
Fax: (11) 332.2639
E-mail: oxford.publ@axcess.net.in

17 Park Street
Calcutta 700016 Tel. 240832

INDONESIA – INDONÉSIE
Pdii-Lipi
P.O. Box 4298
Jakarta 12042 Tel. (21) 573.34.67
Fax: (21) 573.34.67

IRELAND – IRLANDE
Government Supplies Agency
Publications Section
4/5 Harcourt Road
Dublin 2 Tel. 661.31.11
Fax: 475.27.60

ISRAEL – ISRAËL
Praedicta
5 Shatner Street
P.O. Box 34030
Jerusalem 91430 Tel. (2) 652.84.90/1/2
Fax: (2) 652.84.93

R.O.Y. International
P.O. Box 13056
Tel Aviv 61130 Tel. (3) 546 1423
Fax: (3) 546 1442
E-mail: royil@netvision.net.il

Palestinian Authority/Middle East:
INDEX Information Services
P.O.B. 19502
Jerusalem Tel. (2) 627.16.34
Fax: (2) 627.12.19

ITALY – ITALIE
Libreria Commissionaria Sansoni
Via Duca di Calabria, 1/1
50125 Firenze Tel. (055) 64.54.15
Fax: (055) 64.12.57
E-mail: licosa@ftbcc.it

Via Bartolini 29
20155 Milano Tel. (02) 36.50.83

Editrice e Libreria Herder
Piazza Montecitorio 120
00186 Roma Tel. 679.46.28
Fax: 678.47.51

Libreria Hoepli
Via Hoepli 5
20121 Milano Tel. (02) 86.54.46
Fax: (02) 805.28.86

Libreria Scientifica
Dott. Lucio de Biasio 'Aeiou'
Via Coronelli, 6
20146 Milano Tel. (02) 48.95.45.52
Fax: (02) 48.95.45.48

JAPAN – JAPON
OECD Tokyo Centre
Landic Akasaka Building
2-3-4 Akasaka, Minato-ku
Tokyo 107 Tel. (81.3) 3586.2016
Fax: (81.3) 3584.7929

KOREA – CORÉE
Kyobo Book Centre Co. Ltd.
P.O. Box 1658, Kwang Hwa Moon
Seoul Tel. 730.78.91
Fax: 735.00.30

MALAYSIA – MALAISIE
University of Malaya Bookshop
University of Malaya
P.O. Box 1127, Jalan Pantai Baru
59700 Kuala Lumpur
Malaysia Tel. 756.5000/756.5425
Fax: 756.3246

MEXICO – MEXIQUE
OECD Mexico Centre
Edificio INFOTEC
Av. San Fernando no. 37
Col. Toriello Guerra
Tlalpan C.P. 14050
Mexico D.F. Tel. (525) 528.10.38
Fax: (525) 606.13.07
E-mail: ocde@rtn.net.mx

NETHERLANDS – PAYS-BAS
SDU Uitgeverij Plantijnstraat
Externe Fondsen
Postbus 20014
2500 EA's-Gravenhage Tel. (070) 37.89.880
Voor bestellingen: Fax: (070) 34.75.778

Subscription Agency/Agence d'abonnements :
SWETS & ZEITLINGER BV
Heereweg 347B
P.O. Box 830
2160 SZ Lisse Tel. 252.435.111
Fax: 252.415.888

NEW ZEALAND – NOUVELLE-ZÉLANDE
GPLegislation Services
P.O. Box 12418
Thorndon, Wellington Tel. (04) 496.5655
Fax: (04) 496.5698

NORWAY – NORVÈGE
NIC INFO A/S
Ostensjoveien 18
P.O. Box 6512 Etterstad
0606 Oslo Tel. (22) 97.45.00
Fax: (22) 97.45.45

PAKISTAN
Mirza Book Agency
65 Shahrah Quaid-E-Azam
Lahore 54000 Tel. (42) 735.36.01
Fax: (42) 576.37.14

PHILIPPINE – PHILIPPINES
International Booksource Center Inc.
Rm 179/920 Cityland 10 Condo Tower 2
HV dela Costa Ext cor Valero St.
Makati Metro Manila Tel. (632) 817 9676
Fax: (632) 817 1741

POLAND – POLOGNE
Ars Polona
00-950 Warszawa
Krakowskie Prezdmiescie 7 Tel. (22) 264760
Fax: (22) 265334

PORTUGAL
Livraria Portugal
Rua do Carmo 70-74
Apart. 2681
1200 Lisboa Tel. (01) 347.49.82/5
Fax: (01) 347.02.64

SINGAPORE – SINGAPOUR
Ashgate Publishing
Asia Pacific Pte. Ltd
Golden Wheel Building, 04-03
41, Kallang Pudding Road
Singapore 349316 Tel. 741.5166
Fax: 742.9356

SPAIN – ESPAGNE
Mundi-Prensa Libros S.A.
Castelló 37, Apartado 1223
Madrid 28001 Tel. (91) 431.33.99
Fax: (91) 575.39.98
E-mail: mundiprensa@tsai.es
Internet: http://www.mundiprensa.es

Mundi-Prensa Barcelona
Consell de Cent No. 391
08009 – Barcelona Tel. (93) 488.34.92
Fax: (93) 487.76.59

Libreria de la Generalitat
Palau Moja
Rambla dels Estudis, 118
08002 – Barcelona
(Suscripciones) Tel. (93) 318.80.12
(Publicaciones) Tel. (93) 302.67.23
Fax: (93) 412.18.54

SRI LANKA
Centre for Policy Research
c/o Colombo Agencies Ltd.
No. 300-304, Galle Road
Colombo 3 Tel. (1) 574240, 573551-2
Fax: (1) 575394, 510711

SWEDEN – SUÈDE
CE Fritzes AB
S–106 47 Stockholm Tel. (08) 690.90.90
Fax: (08) 20.50.21

For electronic publications only/
Publications électroniques seulement
STATISTICS SWEDEN
Informationsservice
S-115 81 Stockholm Tel. 8 783 5066
Fax: 8 783 4045

Subscription Agency/Agence d'abonnements :
Wennergren-Williams Info AB
P.O. Box 1305
171 25 Solna Tel. (08) 705.97.50
Fax: (08) 27.00.71

Liber distribution
Internatinal organizations
Fagerstagatan 21
S-163 52 Spanga

SWITZERLAND – SUISSE
Maditec S.A. (Books and Periodicals/Livres et périodiques)
Chemin des Palettes 4
Case postale 266
1020 Renens VD 1 Tel. (021) 635.08.65
Fax: (021) 635.07.80

Librairie Payot S.A.
4, place Pépinet
CP 3212
1002 Lausanne Tel. (021) 320.25.11
Fax: (021) 320.25.14

Librairie Unilivres
6, rue de Candolle
1205 Genève Tel. (022) 320.26.23
Fax: (022) 329.73.18

Subscription Agency/Agence d'abonnements :
Dynapresse Marketing S.A.
38, avenue Vibert
1227 Carouge Tel. (022) 308.08.70
Fax: (022) 308.07.99

See also – Voir aussi :
OECD Bonn Centre
August-Bebel-Allee 6
D-53175 Bonn (Germany) Tel. (0228) 959.120
Fax: (0228) 959.12.17

THAILAND – THAÏLANDE
Suksit Siam Co. Ltd.
113, 115 Fuang Nakhon Rd.
Opp. Wat Rajbopith
Bangkok 10200 Tel. (662) 225.9531/2
Fax: (662) 222.5188

TRINIDAD & TOBAGO, CARIBBEAN TRINITÉ-ET-TOBAGO, CARAÏBES
Systematics Studies Limited
9 Watts Street
Curepe
Trinidad & Tobago, W.I. Tel. (1809) 645.3475
Fax: (1809) 662.5654
E-mail: tobe@trinidad.net

TUNISIA – TUNISIE
Grande Librairie Spécialisée
Fendri Ali
Avenue Haffouz Imm El-Intilaka
Bloc B 1 Sfax 3000 Tel. (216-4) 296 855
Fax: (216-4) 298.270

TURKEY – TURQUIE
Kültür Yayinlari Is-Türk Ltd.
Atatürk Bulvari No. 191/Kat 13
06684 Kavaklidere/Ankara
Tel. (312) 428.11.40 Ext. 2458
Fax : (312) 417.24.90

Dolmabahce Cad. No. 29
Besiktas/Istanbul Tel. (212) 260 7188

UNITED KINGDOM – ROYAUME-UNI
The Stationery Office Ltd.
Postal orders only:
P.O. Box 276, London SW8 5DT
Gen. enquiries Tel. (171) 873 0011
Fax: (171) 873 8463

The Stationery Office Ltd.
Postal orders only:
49 High Holborn, London WC1V 6HB

Branches at: Belfast, Birmingham, Bristol, Edinburgh, Manchester

UNITED STATES – ÉTATS-UNIS
OECD Washington Center
2001 L Street N.W., Suite 650
Washington, D.C. 20036-4922
Tel. (202) 785.6323
Fax: (202) 785.0350
Internet: washcont@oecd.org

Subscriptions to OECD periodicals may also be placed through main subscription agencies.

Les abonnements aux publications périodiques de l'OCDE peuvent être souscrits auprès des principales agences d'abonnement.

Orders and inquiries from countries where Distributors have not yet been appointed should be sent to: OECD Publications, 2, rue André-Pascal, 75775 Paris Cedex 16, France.

Les commandes provenant de pays où l'OCDE n'a pas encore désigné de distributeur peuvent être adressées aux Éditions de l'OCDE, 2, rue André-Pascal, 75775 Paris Cedex 16, France.

12-1996

OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 97 26 1 P) ISBN 92-64-15438-8 – No. 49693 1997
ISSN 0304-3363